

Knowledge Mechanisms Across the Lifecycle of Large Language Models

¹Chinese Information Processing Laboratory, Institute of Software, Chinese Academy of Sciences

²The Laboratory of Cognition and Decision Intelligence for Complex Systems,
Institute of Automation, Chinese Academy of Sciences

³Tsinghua University

⁴MiLM Plus, Xiaomi Inc

Abstract

Knowledge is fundamental to the capabilities of large language models. Understanding how these models acquire, store, use, and revise knowledge has therefore become a central concern for interpretability, reliability, and safe deployment. Despite a rapidly growing of research, related studies remain scattered across communities with their own concepts, methods, and standards. There still lacks a unified view of how the mechanisms uncovered at different stages relate to one another. In this survey, we systematically review knowledge mechanisms in LLMs by following knowledge through a lifecycle. Specifically, we organize existing studies into six interconnected stages, tracing how knowledge is acquired, how it is represented, how it is utilized and augmented, how it evolves through editing and unlearning, how it transfers across models, modalities, and languages, and how it is evaluated. Treating these stages as successive states of one underlying object brings their dependencies into view, and one observation recurs throughout. Finally, we distill the main findings and open problems of each stage. Closing the distance between behavioral observation and mechanistic understanding is the key challenge for understanding and steering knowledge in the model.

1 Introduction

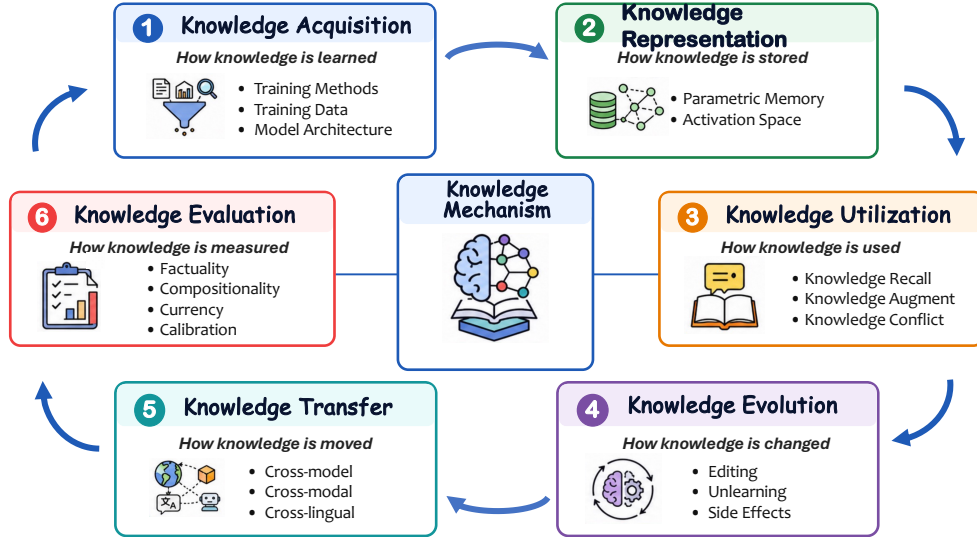


Figure 1: The knowledge lifecycle of large language models.

Large language models (LLMs) have reshaped artificial intelligence. They now achieve strong performance across question answering, commonsense reasoning, code generation, and mathematical problem solving (Brown et al., 2020). These striking capabilities ultimately rest on the knowledge a model acquires during training (Petroni et al., 2019; Roberts et al., 2020a). The internal processes by which a model acquires, encodes, uses, revises, and transfers this knowledge constitute its *knowledge mechanisms*. However, our understanding of these mechanisms has lagged behind the steady advance of the models’ external capabilities. How a model with billions of parameters takes in, organizes, and deploys knowledge is no longer only an engineering concern about reducing hallucinations. It has become a central question for interpretability, and answering it is a prerequisite for putting the study of these models on a scientific footing.

The study of knowledge in LLMs has recently gone through a shift in perspective. Earlier surveys largely took a macro view of the knowledge lifecycle, treating the model as an implicit knowledge base and organizing the field around its observable phenomena (Cao et al., 2023; Wang et al., 2024c). Such a view captures external behavior but not the internal machinery, so failures such as factual fabrication, answer inconsistency, and capability degradation after editing remain hard to explain (Gekhman et al., 2025; Li et al., 2024b; Yao et al., 2023b). Recent work has instead turned to the micro-level mechanisms inside the model, examining knowledge localization, feature superposition in activation space, knowledge conflict in retrieval-augmented generation, and the way multi-step reasoning unfolds (Dai et al., 2022; Meng et al., 2022; Conmy et al., 2023; Templeton, 2024; Lewis et al., 2020; OpenAI, 2024b; Guo et al., 2025). Yet this work remains scattered across communities with their own concepts, methods, and evaluation standards.

To bridge this gap, we examine knowledge mechanisms across the full lifecycle of large language models. We adopt a unified, stage-driven taxonomy that treats knowledge as an object under continuous change, tracing stage by stage how it is transformed as it moves through this lifecycle. Specifically, we organize the survey around six interconnected stages (Figure 1):

1. **Knowledge Acquisition (§3).** How knowledge enters the model, and how training strategies shape the knowledge ultimately acquired.
2. **Knowledge Representation (§4).** How the acquired knowledge is encoded and organized in the model’s parameters and activation space.
3. **Knowledge Utilization (§5).** How the model accesses its internal knowledge and augments it with external sources, and the conflicts that arise among these sources in practice.
4. **Knowledge Evolution (§6).** How knowledge is updated or removed after training, and the side effects of these interventions.
5. **Knowledge Transfer (§7).** How knowledge transfers across different models, modalities, and languages.
6. **Knowledge Evaluation (§8).** How to evaluate what a model knows across the lifecycle.

This survey helps trace where the knowledge-related failures of current models originate and how interventions at one stage propagate to the others. These connections are difficult to see from a single-stage view. This lifecycle perspective points beyond the black-box view toward models whose knowledge is mechanistically transparent, continually adaptable, and trustworthy.

2 Overview

The taxonomy in Figure 2 treats knowledge not as a static store but as an object that is acquired, encoded, used, revised, transferred, and evaluated across a single lifecycle. The first five stages form a progression rather than a set of independent topics. What a model acquires constrains what it can represent, how knowledge is represented governs how it is utilized, evolution revises or removes knowledge already inside the model, and transfer moves that knowledge to other models, modalities, and languages. Evaluation then runs alongside them, checking at every stage whether the knowledge inside the model is what we intend it to be.

Knowledge Acquisition (§3) is the foundational stage that determines what a model can later represent, use, and transfer. Because LLMs absorb knowledge implicitly from large corpora rather than from explicit symbolic records, what they end up knowing is shaped by the whole training pipeline rather than by any single choice. The stage has three dimensions. Training methods divide the labor across pre-training, which establishes the breadth of parametric knowledge, mid-training, which consolidates selected domains and capabilities, and post-training, which controls how knowledge is expressed and accessed rather than adding new facts. Training data governs which knowledge is learned in the first place, since scaling behavior, information density, and corpus composition together leave rare and long-tail facts underacquired. Model architecture sets the ceiling on how much can be stored and, in variants such as mixture-of-experts, changes how knowledge is organized and retrieved. These three dimensions are interdependent, so we should treat them as one system rather than as separate knobs.

Knowledge Representation (§4) examines how the acquired knowledge is encoded inside the model, moving from where it is stored to how it is instantiated during inference. From the parametric-memory perspective, we examine which components carry knowledge and how localization methods point to where a fact resides, separating attribution-based methods that estimate relevance under the current computation from intervention-based methods that give stronger causal evidence. From the activation-space perspective, we follow the line of work built on the linear representation hypothesis and its extension to multidimensional and nonlinear structures, and then turn to the superposition hypothesis, with sparse autoencoders as a tool for recovering interpretable features. The two perspectives are complementary rather than competing. A central open problem is to explain how parameter-level storage gives rise to the activation-space representations that drive behavior, and to validate representational claims through stability and causal intervention.

Knowledge Utilization (§5) concerns whether encoded knowledge is expressed when the model runs, how external evidence supplements it, and how competing signals are resolved. Models often hold factual signals that ordinary generation fails to surface, a mismatch we call the *encode-express gap*, so many factual errors are access failures rather than

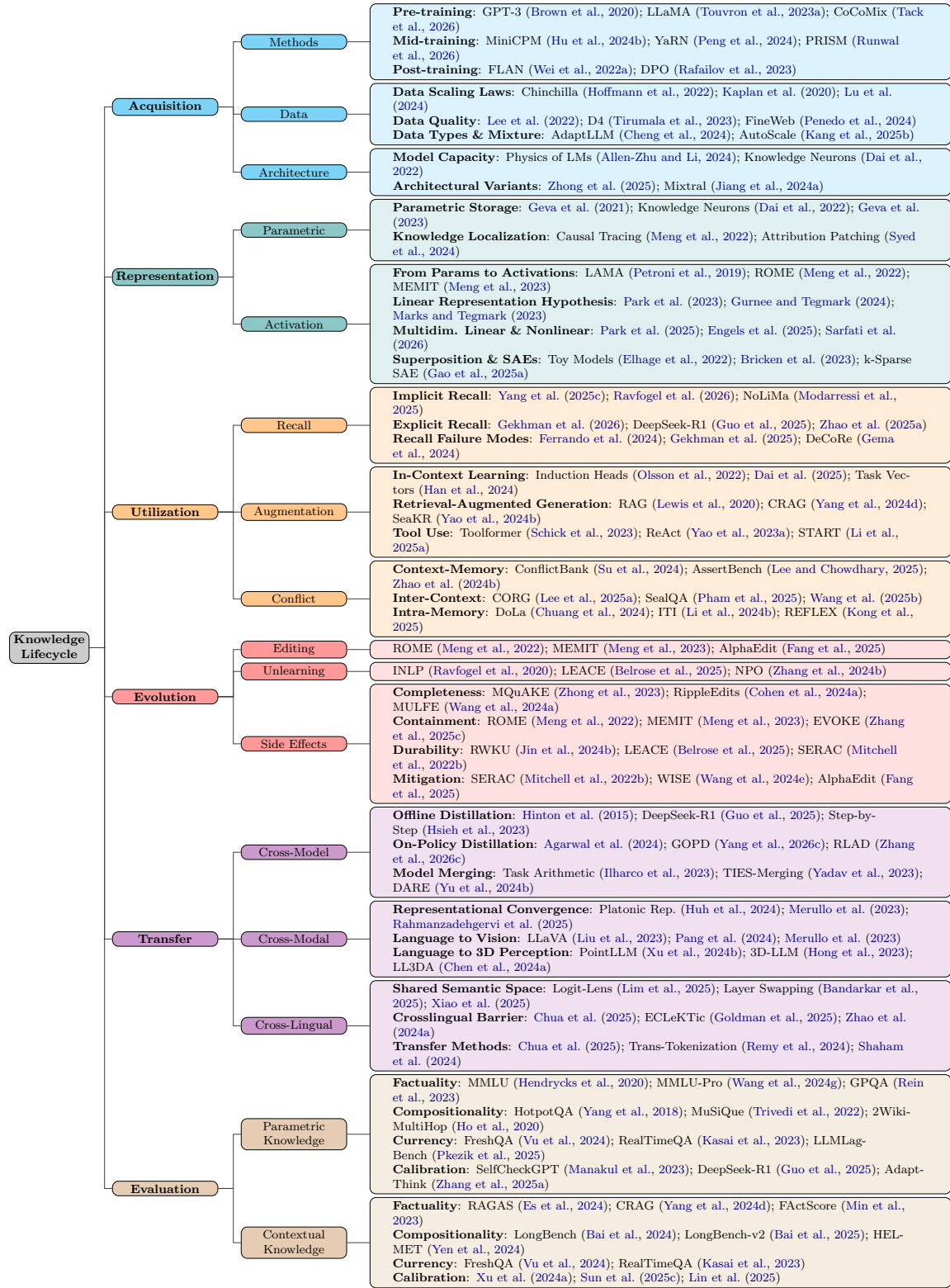


Figure 2: Taxonomy of knowledge mechanisms in large language models, organized along the six stages of the knowledge lifecycle (acquisition, representation, utilization, evolution, transfer, and evaluation).₄

gaps in what the model knows. Utilization is more than access, however. Some failures are compute-limited, where the answer is encoded, but extra reasoning is needed to extract it, while others are information-limited, where the fact is absent from the parameters. We organize utilization into three aspects. Knowledge recall studies how parametric knowledge is reached, through implicit recall or explicit reasoning, and why it fails at recognition, retrieval, or generation. Knowledge augmentation studies how in-context learning, retrieval, and tool use supply or reshape knowledge when parametric recall is not enough. Knowledge conflict studies how the model arbitrates when context disagrees with memory, when retrieved sources contradict each other, or when its own beliefs are internally inconsistent.

Knowledge Evolution (§6) addresses the fact that the knowledge fixed at pre-training is a single snapshot of a world that keeps changing, so models must be updated without the cost of full retraining. Knowledge editing overwrites specific outdated or incorrect facts, while knowledge unlearning removes undesirable content. Both tasks face the same challenge of changing the target while preserving related knowledge. Because knowledge is distributed rather than stored as separate records, a local intervention can spread beyond its target. We analyze these costs along three axes. Completeness asks whether a change reaches all the paraphrases and implications it should. Containment asks whether it leaves unrelated knowledge intact. Durability asks whether it survives stronger and adversarial evaluation rather than merely suppressing an answer that can later resurface. Mitigation strategies then address the scope and auditability of post-training knowledge changes.

Knowledge Transfer (§7) turns the focus to how knowledge moves across boundaries. We organize the section by the boundary being crossed. Cross-model transfer moves knowledge between models of different scales and architectures through offline distillation, on-policy distillation, and model merging in parameter space. Cross-modal transfer carries knowledge from language to vision and to 3D perception. Cross-lingual transfer moves knowledge between languages. Cross-modal and cross-lingual transfer share a common premise. Independently trained systems converge toward shared representational structures, an aligned space across modalities and a shared semantic space across languages. This convergence is what makes transfer across these boundaries possible. The barriers, by contrast, are specific to each boundary. Cross-model transfer contends with distribution mismatch and parameter interference, language-to-3D transfer with missing spatial priors, and cross-lingual transfer with a knowledge barrier that keeps facts tied to the language in which they were learned.

Knowledge Evaluation (§8) measures what a model knows across the lifecycle. Knowledge takes two qualitatively distinct forms. Parametric knowledge is encoded in the weights and reachable only through behavior, whereas contextual knowledge is supplied at inference time and is, in principle, directly checkable. We cross these two sources with four properties (factuality, compositionality, currency, and calibration), yielding a two-by-four matrix in which the same property looks different on each side. Strong scores on static benchmarks can overstate genuine knowledge, since a model may succeed through surface pattern matching, may verbalize reasoning that does not reflect its actual retrieval, or may reject correct evidence that conflicts with its parametric priors.

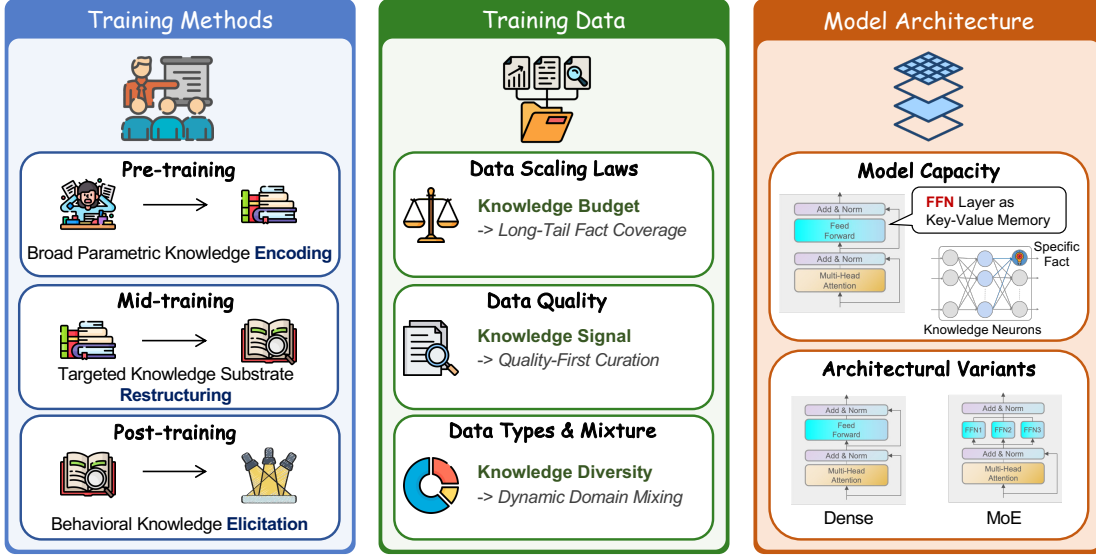


Figure 3: Taxonomy of knowledge acquisition in LLMs, illustrating how training methods (§3.1), training data (§3.2), and model architecture (§3.3) jointly shape acquisition.

3 Knowledge Acquisition

Knowledge acquisition constitutes the foundational stage of the LLM knowledge lifecycle, determining what information a model can subsequently represent, utilize, and transfer. Unlike symbolic AI systems that encode knowledge explicitly, LLMs acquire knowledge implicitly through exposure to large-scale text corpora, with knowledge distributed across billions of parameters. We organize this stage around three dimensions that govern knowledge acquisition, namely training methods (§3.1), training data (§3.2), and model architecture (§3.3). These dimensions are summarized in Figure 3.

3.1 Training Methods

We divide the training pipeline of modern LLMs into three knowledge-acquisition phases, namely pre-training, mid-training, and post-training. Pre-training establishes a broad parametric substrate, mid-training selectively refines this substrate prior to alignment, and post-training shapes how acquired knowledge is elicited and utilized during inference. Although post-training adds little new factual content, we include it here because acquisition is only complete once the acquired knowledge becomes accessible.

3.1.1 PRE-TRAINING

Finding

Pre-training is the main stage for large-scale knowledge acquisition, establishing the breadth of parametric knowledge that subsequent training stages can refine, reorganize, and access.

Pre-training is founded on the hypothesis that self-supervised learning over large text corpora enables models to acquire world knowledge from the statistical structure of language. The Transformer architecture (Vaswani et al., 2017) provided the computational framework for this process, while scaling studies such as GPT-3 (Brown et al., 2020) demonstrated that knowledge coverage and retrieval reliability improve predictably with increased model capacity, data, and compute. Subsequent generations of foundation models, including LLaMA (Touvron et al., 2023a,b; Dubey et al., 2024), Qwen3.5 (Team, 2026), Gemma 2 (Team et al., 2024), and DeepSeek-V4 (DeepSeek-AI, 2026), have retained this basic training paradigm.

Knowledge acquisition during pre-training is shaped by corpus statistics. Because optimization is driven by next-token prediction, frequently occurring concepts receive more learning signal than rare ones (Chang et al., 2024a). As a result, common entities and relations are typically acquired earlier and represented more robustly, whereas long-tail knowledge remains comparatively difficult to encode reliably, even at larger scales. Prior work also suggests that different forms of knowledge exhibit distinct acquisition dynamics. Procedural knowledge, particularly from code and structured reasoning traces, often generalizes beyond the specific training instances from which it is learned (Ruis et al., 2025), whereas factual associations tend to depend more directly on repeated exposure. These observations have motivated research on improving the efficiency of knowledge acquisition during pre-training itself. Recent approaches such as CoCoMix (Tack et al., 2026) and LFR (Prakriya et al., 2025) augment the standard next-token prediction paradigm with alternative representations or curriculum strategies, suggesting that corpus scale alone may not determine knowledge acquisition efficiency.

3.1.2 MID-TRAINING

Finding

Mid-training selectively consolidates knowledge and capabilities acquired during pre-training, providing the basis for downstream behavioral alignment.

Positioned between large-scale pre-training and task-specific post-training, mid-training was historically associated with continued pre-training (CPT), a broad term for additional self-supervised training on shifted or curated corpora (Mo et al., 2025; Tu et al., 2025). As training pipelines have evolved, the scope of mid-training has expanded beyond generic CPT. Building on existing studies and practices (Mo et al., 2025; Tu et al., 2025), we include *learning-rate annealing*, *long-context extension*, *domain-adaptive mid-training*, and *agentic mid-training*. This taxonomy captures the ways mid-training influences knowledge acquisition and consolidation before post-training, a bridging role also emphasized in recent analyses (Liu et al., 2026a).

The first two aspects primarily influence how knowledge is incorporated through training dynamics and context capacity. *Learning-rate annealing* exploits the decay window as a receptive period for targeted data integration. As the learning rate drops toward zero, curated knowledge introduced at this point can be absorbed more effectively than when reserved for later stages and can guide the model toward higher-quality local optima. Across frontier models, mixing supervised data during annealing consistently outperforms reserving the same data for post-training (Hu et al., 2024b; Dubey et al., 2024), and the advantage over naive CPT holds across target capability domains (Ibrahim et al., 2024). Meanwhile, *long-context extension* modifies positional encoding mechanisms and introduces long-document training data to expand the model’s effective receptive field. Because this intervention operates on an already-trained model, context windows can often be extended far beyond the original training horizon using only a small fraction of pre-training compute (Chen et al., 2023; Peng et al., 2024), making long-context adaptation a standard component of many frontier training pipelines (Dubey et al., 2024; Gao et al., 2025b).

The latter two aspects focus more directly on knowledge consolidation and preparation for downstream behaviors. *Domain-adaptive mid-training* addresses the mismatch between broad web-scale pre-training distributions and domains that matter disproportionately for downstream use, using targeted corpus construction to strengthen domain-specific capabilities. This strategy has become a common component of modern foundation-model pipelines and can yield gains in specialized domains (Mo et al., 2025; Tu et al., 2025; Runwal et al., 2026; Abdin et al., 2024). Extending this logic toward agentic behavior, *agentic mid-training* incorporates tool-use trajectories, planning traces, and interaction data before reinforcement learning. Emerging evidence suggests that reinforcement learning operates more effectively when these capabilities have already been partially internalized during mid-training, thereby reducing the amount of downstream alignment supervision required (Huang et al., 2026b; Zeng et al., 2026).

3.1.3 POST-TRAINING

Finding

Post-training acts as a behavioral control period that governs how acquired knowledge is expressed, accessed, and deployed during inference rather than expanding parametric capacity.

Across supervised fine-tuning (SFT), preference alignment, and reinforcement learning, post-training mainly elicits and reorganizes knowledge already present in the parametric substrate rather than injecting substantively new factual content. Instruction tuning on diverse formats teaches the model how to express and apply pre-trained knowledge, with task diversity driving generalization more than data volume (Wei et al., 2022a; Chung et al., 2024). Mechanistic studies suggest that SFT predominantly modifies upper transformer layers associated with task behavior while leaving lower layers that encode factual knowledge comparatively stable (Zhao et al., 2026a). Consistent with this view, knowledge introduced directly through fine-tuning is often less robust and transferable than knowledge acquired during large-scale pre-training (Mecklenburg et al., 2024; Jiang et al., 2026). Preference alignment extends this behavioral shaping process by determining which knowl-

edge is surfaced and how it is expressed, but the resulting alignment tax can inadvertently suppress access to nuanced or specialized knowledge that conflicts with common preference patterns (Ouyang et al., 2022; Rafailov et al., 2023; Lin et al., 2024; Barnhart et al., 2025).

Reinforcement learning with verifiable rewards (RLVR) extends this principle from knowledge expression to complex problem solving. Unlike preference-based alignment, RLVR relies on objective reward signals derived from verifiable outcomes such as mathematical correctness, code execution, or environmental feedback. A growing body of evidence suggests that RLVR is effective at eliciting capabilities that are already latent within the model. Early evidence came from the inference stage, where chain-of-thought prompting demonstrated that multi-step reasoning can often be activated without introducing new knowledge (Wei et al., 2022b). Systems such as o1 (OpenAI, 2024a) and DeepSeek-R1 (Guo et al., 2025) later showed that process-level reinforcement learning can improve the deployment of existing knowledge through extended reasoning traces. Similar patterns have been observed in mathematics (Shao et al., 2024), code generation, and agentic environments (Wang et al., 2024f), where reinforcement learning improves the utilization of capabilities that appear to be at least partially present before post-training. Recent theoretical analyses of GRPO (Zhou et al., 2026b) provide a principled account of this phenomenon, suggesting that reinforcement learning primarily reorganizes and amplifies existing representations rather than replacing them. These findings collectively raise the question of where the boundary lies between knowledge that can be elicited through post-training and knowledge that must first be acquired during earlier stages.

3.2 Training Data

Training data is the primary medium through which LLMs acquire world knowledge. Knowledge acquisition depends on data scale as well as the quality and composition of the corpus, and we examine these three factors in turn.

3.2.1 SCALING LAWS

Finding

Knowledge acquisition follows a steeper scaling law than general language modeling, so even compute-optimal training can leave rare and long-tail facts underacquired.

A natural first question is how knowledge acquisition scales with data, model capacity, and compute. The scaling laws introduced by Kaplan et al. (2020) and later refined by the Chinchilla analysis (Hoffmann et al., 2022) established a quantitative framework for balancing model size, data volume, and compute, motivating the widely adopted regime of approximately 20 training tokens per parameter. However, these laws are derived from average language modeling performance and do not directly characterize knowledge acquisition. Lu et al. (2024) show that factual memorization follows a steeper scaling law than perplexity, suggesting that compute-optimal training may still leave rare and long-tail facts underrepresented even as overall linguistic competence improves. Complementing this observation, the Physics of Language Models series (Allen-Zhu and Li, 2025) shows that knowledge storage is also bounded by parametric capacity itself, a constraint we quantify in §3.3.1. These findings imply that scaling data uniformly is not necessarily the most effective

tive strategy for knowledge acquisition. In data-constrained settings, controlled repetition can partially compensate for limited token budgets (Muennighoff et al., 2023; Xue et al., 2023), particularly when additional exposure is allocated to informative or underrepresented content.

3.2.2 DATA QUALITY

Finding

Knowledge acquisition depends more on information density than on token volume, creating a quality–quantity tradeoff in data filtering.

Beyond raw quantity, what a model learns from a corpus depends on how much usable information the corpus carries. Quality curation increases usable signal by removing redundant, noisy, or low-value content, allowing a larger fraction of the training budget to support knowledge acquisition. Deduplication (Lee et al., 2022) is the most widely adopted form of this process, consistently improving model performance while reducing brittle memorization. Subsequent work has shown that quality gains are largest when deduplication is combined with diversification, preserving semantic coverage while removing repetitive content (Tirumala et al., 2023). At web scale, carefully filtered corpora outperform raw crawls on knowledge-intensive benchmarks (Penedo et al., 2024), and LLM-based quality assessment can identify factual and structural signals that are difficult to capture with heuristic filtering alone (Zhang et al., 2025b). Quality considerations also motivate the growing use of synthetic data. By generating targeted content for underrepresented domains, synthetic corpora offer a controllable mechanism for increasing knowledge coverage (Liu et al., 2024b; Kang et al., 2025a; Ma et al., 2025). However, their effectiveness remains closely tied to the quality and diversity of the generating model, making synthetic data a complement to, rather than a replacement for, high-quality natural corpora. More broadly, the literature highlights a persistent tradeoff in which aggressive filtering improves signal density and factual reliability but may simultaneously reduce coverage by removing rare or unconventional knowledge sources. As a result, data quality cannot be considered independently of corpus composition, which determines what knowledge remains available for acquisition.

3.2.3 DATA TYPES AND MIXTURE

Finding

Knowledge acquisition is shaped by data composition, where different data sources, formats, and mixing strategies contribute distinct types of knowledge.

While data scale determines exposure and data quality determines signal density, data composition determines which forms of knowledge are available for acquisition. Different data sources contribute complementary capabilities. Web corpora provide broad factual and commonsense knowledge, code contributes procedural and logical knowledge, scientific literature supplies specialized expertise, and books support long-range reasoning and narrative understanding. Beyond domain origin, the format of the training data also matters. Studies such as AdaptLLM (Cheng et al., 2024) show that reformulating domain content into task-oriented formats can improve knowledge acquisition relative to training on raw

text alone, while combining domain-adaptive pre-training with targeted supervision often yields stronger domain expertise without sacrificing general capabilities (Song et al., 2025b). Recent work suggests that temporal composition matters as well. The influence of training data decays over time (Sun et al., 2025a), and curriculum strategies that prioritize informative examples can accelerate the acquisition of rare or difficult knowledge (Fan and Jaggi, 2023; Elgaar and Amiri, 2026).

A growing body of research therefore treats data mixture design as an optimization problem rather than a fixed heuristic. Methods based on distributionally robust optimization, scaling-aware mixture adaptation, and learning-signal-guided reweighting have shown that optimal domain proportions vary across model scales and training stages (Xie et al., 2023; Kang et al., 2025b; Diao et al., 2025). Data influence approaches also enable the targeted selection of examples that contribute most directly to downstream knowledge acquisition (Xia et al., 2024). Together, these findings suggest that effective corpus construction depends on the quantity and quality of data as well as how different knowledge sources are combined throughout training. Whether principled and scale-invariant mixture strategies can be derived from first principles remains an open question (Liu et al., 2025b).

3.3 Model Architecture

Model architecture shapes both a model’s capacity to acquire knowledge and how that knowledge is represented and retrieved. Parameter capacity bounds how much knowledge can be encoded, whereas architectural design determines how that knowledge is organized and accessed.

3.3.1 MODEL CAPACITY

Finding

Model capacity sets a hard limit on factual knowledge storage, with FFN layers serving as the main storage substrate.

Recent theoretical work provides a mechanistic account of how transformers store knowledge and where the resulting capacity limits come from. The Physics of Language Models series argues that feed-forward network (FFN) layers function as key-value memory systems in which conceptual patterns act as retrieval keys and factual content is encoded in the corresponding values (Allen-Zhu and Li, 2024). Building on this view, Allen-Zhu and Li (2025) derive an information-theoretic storage bound of approximately $2N$ bits for a model with N parameters, suggesting that knowledge capacity scales predictably with model size. Complementary studies indicate that attention key-value states also contribute to knowledge storage (Wong, 2025), while analyses of large-scale models identify increasing memorization and reduced generalization as signs of approaching capacity limits (Barron and White, 2025).

Interpretability research has also examined how this stored knowledge is organized internally. Early studies localized factual associations to specific neurons and MLP layers (Dai et al., 2022; Meng et al., 2022), supporting the view that identifiable model components contribute disproportionately to factual recall. Subsequent work, however, suggests a more distributed picture. Knowledge retrieval appears to emerge from coordinated circuits span-

ning multiple layers and computational stages (Yao et al., 2024a; Lv et al., 2024), while individual facts are often represented across multiple neurons rather than isolated locations (Niu et al., 2024). Transformer knowledge storage thus appears to be simultaneously localized and distributed, with factual recall arising from structured interactions among many components, an organization we examine in depth in §4. The gap between theoretical storage capacity and practically usable, generalizable knowledge remains an open problem linking architecture, data, and training dynamics.

3.3.2 ARCHITECTURAL VARIANTS

Finding

Architecture determines how knowledge is stored and retrieved. MoE extends capacity via sparsely activated experts, while meaningful expert specialization is not guaranteed and depends critically on routing.

Beyond parameter count, architectural design also shapes how knowledge is represented, organized, and retrieved. The decoder-only transformer remains the dominant foundation-model architecture. Under the associative-memory view, its attention layers support content-based retrieval while FFN layers serve as the primary substrate for storing factual associations (Zhong et al., 2025). Architectural choices within this framework influence both storage capacity and knowledge organization. In particular, wider FFN layers increase the amount of information that can be represented, whereas greater depth supports the composition of increasingly complex concepts and reasoning patterns (Jin et al., 2024a; Gromov et al., 2024). Mechanistic studies further suggest that factual recall emerges from coordinated interactions between MLP and attention components rather than from any single module in isolation (Choe et al., 2025).

Mixture-of-Experts (MoE) architectures provide an alternative scaling strategy by replacing dense FFN layers with sparsely activated expert networks (Artetxe et al., 2022). Because only a subset of experts is activated for each token, MoE models can increase parameter count and theoretical storage capacity without proportional increases in computation. Empirical results from systems such as Mixtral (Jiang et al., 2024a) and DeepSeekMoE (Dai et al., 2024) demonstrate that this approach can achieve performance comparable to much larger dense models. However, a central question is whether experts develop meaningful knowledge specialization. Recent analyses suggest that MoE models exhibit more modular knowledge organization than dense transformers (Wang et al., 2026a), but that expert specialization is often weaker and less domain-specific than simple intuition would predict (Lo et al., 2025). Existing evidence points to routing mechanisms as a key determinant of specialization quality (Cai et al., 2025). How routing objectives shape knowledge modularity, and whether modular specialization can further expand effective knowledge capacity, remain open questions.

3.4 Discussion

Finding

Progress in knowledge acquisition depends on the interaction between training objectives, data design, and model architecture, suggesting the need for more integrated optimization frameworks.

The objectives used to train language models remain only indirect proxies for knowledge acquisition. While next-token prediction has proven highly effective at learning broad statistical regularities, evidence suggests that rare and long-tail knowledge follows different acquisition dynamics (Lu et al., 2024). Mid-training strategies partially address this gap by reallocating training signal toward underrepresented domains (Ibrahim et al., 2024; Hu et al., 2024b), and advances in data curation improve the density of useful learning signals (Penedo et al., 2024; Zhang et al., 2025b). Nevertheless, it remains unclear whether training objectives, data quality criteria, and knowledge acquisition goals can be more directly aligned, particularly for factual and verifiable knowledge (Tack et al., 2026; Prakriya et al., 2025).

At the same time, the effectiveness of improved training signals ultimately depends on the model architectures that absorb them. The gap between theoretical storage capacity and practically useful knowledge suggests that scaling parameters alone does not guarantee efficient knowledge utilization (Allen-Zhu and Li, 2025; Barron and White, 2025). Moreover, many architectural phenomena are closely coupled to data and training dynamics. For example, expert specialization in MoE models depends on the distributions on which experts are trained (Lo et al., 2025; Cai et al., 2025), while the effectiveness of mid-training interventions depends on the available representational capacity of the underlying model. Together, these observations suggest we treat training objectives, data composition, and architectural design as components of a unified knowledge acquisition system.

4 Knowledge Representation

Once large language models (LLMs) acquire knowledge from large training corpora, a central question is how this knowledge is internally represented. We examine knowledge representation in large language models from two complementary perspectives. **Knowledge Storage in Parametric Memory** (§4.1) concerns how knowledge may be encoded within model parameters and how such knowledge can be localized. **Knowledge Representation in Activation Space** (§4.2) shifts the focus from static storage to the dynamic representations formed during inference. Together, these perspectives trace the progression from where knowledge is stored to how it is instantiated during inference.

4.1 Knowledge Storage in Parametric Memory

A natural hypothesis is that the knowledge acquired during training is stored *parametrically*, encoded directly in the model’s weights. Based on this view, prior studies have explored which architectural components may function as storage structures and how specific pieces of knowledge can be localized and validated within the network. We group these studies into three perspectives. **Storage Components** (§4.1.1) focuses on the model components that

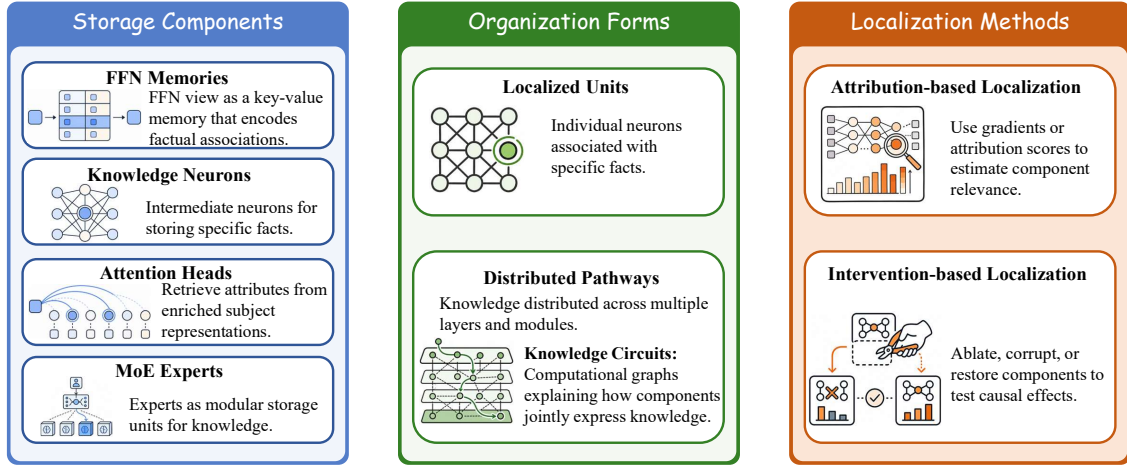


Figure 4: Overview of knowledge representation in parametric memory. This taxonomy organizes prior studies on parametric knowledge representation into two perspectives. The storage perspective covers components that potentially encode knowledge and organizational forms that describe how knowledge is structured across model internals. The localization perspective covers methods that identify knowledge-related components through attribution or intervention.

potentially store knowledge. **Storage Organization** (§4.1.2) examines how knowledge is organized across these components. **Knowledge Localization** (§4.1.3) reviews methods for locating the internal sites associated with particular knowledge. Figure 4 gives the corresponding taxonomy.

4.1.1 STORAGE COMPONENTS

Finding

Parametric knowledge is not stored in a single type of component. FFN modules provide a major locus for associative storage, but factual recall also depends on attention heads. In MoE models, sparse experts make knowledge storage more modular.

A prominent line of work on parametric knowledge storage is built upon the key-value memory view of Transformer feed-forward networks (FFNs). This idea is partly motivated by earlier work on persistent memory in self-attention (Sukhbaatar et al., 2019), which observed a close formal similarity between FFNs and attention mechanisms and proposed replacing FFNs with persistent memory vectors inside attention layers. Building on this observation, Geva et al. (2021) analyze the functional role of FFNs in Transformer language models. Under this view, the first linear layer in an FFN provides keys that are associated with a set of human-interpretable input patterns, while the second linear layer provides values that induce a distribution over the output vocabulary. Their analysis also investigates how multiple memories are composed, and how their outputs are progressively refined across layers through residual connections to shape the final prediction. Geva et al. (2022) examine

FFN memories at a finer granularity, decomposing the forward computation to show that an FFN’s update can be formalized as a collection of sub-updates. Their analysis reveals that these sub-updates typically encode human-interpretable concepts and directly influence the evolution of the final output distribution.

The key-value memory view is extended by studies on knowledge neurons (Dai et al., 2022). Under this interpretation, the first linear layer of an FFN computes the activation of intermediate neurons through inner products with the input representation, while the second linear layer uses these activations as coefficients to aggregate the corresponding value vectors. Based on this formulation, knowledge neuron studies regard certain intermediate neurons as potential carriers of factual associations, since their activations contribute to the expression of specific relational facts. Following this line of work, subsequent studies have identified neurons with specialized functions under different settings, such as multilingual neurons (Chen et al., 2024b; Cao et al., 2026) and privacy-related neurons (Wu et al., 2023), suggesting that FFN neurons may encode not only factual associations but also more specific linguistic, functional, or behavioral properties.

Beyond FFN neurons, attention heads also contribute to factual recall. Rather than serving merely as generic context-mixing modules, certain attention heads can retrieve attributes from enriched subject representations. For example, Geva et al. (2023) show that factual recall in autoregressive language models involves a multi-step process, in which the prediction position queries an enriched subject representation through attention heads to extract the target attribute. In a related study, Wei et al. (2024b) investigate knowledge localization from entity and relation perspectives and show that relational knowledge is also encoded in attention modules. Lv et al. (2024) further identify attention heads that extract topic tokens and, in some cases, map them directly to the desired outputs. These findings suggest that attention heads should also be considered components in factual knowledge recall, especially in retrieving and routing information needed for the final prediction.

A further line of work examines modular structures within FFNs and expert-based architectures. Recent work suggests that dense FFNs may themselves contain implicitly modularized structures (Zhao et al., 2024d). Specifically, dense FFNs can be factored into sparse expert-like subnetworks, indicating that knowledge computation may be organized in a more modular manner even within conventional dense architectures. This perspective naturally connects to Mixture-of-Experts (MoE) architectures (Shazeer et al., 2017; Fedus et al., 2022; Du et al., 2022), which have been widely adopted by recent large language models. By replacing the feed-forward networks with sparsely activated experts, MoE models can scale the total number of parameters while keeping computational cost limited. This architectural design makes experts natural candidates for analyzing how knowledge is stored and retrieved. For instance, Bandarkar et al. (2026) identify experts associated with specific factual knowledge by contrasting cases where the model successfully recalls a fact with cases where it fails. To enhance knowledge-level expert specialization, DeepSeekMoE (Dai et al., 2024) improves the MoE architecture by introducing fine-grained expert segmentation and shared expert isolation, thereby encouraging more specialized routed experts, a finer division of labor that also makes expert-level knowledge attribution more analyzable.

4.1.2 STORAGE ORGANIZATION

Finding

Neuron-level attributions can be unstable under paraphrasing, and knowledge circuits spanning layers and modules offer a more faithful account of how models express specific facts.

Beyond identifying individual storage components, prior studies also examine how knowledge is organized across them. The overall understanding has shifted from highly localized units to more distributed computational pathways. Although the knowledge neuron perspective provides valuable insights into the parametric storage of factual knowledge, its limitations have also become increasingly evident. A growing body of work suggests that knowledge cannot always be attributed to a fixed set of individual neurons (Chen et al., 2025a,b). For example, when a factual query is paraphrased, the neurons identified as responsible for the same knowledge may change, indicating that neuron-level localization can be sensitive to input formulation. This observation reveals a fundamental limitation of highly localized storage views exemplified by knowledge neurons. To address this issue, recent studies have turned to more distributed accounts of knowledge storage, where knowledge is understood as being encoded across multiple layers, modules, and computational pathways. The notion of circuits (Conmy et al., 2023; Wang et al., 2023), developed in mechanistic interpretability, has also been introduced into knowledge analysis, leading to the study of knowledge circuits (Yao et al., 2024a). Unlike neuron-level analyses that focus primarily on individual neurons or FFN layers, knowledge circuits aim to construct a broader computational graph that captures how different components jointly contribute to the expression of specific knowledge. Such circuits may involve components or parameters distributed across different layers and modules, thereby providing a more faithful account of model behavior on particular inputs. However, knowledge circuits are often highly complex, and identifying them does not necessarily imply that their internal mechanisms are fully understood.

4.1.3 KNOWLEDGE LOCALIZATION

Finding

Attribution-based knowledge localization estimates component relevance under the current computation, while intervention-based localization provides causal evidence by testing how model behavior changes under controlled modifications.

From the perspective of parametric knowledge storage, another important question is how to identify where specific knowledge resides within a model. This problem is commonly referred to as **knowledge localization**. The central goal of knowledge localization is to attribute the expression of a target piece of knowledge to specific internal components or parameters, which are regarded as likely carriers or mediators of that knowledge. Although existing localization methods differ in their technical implementations, they share a common rationale. Each estimates the relevance or causal contribution of model components to a factual prediction and selects the highest-scoring components as candidates associated with the target knowledge. Knowledge localization methods can therefore be grouped by how

they assess internal components, distinguishing between **attribution-based localization** and **intervention-based localization**.

Attribution-based localization assesses internal components, such as knowledge neurons, by estimating their contribution to the model’s prediction. In other words, it asks how much a specific neuron contributes to the final prediction under the current computation. A representative approach in this category is Integrated Gradients (IG) (Sundararajan et al., 2017; Dai et al., 2022; Chen et al., 2024b). Given an input prompt and a target answer, IG measures how the model’s predicted probability of the target answer changes as the activation of a component moves from a predefined baseline value to its actual value. Rather than relying only on the gradient at the final activation, IG accumulates gradient information along the interpolation path from the baseline activation to the actual activation, thereby providing an attribution score that reflects the component’s contribution to the prediction. After obtaining attribution scores, a filtering procedure is often required to identify relevant neurons. A common strategy first filters neurons using a predefined threshold and then evaluates their consistency across semantically equivalent paraphrases of the input. Attribution scores are computed for each paraphrased prompt, and neurons that are consistently selected across these paraphrases are retained as the final localized neurons (Dai et al., 2022). However, since LLMs are highly sensitive to input variations, later studies refine this selection process by quantifying the uncertainty of localization results, thereby improving their stability and reliability (Cao et al., 2026).

While attribution-based localization offers useful insights, it mainly uncovers correlational patterns. Since gradients only capture the local sensitivity of the model output around the current computation, a high attribution score does not necessarily imply that the corresponding neuron is causally necessary for expressing the target knowledge. Moreover, neural representations in LLMs are often distributed, redundant, or degenerate (Chen et al., 2025b). The same factual association may be supported by multiple partially overlapping components, and the effect of a single neuron may be masked or substituted by others. Therefore, attribution scores provide useful clues about potentially relevant neurons, but they may not fully reveal the causal structure underlying knowledge expression.

In contrast, **intervention-based localization** directly measures the causal importance of a component by comparing the model’s behavior before and after an intervention on its internal state. Instead of asking how much a component contributes under the current computation, it asks *whether the model’s knowledge expression would change if this component were removed, corrupted, or restored*. Given an internal component, such as the hidden state at a particular layer and token position, the method first records the model’s original output under the normal computation, then forces this component to take a modified value, such as a corrupted, ablated, or restored activation, and observes the resulting output. If this modification causes a large change in the target prediction or the evaluation metric, the component is considered to play a strong causal role in the corresponding knowledge expression.

A representative family of intervention-based methods is based on causal tracing (Meng et al., 2022). These methods first construct a corrupted run in which the model’s ability to predict the target answer is disrupted, for example by adding noise to the subject token representations. They then restore a specific internal state from the clean run into the corrupted computation. If restoring a particular component recovers much of the probability

of the correct answer, that component is regarded as causally important for expressing the target knowledge. Another common intervention strategy is ablation, which removes or masks a component and measures the resulting degradation in model behavior. For example, one may set the activation of a neuron, an attention head, or an MLP submodule to zero (Geva et al., 2023) or to a mean activation (Ghorbani and Zou, 2020), and then evaluate how much the target prediction is affected. If ablating a component causes a large drop in the metric, this provides evidence that the component is functionally involved in the corresponding knowledge expression.

Compared with attribution-based localization, intervention-based localization offers stronger causal evidence because it directly tests the effect of modifying internal components. However, intervention results may depend on the choice of corrupted input, intervention strength, and the metric used to quantify recovery. Thus, intervention-based localization still requires careful experimental design and interpretation. Additionally, intervention-based localization is often computationally expensive. Exhaustively testing a large number of layers, token positions, neurons, or attention heads requires a separate forward pass for each intervention. To mitigate this issue, later work introduces *attribution patching* (Syed et al., 2024; Kramár et al., 2024; Marks et al., 2025). Instead of explicitly performing every intervention, attribution patching uses a first-order Taylor approximation to estimate the effect of replacing an activation in one run with its counterpart from another run. In this way, it approximates the effect of activation patching with fewer forward passes.

Overall, knowledge localization provides an entry point for understanding parametric knowledge storage in large language models. By associating factual predictions with specific neurons, layers, attention heads, or hidden states, localization methods make the otherwise opaque process of knowledge expression more analyzable. At the same time, localization results remain sensitive to the choice of method and the formulation of the input.

4.2 Knowledge Representation in Activation Space

Knowing where knowledge may be stored does not yet explain how it is expressed and organized during inference, because the use of knowledge is often conditioned by input contexts rather than determined solely by static parameter locations. This motivates a shift of analysis from parametric memory to activation space (§4.2.1). The next question is what geometric form knowledge takes in activations, which leads to the linear representation hypothesis (§4.2.2) and its extensions to multidimensional and potentially nonlinear structures (§4.2.3). A further question is how a limited number of representational dimensions can encode an overcomplete set of linear features, which is addressed by the superposition hypothesis and sparse autoencoders (§4.2.4). Figure 5 depicts the relationship among these ideas.

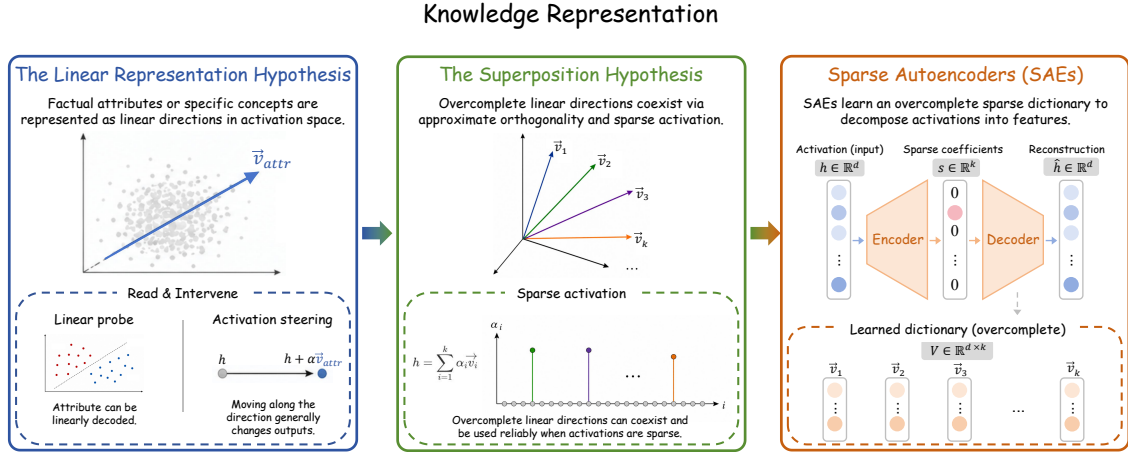


Figure 5: Overview of knowledge representation in activation space. The linear representation hypothesis characterizes knowledge as readable and intervenable directions. The superposition hypothesis explains how overcomplete linear features can co-exist through sparse activation. Sparse autoencoders (SAEs) provide a practical method for decomposing activations into sparse latent features.

4.2.1 FROM PARAMETRIC MEMORY TO ACTIVATION SPACE

Finding

Knowledge localization is unstable across contexts, and parameter-level edits do not guarantee behavioral use. The activation-space perspective responds to these limitations by studying how parametric knowledge is expressed during inference.

The idea of “language models as knowledge bases” serves as the starting point for research on parametric memory. [Petroni et al. \(2019\)](#), through LAMA, demonstrate that pre-trained language models can complete relational facts without relying on external knowledge bases. [Roberts et al. \(2020b\)](#) ask “how much knowledge can be stored in model parameters,” thereby helping to establish parametric memory as a mainstream problem setting. Within this framework, subsequent work on knowledge localization and knowledge editing has reinforced the intuitive picture that knowledge can be both localized and modified. Methods such as ModifyingMemories ([Zhu et al., 2020](#)), KnowledgeEditor ([De Cao et al., 2021](#)), ROME ([Meng et al., 2022](#)), and MEMIT ([Meng et al., 2023](#)) show that, by intervening on localized parameters ([Liu et al., 2025a](#); [Zhou et al., 2026a](#)), the model’s outputs for individual or multiple factual associations can be rewritten in a targeted manner.

However, the limitations of the parametric-memory perspective have gradually become apparent. First, knowledge localization itself is not stable. Different localization methods often point to different parameter regions ([Meng et al., 2022](#); [Dai et al., 2022](#)), and even the same fact may depend on different parameter regions under different prompt formulations ([Chen et al., 2025a](#)). Therefore, the so-called “storage location” of knowledge is more likely to be a causally sensitive region under specific contextual conditions, rather than a

fixed and unique knowledge address. Second, knowledge editing alone cannot fully explain how knowledge is expressed and used during a concrete forward propagation. This issue is particularly salient in multi-hop reasoning or compositional queries, where edits to individual facts can succeed while multi-hop queries that depend on them still fail. [Cohen et al. \(2024a\)](#) describe this as a ripple effect of knowledge editing. [Zhong et al. \(2023\)](#) propose MQuAKE to evaluate editing methods on multi-hop questions, and their results show that existing editing methods still often fail in this setting. Thus, the fact that knowledge is modifiable at the parameter level is insufficient to explain whether it can be used at the behavioral level. A more complete treatment of knowledge editing and its side effects is given in §6.1 and §6.3.

These limitations have shifted attention from static parametric memory to dynamic activation space. Parametric memory addresses whether and how knowledge is stored in parameters, whereas the activation-space perspective asks how such knowledge is represented under specific input conditions and transformed into output behavior, thereby extending rather than negating the parametric-memory view.

4.2.2 THE LINEAR REPRESENTATION HYPOTHESIS

Finding

Some knowledge manifests as linear directions in activation space that can be reliably decoded, generalize across datasets, and causally steer model outputs when intervened upon.

After the shift from parametric memory to activation space, the activation-space representation perspective must address a central question. When a factual attribute or a specific concept is activated by an input, what form does it take in the model’s intermediate activations? This form could be a complex pattern distributed across high-dimensional representations, or it could be simplified into a readable and intervenable geometric object.

The linear representation hypothesis offers the most concise geometric answer, proposing that factual attributes or specific concepts can be viewed as linear directions in representation space, and that the model performs related tasks by reading, combining, and manipulating these directions. Accordingly, activation-space representations of knowledge are no longer merely high-dimensional vectors that are difficult to interpret. Instead, they can be understood as directional geometric structures. If a given attribute can be stably decoded by a linear probe, and interventions along the corresponding direction systematically alter the model’s outputs, then this direction may constitute the representational carrier through which the model uses that knowledge.

[Park et al. \(2023\)](#) provide a systematic formalization of the linear representation hypothesis. Using counterfactual language, they define linear representations in both output space and input space, clarify the relationship between linear representations and methods such as linear probing and activation steering, and point out that geometric analyses of language-model representations should not simply assume the Euclidean inner product, but should instead adopt inner-product structures better aligned with model structure. Their contribution lies not in proving that all knowledge is linear, but in elevating linear representation from an empirical observation to a testable and operational geometric proposition. In

this sense, the linear representation hypothesis becomes a natural starting point for research on activation-space representation.

Subsequent studies provide supporting evidence for this hypothesis across different types of knowledge. [Gurnee and Tegmark \(2024\)](#) find relatively robust linear representations of space and time in the Llama-2 family, and localize the associated space and time neurons. Beyond spatial and temporal knowledge, [Marks and Tegmark \(2023\)](#) observe a salient linear structure in truthfulness-classification data, and show through cross-dataset generalization and causal intervention that a truth direction can not only be decoded but also be used to control model outputs. [Merullo et al. \(2024\)](#), in their study of relational tasks, suggest that, in certain knowledge-retrieval tasks, key internal operations of language models can be approximated as Word2Vec-style vector arithmetic. Taken together, these studies indicate that, at least for spatial, temporal, truth-related, and some relational knowledge, the activation space of language models indeed contains low-dimensional linear structures that are readable, generalizable, and intervenable.

4.2.3 MULTIDIMENSIONAL LINEAR AND NONLINEAR REPRESENTATIONS

Finding

Complex knowledge often exhibits multidimensional rather than one-dimensional linear geometry. Strictly nonlinear representations add explanatory power only where linear geometry falls short.

The linear representation hypothesis provides a concise entry point for understanding activation space. In its most direct form, however, it typically relies on one-dimensional linear geometry, that is, characterizing variations in a concept or attribute by projecting activations onto a particular direction. This formulation is well suited to relatively simple binary or continuous attributes, such as true/false, presence/absence, or graded attribute intensity. However, for more complex forms of knowledge, such as category concepts, hierarchical relations, periodic concepts, and compositional knowledge, knowledge representations often exhibit a multidimensional linear geometry.

[Park et al. \(2025\)](#) extend binary attributes to category concepts and hierarchical concepts, proposing that categories can be represented as polyhedral structures, while hierarchical relations are associated with geometric containment relations. This extension suggests that, even if one-dimensional linear geometry holds for local concepts, knowledge representation may still involve rich compositional relations, thereby giving rise to a multidimensional linear geometry at a higher level. Similarly, contemporaneous work by [Engels et al. \(2025\)](#) explicitly argues that not all features in language models are one-dimensionally linear, and provides a definition of irreducibly multidimensional features. Empirically, they identify circular representations of days of the week and months of the year in GPT-2 and Mistral 7B, which are effectively multidimensional linear structures confined to a two-dimensional plane, and validate these representations through intervention experiments. This indicates that periodic concepts exhibit a multidimensional linear geometry. Overall, the move from one-dimensional linearity to multidimensional linearity is not a denial of the linear representation hypothesis, but rather an extension of its expressive capacity. One-dimensional

linear geometry explains many separable abstract attributes, while multidimensional linear geometry further accounts for more complex forms of knowledge.

A natural further question is whether activation-space representations can be nonlinear. At present, the mainstream evidence on knowledge representation in LLMs still leans toward linearity, including both one-dimensional and multidimensional linearity. Multidimensional linearity is not equivalent to nonlinearity. Rather, it is an extension of one-dimensional linearity. By contrast, nonlinear representations in the stricter sense emphasize that concepts cannot be globally represented by a fixed direction or fixed subspace, but instead appear as piecewise-linear regions, local geometric structures, curved manifolds, or more complex geometric forms. Several studies have begun to explore this direction. For example, [Black et al. \(2022\)](#) shift the representational unit from directions to polyhedral regions induced by nonlinear activations. [Shafran et al. \(2026\)](#) characterize global nonlinear concepts through local regions and their covariance structures. [Sarfati et al. \(2026\)](#) show that a model’s context-conditioned beliefs may form curved manifolds. These works suggest that models may employ more complex nonlinear geometry.

In summary, current research remains largely grounded in the linear representation hypothesis, extending the study of representation space from one-dimensional to multidimensional linearity rather than directly shifting toward nonlinear representations. Linear representations provide a concise and testable geometric characterization of how knowledge exists in activation space, whereas nonlinear representations in the stricter sense are more difficult to formalize and introduce greater explanatory freedom. Following the principle of Occam’s razor, nonlinear representations constitute a meaningful theoretical extension only when a certain type of knowledge cannot be adequately explained by one-dimensional or multidimensional linear geometry, and when the nonlinear structure provides additional predictive, interventional, or mechanistic explanatory power.

4.2.4 THE SUPERPOSITION HYPOTHESIS AND SPARSE AUTOENCODERS

Finding

Superposition lets models represent more features than dimensions via approximately orthogonal, sparsely activated directions, and SAEs decompose these features at scale. However, SAEs remain a practical approximation, and their decompositions vary with seeds, dictionary sizes, and training settings.

Another core concept closely related to the linear and nonlinear representation hypothesis is the superposition hypothesis ([Elhage et al., 2022](#); [Hu et al., 2025a](#); [Liu et al., 2026b](#)). If the linear representation hypothesis concerns whether knowledge can be represented as linear directions in activation space, the superposition hypothesis asks how these directions can coexist when the number of features a model needs to represent far exceeds the dimensionality of its activation space. Because models must represent far more features than the dimensionality of their activation space, the corresponding directions form an overcomplete set and cannot be mutually orthogonal, but can only be approximately orthogonal. Such approximate orthogonality implies potential interference among different features. The superposition hypothesis suggests that this interference remains controllable as long as fea-

tures are sparsely activated, and the model can still functionally distinguish, combine, and use these features.

This idea was systematically developed by [Elhage et al. \(2022\)](#). Starting from the phenomenon of neuronal polysemanticity, the tendency of a single neuron to respond to multiple semantically unrelated inputs, they argue that such polysemanticity may arise because models represent features in superposition within a finite-dimensional space. Through controlled toy models, they show that when features are sufficiently sparse, a model can represent more features than its available dimensions. Subsequently, [Hu et al. \(2025b\)](#) unify the linear representation hypothesis and the superposition hypothesis under Atom Theory, thereby characterizing how approximately orthogonal and sparsely activated linear directions can constitute faithful and stable representations. This suggests that approximate orthogonality does not necessarily lead to uncontrollable interference. As long as certain geometric conditions, which [Hu et al. \(2025b\)](#) refer to as stability, are satisfied, these overcomplete linear features can still be clearly distinguished and stably composed in activation space.

Sparse autoencoders (SAEs) were proposed and developed precisely in this context, and can be viewed as an operationalization of the linear representation hypothesis and the superposition hypothesis. The underlying intuition is straightforward. If the internal features of a model are linear directions superposed within activation space, one should be able to recover an overcomplete sparse feature dictionary from its hidden representations. Accordingly, [Bricken et al. \(2023\)](#) and [Cunningham et al. \(2023\)](#) first introduce SAEs into the decomposition of hidden representations in language models, aiming to learn interpretable sparse features from high-dimensional representations. A large body of experimental results shows that the features learned by SAEs are more interpretable than raw neurons and several alternative decomposition methods ([Chen et al., 2025c](#)).

Subsequently, SAE research began to move toward engineering scalability, which expanded its value for studying knowledge mechanisms in LLMs. [Gao et al. \(2025a\)](#) introduce k-sparse SAEs, which improve the reconstruction-sparsity frontier through a TopK objective and make it possible to train much larger sparse dictionaries, reaching 16 million latents on GPT-4 activations at the largest scale. Gemma Scope ([Lieberum et al., 2024](#)) extends SAEs into a publicly reusable, all-layer and all-site resource, enabling systematic analysis of how knowledge-related features are distributed across layers, activation sites, and model scales. [Templeton \(2024\)](#) extend SAEs to production models, reporting millions of conceptual features in Claude Sonnet and showing that SAE features can support the analysis of high-level concepts, factual associations, and behaviorally relevant representations. Building on these scalable decompositions, later methods move from identifying individual features to analyzing feature-level computation. Cross-layer transcoders ([Minder et al., 2026](#)) aim to trace how sparse features are transformed across layers, thereby helping analyze how representations are progressively processed during inference. Model diffing ([Chaudhari et al., 2026](#)) uses SAE features to compare internal representations between models, making it possible to study how training affects model behavior. Attribution graphs ([Ameisen et al., 2025](#); [Lindsey et al., 2025](#)) connect sparse features into causal computational structures, offering a way to analyze how knowledge-relevant features interact and contribute to downstream outputs. Overall, this line of work has transformed SAEs

from an interpretability-oriented decomposition technique into infrastructure for studying the internal knowledge representations of LLMs.

Meanwhile, SAEs still have several limitations that should be treated with caution. First, SAE decompositions are not inherently canonical or unique. [Paulo and Belrose \(2025\)](#) find that, even on the same model and the same data, merely changing the random seed can lead SAEs to learn different sets of features. For example, in certain layers of Llama-3-8B, the proportion of shared features between SAEs trained with different seeds is limited. This suggests that SAE latents are better understood as a practical decomposition of activation space, rather than as a unique and exhaustive inventory of the features inside the model. Second, SAE features do not necessarily constitute indivisible semantic atoms. Through SAE stitching and meta-SAEs, [Leask et al. \(2025\)](#) show that smaller SAEs may miss novel representations found in larger SAEs, while some representations in larger SAEs can themselves be decomposed into combinations of smaller or more basic representations. Therefore, the granularity of SAE features depends on dictionary size and training setup, and does not naturally correspond to a canonical atomic unit. Third, the reconstruction objective of SAEs often deviates from the ideal case in practice. [Engels et al. \(2024\)](#) show that the residuals left unexplained by SAEs are not merely random noise. A considerable portion of the error is structured and affects downstream loss when SAE-reconstructed activations are inserted back into the model. Fourth, [Farrell et al. \(2024\)](#) show that, in applications such as unlearning, interventions on SAE features can achieve a certain degree of concept suppression, but may also introduce side effects. In such applications, current SAE quality and intervention techniques are still insufficient to serve as stable replacements for more mature fine-tuning-based methods.

However, the limitations discussed above do not necessarily invalidate the linear representation hypothesis, the superposition hypothesis, or the SAE approach itself. More precisely, they indicate that existing SAE decompositions are highly dependent on specific training paradigms, model architectures, dictionary sizes, and data distributions. Therefore, many current negative findings should be understood as empirical results under particular training settings, rather than being directly extrapolated into the stronger claim that SAEs cannot recover the true representational structure. In this regard, [Hu et al. \(2025b\)](#) provide a theoretical refinement. Building on the linear representation hypothesis and the superposition hypothesis, they examine the identifiability conditions of representational atoms under threshold-activated SAEs (TSAEs), and point out that the model scale of TSAEs must be properly matched to the data scale in order to obtain faithful and stable feature decompositions.

Thus, evaluations of the limitations of SAEs should avoid two extremes. On the one hand, SAE latents should not be directly equated with unique, complete, and indivisible units of knowledge. On the other hand, the theoretical potential of SAEs as tools for decomposing superposed linear representations should not be dismissed solely because early SAEs exhibit issues related to randomness, atomicity, or reconstruction error. A more cautious conclusion is that SAEs remain an approximate decomposition method with explicit modeling assumptions and training dependencies, and their reliability must be jointly validated through scale matching, stability tests, causal interventions, and cross-task generalization.

4.3 Discussion

Knowledge representation in LLMs involves two closely related perspectives, **Knowledge Storage in Parametric Memory** and **Knowledge Representation in Activation Space**. The former examines where knowledge may be encoded within model parameters and how such knowledge can be localized, while the latter extends this view by connecting model parameters, input contexts, and observable model behavior, focusing on how knowledge is expressed and organized during inference.

Several challenges remain. Studies on parametric memory can identify parameters or components associated with specific facts, but they often do not fully explain how such stored knowledge is transformed into activation-space representations during a concrete forward pass. Conversely, studies on activation-space geometry provide useful tools for characterizing representations, but the causality, stability, and robustness of these representations remain insufficiently established. For example, linear directions, multidimensional structures, or SAE features may be decodable or interpretable, yet this alone does not guarantee that they are robust across contexts or functionally necessary for model behavior.

Future research therefore needs a more integrated and causally grounded analysis framework of knowledge representation. Instead of treating parameter localization, activation geometry, and behavioral evaluation as separate problems, this framework should examine how parameter-level knowledge is activated by specific inputs, how it evolves across layers and token positions, and how it eventually contributes to observable outputs. At the same time, representational claims should meet the validation criteria discussed for SAEs (§4.2.4), so that the parametric-storage and activation-geometry lines of research can corroborate each other. Such efforts would help move the study of knowledge representation from identifying interpretable structures toward explaining how LLMs actually use knowledge during inference.

5 Knowledge Utilization

After knowledge is acquired, represented, and stored in model parameters, a central question is whether it can be effectively used during inference. A model may hold knowledge in its parameters yet fail to express it in its outputs, a discrepancy we refer to as the *encode-express gap*. As shown in Figure 6, we organize knowledge utilization into three complementary aspects: *knowledge recall* (§5.1) examines how parametric knowledge is accessed and why recall may fail; *knowledge augmentation* (§5.2) studies how demonstrations, retrieval, and tools supplement insufficient parametric knowledge; and *knowledge conflict* (§5.3) addresses how models arbitrate among inconsistent knowledge sources.

5.1 Knowledge Recall

Knowledge recall refers to the process by which a model accesses and expresses knowledge already stored in its parameters. This process can operate implicitly, where the model produces an answer without generating intermediate steps (§5.1.1), or explicitly, where chain-of-thought (CoT) tokens externalize intermediate reasoning (§5.1.2). Both modes face systematic limitations. When recall fails, the failure can often be localized to a specific stage along the pathway from stored knowledge to generated output (§5.1.3).

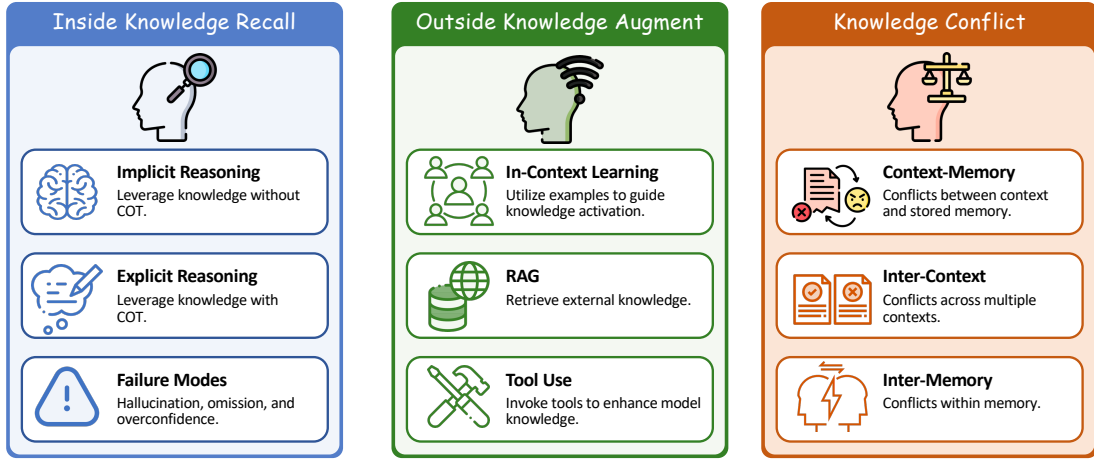


Figure 6: Overview of knowledge utilization, organized into knowledge recall from parametric memory, knowledge augmentation through demonstrations, retrieval, and tools, and knowledge conflict among competing knowledge sources.

5.1.1 IMPLICIT RECALL

Finding

Implicit recall can retrieve directly stored facts but fails systematically when answers require composing multiple knowledge steps. Without generating intermediate reasoning tokens, models struggle to chain facts together even when each individual fact is accessible.

Implicit recall occurs when a model produces an answer directly from its parameters without generating intermediate reasoning steps. When a model answers a compositional knowledge question in this way, the correct response may stem from genuine internal knowledge composition, memorized composite facts, or distributional shortcuts. Mechanistic probes help distinguish these sources. In two-hop factual prompts (e.g., “Where was the birthplace of X’s spouse?”), models can internally activate the intermediate entity (the spouse), and this first-hop retrieval improves with scale. The bottleneck lies in the second hop, where the activated intermediate must serve as a cue to access the next piece of knowledge (Yang et al., 2025c). This asymmetry between accessing a single fact and chaining it with the next explains why knowledge edits to base facts frequently fail to propagate to dependent facts, a phenomenon examined further in §6. A theoretical analysis formalizes the bottleneck by showing that single-hop facts compress efficiently into learned embeddings, but multi-hop queries reintroduce a capacity-depth tradeoff that externalizing intermediate steps circumvents (Ravfogel et al., 2026).

The same compositional limitation appears across different evaluation settings. Long-context evaluations show that models can locate individual knowledge fragments scattered across contexts yet fail when answering requires integrating several interdependent facts (Li et al., 2024c). At the representation level, successful multi-step composition depends on

how precisely entity relations are organized in activation space. When this organization is only approximate, multi-step inference can accumulate errors and produce plausible but incorrect answers (Peng et al., 2025).

Several approaches aim to strengthen implicit recall, from latent-space reasoning to activation-space interventions (Qu et al., 2025). However, probing studies find little evidence that models perform sequential internal computation during implicit reasoning. Hidden states encode the input and final answer but skip intermediate knowledge steps (Yu, 2024), suggesting that answers are reached through shortcut associations rather than step-by-step composition. Even models specifically trained to reason without intermediate tokens remain brittle under format shifts and face depth limits as the required number of steps grows. Another constraint arises from surface-cue dependence. When queries and relevant knowledge lack lexical overlap, models fail to locate the target knowledge even at moderate context lengths (Modarressi et al., 2025). These limitations motivate explicit reasoning approaches that externalize intermediate steps through generated tokens.

5.1.2 EXPLICIT RECALL

Finding

Explicit reasoning improves knowledge recall by adding inference-time computation and externalizing intermediate states that prime further retrieval, but the reasoning trace should not be treated as a faithful account of how the model internally accesses knowledge.

When implicit knowledge composition is unreliable, CoT allows models to decompose complex recall into a sequence of simpler sub-retrievals. Several mechanisms drive this improvement. First, additional reasoning tokens create extra forward passes that allow latent computation even when the generated text is not semantically meaningful. Second, intermediate facts generated during reasoning can prime related knowledge by acting as self-produced retrieval cues (Gekhman et al., 2026). Third, restructuring intermediate steps so that the relevant item sits adjacent to the next operation avoids long-range positional lookups that Transformer attention handles poorly (Ebrahimi et al., 2024). Together, these mechanisms help explain why CoT can recover much encoded-but-inaccessible knowledge (Calderon et al., 2026).

Large reasoning models scale up this recall amplification by expanding inference-time computation. Evidence from long-CoT training and reinforcement learning suggests that many self-correction behaviors already exist in the base model, and reasoning training mainly improves the model’s ability to surface existing knowledge rather than adding new factual content (Yeo et al., 2025; Guo et al., 2025). However, this effect does not increase monotonically with reasoning length. Performance peaks at an intermediate reasoning length, beyond which additional deliberation repeats settled assumptions rather than accessing new knowledge (Marjanović et al., 2025). For knowledge-intensive tasks, longer thinking can increase hallucination by encouraging the model to attempt questions whose answers are absent from parameters (Zhao et al., 2025b), and models may initially retrieve the correct answer but overturn it after excessive deliberation (Yang et al., 2025b). Stan-

dard reasoning rewards can reinforce this tendency by prioritizing answer production over declining to answer.

The features that make CoT useful as a recall amplifier also compromise its interpretability. Linear probes can decode decisions from pre-generation activations before any reasoning token is emitted, and activation steering changes the outcome while the generated CoT provides a post-hoc justification for the new answer (Esakkiraja et al., 2026). Counterfactual audits reveal a similar dissociation. In factual domains, answers often remain unchanged even after intermediate reasoning steps are inverted, suggesting that factual knowledge is accessed through internal routes independent of the visible chain (Khanzadeh, 2026). Mathematical tasks, by contrast, show stronger dependence on the intervened steps, indicating that faithfulness varies by knowledge type. Distribution-shift experiments also show that CoT behaves as interpolation over training-time reasoning templates, collapsing when task structure or format departs from the training distribution (Zhao et al., 2025a; Yan et al., 2025). A useful framing is to view reasoning tokens as externalized computational state that must simultaneously carry internal processing and communicate meaning to humans, making semantic mismatch structurally expected (Levy et al., 2025). For knowledge utilization, CoT can improve recall and expose useful intermediate artifacts, but applications requiring reliability should verify final answers independently rather than treating the reasoning trace as evidence of correctness.

5.1.3 RECALL FAILURE MODES

Finding

Many factual errors are stage-specific access failures rather than missing knowledge. A model may fail to recognize that stored knowledge applies, fail to retrieve the relevant signal, or fail to decode an internally available answer.

Recall can fail at any point along a processing pipeline. The model must first recognize that stored knowledge is relevant to the current query (recognition), then access the specific parametric signal among competing associations (retrieval), and finally express it faithfully through token generation (generation). These stages interact, as failed recognition can prevent retrieval, weak retrieval can leave decoding dominated by priors, and generation can corrupt a useful latent signal after it has formed.

Recognition failures. The model must decide whether an entity is familiar enough to support recall and whether a situation calls for a particular body of knowledge. Sparse-autoencoder analyses reveal causal directions in activation space that gate factual recall. Strengthening an “entity-unknown” direction suppresses recall of entities the model does know, while strengthening the “entity-known” direction triggers hallucination about fabricated entities (Ferrando et al., 2024). The result connects knowledge utilization to the representation mechanisms of §4, showing that recall depends on representation-level gating signals rather than on the mere presence of stored facts. A similar pattern appears in reasoning models, which can detect missing premises early in their chain yet continue with assumptions instead of abstaining (Fan et al., 2025). Recognition can also fail at the level of applicability, where a model defines a concept correctly when asked directly but fails

to apply that knowledge when the same concept appears in an unfamiliar context (Maloo, 2026).

These recognition failures motivate boundary-aware training strategies. Dedicated uncertainty tokens allow explicit expression of knowledge limits (Cohen et al., 2024b), while ternary reward schemes, which rank correct answers above abstention and abstention above hallucination, preserve uncertainty awareness that standard SFT suppresses (Wei et al., 2025).

Retrieval failures. The relevant knowledge may exist in parameters but remain hard to access under a given query formulation. Systematically probing model recall across millions of factual queries reveals that retrieval errors cluster semantically, with failures in one knowledge region predicting related failures nearby (Song et al., 2025a). Such clustering suggests that retrieval quality reflects how well a knowledge region was consolidated during acquisition (§3). The mechanisms behind retrieval failure also differ. Tracing hallucinations to their pre-training sources distinguishes pure recall failures, where the model was trained on the correct fact but generates the wrong answer, from noisy-data errors where training data itself contained incorrect information, and from fabrications composed of plausible but ungrounded associations (Ravichander et al., 2025).

Generation failures. The model may have a useful internal signal but fail to express it faithfully through decoding. Internal-consistency analyses reveal an hourglass pattern in which coherence rises through latent layers but can collapse during autoregressive generation as early token choices compound (Liang et al., 2024). Hidden-knowledge probes confirm that correct answers can be decoded from internal representations even when the model fails to produce them in its output (Gekhman et al., 2025). One mechanism behind this collapse is representational drift toward high-frequency associations. When a prompt describes an infrequent entity using generic descriptors, internal representations can be progressively pulled toward the dominant entity sharing those descriptors, overwriting the correct knowledge signal (Bhatia et al., 2025). Post-training can widen the gap further. After RLHF, models may produce fluent answers decoupled from their own internal truth judgments, effectively encoding the correct answer while suppressing it at decoding (Liang et al., 2025a).

Generation failures can also be partial. A model may encode a coarser but correct answer while decoding favors an over-specific completion. Evaluation at multiple granularity levels can reveal such partial knowledge that exact-match metrics miss (Yona et al., 2024). Two intervention strategies target this problem from complementary angles. Factual RL shifts probability mass toward suppressed correct answers so that greedy decoding can surface them (Yang et al., 2026b), while contrastive decoding amplifies the parametric knowledge signal carried by retrieval heads over contextual noise (Gema et al., 2024). Even when generation fails, the correct knowledge signal often persists internally. Both strategies work by amplifying this residual signal relative to competing outputs, consistent with the encode-express gap introduced above.

5.2 Knowledge Augmentation

When parametric recall is insufficient, whether because a fact is long-tailed, temporally stale, or absent from training, models can draw on information supplied at inference time. We distinguish the three augmentation mechanisms by what they supply. Demonstrations provide transient task structure (§5.2.1), retrieval supplies external evidence (§5.2.2), and tools enable model-driven access to external operations (§5.2.3).

5.2.1 IN-CONTEXT LEARNING

Finding

In-context learning adapts model behavior without parameter updates by inducing a task strategy from demonstrations rather than by adding new factual knowledge.

In in-context learning (ICL), a model performs a task using examples placed in the prompt without updating parameters, making this the most lightweight form of knowledge augmentation. The supplied examples may contain facts, but their primary role is to specify how the model should access and format its existing knowledge rather than to introduce new information.

Mechanistic studies have identified several complementary processes underlying this adaptation. Induction heads, attention heads that detect and copy patterns previously seen in the context, provide a circuit-level substrate for ICL and emerge abruptly at a specific point during training (Olsson et al., 2022). At a more abstract level, the forward pass over demonstrations can be understood as performing implicit gradient descent, creating a temporary task-specific adjustment to the model’s computation without modifying weights (Dai et al., 2025; Vladymyrov et al., 2024). This adjustment has a geometric signature in activation space. Task vectors extracted from demonstrations are linearly separable and can be transplanted between contexts to reproduce ICL behavior, suggesting that ICL compresses task information into a low-dimensional subspace that redirects how parametric knowledge is accessed (Han et al., 2024).

The limits of ICL follow from these same mechanisms. Because the task strategy is inferred from prompt evidence rather than learned from gradient updates, performance is sensitive to demonstration coverage, ordering, and formatting (Baek et al., 2024), and degrades under distribution shift (Elmoznino et al., 2024). In many-shot settings, once demonstrations sufficiently cover the target distribution, the specific selection of individual examples becomes less important, but ICL still degrades when the label space is large or demonstrations are poorly arranged (Li et al., 2024e). Theoretical results show that ICL can in principle approximate fine-tuned behavior, but a gap remains between this guarantee and practical performance (Sharma, 2025a).

5.2.2 RETRIEVAL-AUGMENTED GENERATION

Finding

RAG extends the knowledge boundary beyond parametric memory, but retrieved context does not simply supplement parametric knowledge. Because context may suppress parametric recall, evidence selection and grounding verification require as much attention as retrieval quality.

RAG couples a parametric generator with non-parametric retrieval indices, allowing models to condition generation on external evidence (Guu et al., 2020; Lewis et al., 2020; Izacard and Grave, 2021; Borgeaud et al., 2022; Sharma, 2025b). The motivation is clearest where the required knowledge is absent from parameters. On queries involving recently updated or long-tail facts, direct inference yields near-zero accuracy even for strong models (Li et al., 2025b), showing that such queries are information-limited rather than compute-limited, since no amount of additional reasoning can recover the missing facts.

From a knowledge-mechanism perspective, RAG does not simply combine parametric and contextual sources additively. Causal tracing reveals that retrieved context can suppress the model’s normal parametric recall pathway, redirecting generation to rely on the retrieved passage rather than on stored knowledge (Wadhwa et al., 2024). This shortcut helps when the retrieved content is correct but creates vulnerability when it is noisy or conflicting. Retrieval can therefore improve accuracy on well-covered questions while increasing hallucination on dynamic or complex ones, so the net effect of RAG varies across question types (Yang et al., 2024d). Moreover, because RAG selects a limited set of passages, it risks missing dispersed relevant information that long-context models can cover (Jiang et al., 2024b; Laban et al., 2024). One response is to re-rank retrieved snippets by how much they reduce the model’s internal-state uncertainty rather than by surface relevance, directly coupling passage selection with the model’s knowledge needs (Yao et al., 2024b).

Beyond improving single-pass retrieval, a growing line of work replaces static retrieve-then-generate pipelines with iterative loops in which the model interleaves reasoning and retrieval (Li et al., 2025d). The key shift is that the model itself decides when additional knowledge is needed, what to search for, and how to integrate the results. This paradigm appears in tree-search RAG that dynamically triggers retrieval at reasoning branch points (Hu et al., 2025c), hierarchical decomposition of complex queries into sub-retrievals (Jin et al., 2025b), and RL-trained search strategies that emerge from reward signals (Jin et al., 2025a; Li et al., 2025c). Whether to retrieve or to rely on parametric knowledge can also be learned at the subquery level, avoiding unnecessary retrieval while maintaining accuracy (Guan et al., 2025). These approaches reflect the compute-limited versus information-limited distinction introduced earlier. At each step, the model must judge whether further retrieval or further reasoning will be more productive.

Verifying that the model actually used retrieved evidence, rather than falling back on parametric memory, remains a core challenge. Surface-level validators based on natural language inference or self-citation can be fooled by plausible textual overlap, motivating internals-based attribution methods that identify tokens whose generation probabilities genuinely depend on retrieved documents (Qi et al., 2024). Citation training also shows that faithful attribution is a learned behavior that must be explicitly cultivated (Chuang et al.,

2025; jiajie Zhang et al., 2024). RAG thus exposes a persistent tension in knowledge utilization, as answer correctness and evidence faithfulness are related but not identical objectives.

5.2.3 TOOL USE

Finding

Tool use lets models actively acquire, verify, and integrate external knowledge rather than passively receive it. A core agentic challenge is knowledge boundary awareness, namely recognizing when parametric knowledge suffices and when external acquisition is needed.

Tool use extends the augmentation paradigm beyond passively receiving retrieved text. The model actively constructs queries, executes external operations, and integrates heterogeneous results into ongoing reasoning (Mirza et al., 2024). Models can learn with minimal supervision to select tools and construct arguments (Schick et al., 2023), and interleaving reasoning traces with tool actions allows returned observations to steer subsequent steps (Yao et al., 2023a). Building on these capabilities, the agent paradigm treats the LLM as a central reasoning engine that orchestrates search engines, code interpreters, domain APIs, and standardized tool protocols to acquire knowledge on demand. As tool ecosystems grow, hierarchical architectures decouple strategic planning from tool execution, using dedicated planner, coordinator, and executor modules to manage multi-step knowledge acquisition without disrupting the reasoning chain (Jin et al., 2025b).

From the perspective of knowledge mechanisms, tools do not merely supplement missing information but can activate knowledge that is latent in the model’s parameters. Fine-tuning on tool-using trajectories improves performance beyond what additional reasoning data alone provides, because tool execution opens computation paths that pure text generation cannot reach (Li et al., 2025a). Theoretical analysis also shows that without tools, verifying certain computations requires the model to memorize an impractically large volume of facts, whereas access to a simple calculator eliminates this requirement entirely (Kang et al., 2026). Tool-integrated reasoning also reduces overthinking by replacing verbal speculation with executable verification, although poorly integrated tool feedback can corrupt initially correct reasoning traces (Zhao et al., 2025c).

The hardest decision in tool use is locating the knowledge boundary itself. Evaluations of agentic search reveal that LLMs systematically default to parametric memory even in tool-augmented settings, producing fabricated URLs and missing citations rather than invoking available retrieval (Gou et al., 2026). Similarly, agents in interactive environments rarely use provided tools due to misestimated self-competence, allowing parametric knowledge to override grounded observations (Miyai et al., 2025). Reinforcement learning can train this awareness by teaching the model to distinguish internally answerable questions from those requiring external retrieval, reducing unnecessary tool calls while improving accuracy through elimination of retrieval redundancy and knowledge conflicts (Huang et al., 2025). Beyond individual decisions, long interaction histories cause information loss across tool-call sequences, and coordinating heterogeneous outputs from diverse tools adds complexity to multi-step reasoning (Wang et al., 2024b; Luo et al., 2025). Scaling agentic knowledge acquisition thus requires progress on both the awareness problem (when to use tools) and the orchestration problem (how to manage knowledge across extended interactions).

5.3 Knowledge Conflict

Knowledge conflict occurs when the model faces incompatible knowledge signals. Parametric memory may disagree with the current context, retrieved sources may contradict one another, or the model may express inconsistent beliefs across equivalent prompts. We divide knowledge conflict into context-memory conflict (CM), inter-context conflict (IC), and intra-memory conflict (IM). These three conflict types require different responses rather than a single robustness policy. Stale parametric memory should yield to updated context, but corrupted or adversarial context should be resisted. A blanket policy such as “always trust the context” therefore fails whenever the context itself is unreliable.

5.3.1 CONTEXT-MEMORY CONFLICT

Finding

When context contradicts parametric memory, models do not neutrally weigh both sources but preferentially accept evidence that confirms their existing beliefs while resisting corrections. User assertions can override correct model judgments, making it difficult to distinguish cases where parametric memory should yield from cases where context should be resisted.

CM conflict has two opposite regimes. When parametric knowledge is outdated, the desirable behavior is contextual faithfulness. When the supplied context is misleading or poisoned, the desirable behavior is resistance. Strengthening context reliance cannot be a universal solution, because the same tendency that repairs stale-memory errors can amplify misinformation when context is corrupted (Sun et al., 2025c).

Models facing CM conflict exhibit systematic biases that shape how they arbitrate between sources. The most pervasive bias is confirmation bias, where models accept evidence aligning with parametric beliefs while resisting corrections (Su et al., 2024). Surface-level cues further modulate this bias. When a user explicitly claims that a fact is wrong, models sycophantically reverse correct assessments to agree with the user (Lee and Chowdhary, 2025). In multilingual settings, the query language alone can shift how the model interprets identical retrieved text (Li et al., 2024a). Even when models appear to update, the change may be superficial. Models may verbally acknowledge corrections while remaining behaviorally unchanged (Jiang et al., 2025), or generate rationalizations that make incompatible claims appear compatible (Huo et al., 2025). At the representation level, hallucinated outputs align more strongly with the parametric knowledge direction than with contextual evidence (Islam et al., 2025), indicating that conflict resolution failures often reflect insufficient context integration rather than deliberate rejection.

These biases connect to the knowledge representation mechanisms examined in §4. Probing the residual stream reveals that conflict detection and source selection are layer-separable processes (Zhao et al., 2024c), enabling representation engineering that steers the model toward the desired source at inference time (Zhao et al., 2024b). At the training level, approaches range from teaching models to prioritize context when parametric knowledge is stale (Si et al., 2025) to defensive cross-checking that verifies retrieved claims against parametric knowledge (Wang et al., 2025d). Adaptive approaches avoid a fixed source hierarchy altogether, instead comparing internal and retrieved knowledge by estimated reliabil-

ity (Wang et al., 2025a), decomposing conflicting claims into comparable units (Huo et al., 2025), or using asymmetric debate between contextual and parametric arguments (Zhou et al., 2025b). The remaining challenge is that these methods still require external signals to determine which source is trustworthy. A model that can autonomously infer whether a conflict reflects stale memory, corrupted context, or genuine ambiguity remains an open goal.

5.3.2 INTER-CONTEXT CONFLICT

Finding

When external sources disagree, the primary bottleneck is diagnosing the nature of the disagreement rather than resolving it. Sources may conflict due to genuine ambiguity, misinformation, or information staleness, and each type demands a different response.

IC arises when multiple external sources disagree, and is often harder to resolve than CM because the model may lack reliable parametric knowledge for adjudicating between sources (Liang et al., 2025b). When benchmarks provide ground-truth conflict type labels, response quality improves (Cattan et al., 2025), and adapting the resolution strategy to the identified category outperforms a one-size-fits-all policy (Lee et al., 2025a), suggesting that the primary bottleneck is diagnosis rather than resolution capacity. In agentic settings with multi-step evidence gathering, IC becomes harder because agents may recall individual evidence pieces but fail to correctly revise earlier conclusions when later evidence contradicts them (Ji et al., 2026).

The difficulty of IC is compounded by the fact that retrieval corpora are not internally clean. Naturally occurring inconsistencies in sources like Wikipedia challenge the assumption that retrieved evidence forms a coherent authority (Semnani et al., 2025), and adversaries can exploit RAG pipelines through evidence that is partially correct but misleading (Liang et al., 2025b). Stronger reasoning does not automatically help with these challenges. Search-augmented benchmarks show that advanced reasoning models can become more vulnerable to noisy results, because increased test-time compute may amplify misleading evidence rather than filter it (Pham et al., 2025). This connects to the non-monotonic reasoning gains observed in §5.1.2, where additional computation can worsen outcomes when the underlying knowledge signal is unreliable.

Resolution methods increasingly target type identification as a prerequisite for effective arbitration. Multi-agent debate assigns retrieved documents to separate reasoning threads, allowing legitimate ambiguity to be preserved while misinformation is challenged through adversarial verification (Wang et al., 2025b). Type-aware methods classify inter-document relationships and apply tailored generation strategies for each conflict category (Lee et al., 2025a). When conflicts cannot be fully resolved, span-level explanations can localize which parts of the evidence drive model uncertainty (Sun et al., 2025b), and traceback methods can attribute outputs to specific influential context segments (Wang et al., 2025e, 2026c). Without such identification, models tend to collapse conflicting evidence into a single answer, presenting one viewpoint when multiple valid perspectives exist and discarding legitimate ambiguity along with genuine errors (Cattan et al., 2025; Wang et al., 2025b).

5.3.3 INTRA-MEMORY CONFLICT

Finding

Intra-memory conflict often reflects unstable internal routing rather than absent knowledge. Methods that contrast layer-wise representations or steer activations along truth-correlated directions can improve factuality without adding new information.

IM conflict arises when the same model expresses inconsistent factual claims across paraphrased prompts, different languages, or successive parts of a long generation. No external evidence is needed to reveal the conflict. It is exposed simply by varying how the model is queried (Xu et al., 2024a). At the most basic level, stochastic decoding means that repeated sampling from the same prompt can yield contradictory answers. However, inconsistency persists even under deterministic decoding, where semantically equivalent prompts elicit different factual claims, indicating causes at the representation level. Its causes trace back to earlier lifecycle stages, including contradictory pre-training data, uneven representation of equivalent relations during knowledge acquisition (§3), and editing side effects that fail to propagate across related facts in knowledge evolution (§6). Long-form generation exposes a related pattern. Models increasingly contradict facts and timelines they previously established as the generated context grows, with consistency errors scaling approximately linearly with output length (Li et al., 2026a).

If many IM conflicts stem from unstable routing rather than missing knowledge, they should be correctable by intervening on internal computation rather than by supplying new information. Several methods support this prediction by exploiting the representational structure examined in §4. Layer-contrastive decoding (DoLa) exploits the observation that factual knowledge matures in deeper layers. By contrasting later-layer logits with earlier-layer logits, it amplifies the factual signal that shallower layers have not yet resolved, improving truthfulness without external knowledge (Chuang et al., 2024). Inference-time intervention (ITI) takes a complementary approach by identifying attention heads whose activations correlate with truthful outputs and shifting activations along these directions. This can improve truthfulness, indicating that models possess internal representations distinguishing true from false statements but do not always follow them during generation (Li et al., 2024b). REFLEX extends this logic to the post-training setting. Post-training improves reasoning but can simultaneously introduce factual inconsistencies, and these two effects are separable in activation space. By isolating the activation directions responsible for factual drift and suppressing them while preserving reasoning-style directions, factual accuracy improves (Kong et al., 2025). Together with the SAE-based gating mechanisms discussed in §5.1.3, these approaches indicate that many IM failures are access problems rather than storage problems. It remains open whether training objectives can make equivalent prompts consistently activate the same stored knowledge, or whether some degree of inconsistency is inherent to distributed knowledge representations.

5.4 Discussion

Across recall, augmentation, and conflict, knowledge utilization failures share a common signature. Recall failures are mainly access failures, where encoded knowledge is not reliably surfaced during recognition, retrieval, or generation. Augmentation addresses the comple-

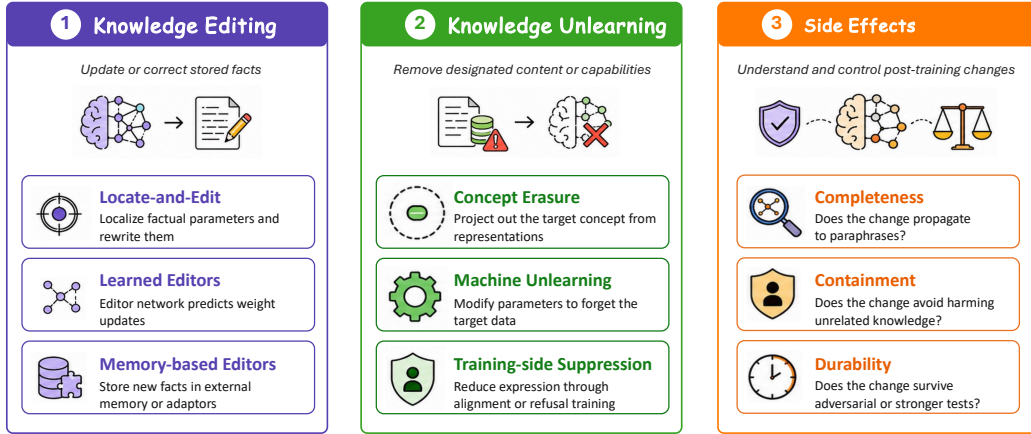


Figure 7: Overview of knowledge evolution, organized into knowledge editing, knowledge unlearning, and the side effects of both operations, which we analyze along the axes of completeness, containment, and durability together with the corresponding mitigation strategies.

mentary availability problem, where required knowledge is absent from parameters and must be supplied externally. Conflict is the point where access and availability intersect, requiring the model to arbitrate between sources of uncertain reliability and to stabilize its own internal disagreements. The encode-express gap introduced at the beginning of this section recurs in analogous forms beyond recall. Recall failures frequently reflect routing instabilities rather than absent information. Augmentation can elicit latent parametric knowledge rather than merely supplementing it. Even in conflict scenarios, the dominant bottleneck is diagnosing the nature of disagreement rather than resolving a known one. In many of these cases, the limiting factor is not whether the relevant knowledge exists but whether the model selects, routes, or arbitrates it correctly, a failure pattern that spans the utilization pipeline from internal retrieval through external integration to source arbitration.

6 Knowledge Evolution

Once pre-training is complete, the knowledge encoded in the parameters of a large language model is essentially frozen. New facts emerge, existing facts become outdated, and copyrighted, private, or harmful content may need to be removed. **Knowledge evolution** studies how to update what an LLM knows without paying the prohibitive cost of a full retraining pass (Yao et al., 2023b; Liu et al., 2024c; Wang et al., 2024c). We focus on **Knowledge Editing** (§6.1), which corrects individual facts, **Knowledge Unlearning** (§6.2), which removes designated content, and the **Side Effects** (§6.3) that both operations incur.

6.1 Knowledge Editing

Finding

Knowledge editing is difficult because locating where a fact is stored does not guarantee that it can be safely overwritten, and chaining many edits can drive the model toward collapse.

Knowledge editing aims to correct outdated or incorrect facts in a pre-trained model. Given a fact described by a subject, a relation, and an object, the goal is to make the model produce the new target object for that subject and relation, while suppressing the previous answer and preserving behavior on unrelated inputs. Overwriting an existing association and protecting the surrounding knowledge are two requirements that pull in opposite directions, and together they shape both method design and evaluation.

Locate-and-edit methods build on the knowledge-neuron hypothesis (Dai et al., 2022) and causal tracing. They first locate the parameters that store the target fact and then rewrite them through a closed-form or low-rank update. ROME (Meng et al., 2022) models a mid-layer MLP as a key-value memory and formulates a single-fact write as a constrained least-squares problem. MEMIT (Meng et al., 2023) extends this to thousands of sequential overwrites, and PMET (Li et al., 2024f) jointly optimizes attention and MLP outputs. AlphaEdit (Fang et al., 2025) confines updates to the null space of preserved keys, reducing the model collapse observed when many edits are chained.

Meta-learning and hyper-network editors instead train a separate editor network that maps an edit request to a weight update. KnowledgeEditor (De Cao et al., 2021) first formulated this as constrained optimization, and MEND (Mitchell et al., 2022a) made it tractable for very large models through low-rank gradient decomposition. MALMEN (Tan et al., 2024) and InstructEdit (Tian et al., 2024) extend this paradigm to large edit batches and natural-language edit instructions.

Recent probing studies reveal the depth of the editing challenge. Localization results are highly sensitive to the choice of prompt and metric, and where a fact is stored is not necessarily where it can be safely overwritten (Hase et al., 2023). Chaining tens of thousands of sequential edits typically causes exploding perplexity and catastrophic forgetting (Gupta et al., 2024; Fang et al., 2025), and even a single overwrite can propagate through semantically related facts and produce logical inconsistencies (Cohen et al., 2024a). These pathologies constitute core open problems of knowledge editing and motivate the side-effect analysis in §6.3.

The classical evaluation metrics for editing are reliability, generality, and locality (Meng et al., 2022; Yao et al., 2023b). Recent work further asks whether the modified fact remains usable in multi-hop and paraphrased queries (Cohen et al., 2024a; Zhang et al., 2024a), while the stability of the model under long sequences of edits remains poorly understood.

6.2 Knowledge Unlearning

Finding

Unlearning methods range from representation-space projection to parameter-level modification to surface-level suppression. Interventions that leave the underlying representation intact are the easiest to bypass.

Knowledge unlearning aims to remove the influence of designated training content, often called the forget set, from a trained model. The unlearned model should behave on the forget set as if it had never been trained on it, while preserving utility on the retained data and on downstream benchmarks (Bourtole et al., 2021; Liu et al., 2024c). Typical use cases include copyright removal, compliance, suppression of harmful capabilities, and bias mitigation. Ideally, the removed content should neither surface in ordinary use nor be recoverable through adversarial probing. Existing methods intervene at different levels, ranging from the representation space to the parameters to the training objectives that shape output behavior.

Concept erasure methods operate in the representation space. They apply a linear or low-rank projection that removes the target concept from the hidden states. INLP (Ravfogel et al., 2020) iteratively trains linear probes and projects representations into their null space, and LEACE (Belrose et al., 2025) provides a closed-form projection with minimal distortion. Representation engineering (Zou et al., 2023) extends this idea to nonlinear directions such as honesty or refusal. A known limitation is that concept erasure typically removes only the linear readability of the target attribute, and downstream nonlinear heads can still recover it (Belrose et al., 2025).

Machine unlearning methods instead modify the parameters directly. Gradient ascent on the forget set (Jang et al., 2023) is the simplest variant but is prone to catastrophic degradation, and NPO (Zhang et al., 2024b) derives a more stable objective from negative preference optimization. Other representative methods include task-vector negation (Ilharco et al., 2023), the teacher-student SCRUB formulation (Kurmanji et al., 2023), and the SISA sharded-training architecture (Bourtole et al., 2021), which enables exact unlearning by retraining only the affected shards.

A third line of work suppresses the expression of knowledge rather than removing it, through refusal training with RLHF or DPO (Rafailov et al., 2023), safety fine-tuning, and preference-level unlearning (Eldan and Russinovich, 2023; Maini et al., 2024). Although such suppression is easy to integrate with existing alignment pipelines, the resulting forgetting has repeatedly been shown to be shallow, and it can be bypassed by jailbreaks, re-elicitation, or a small amount of fine-tuning (Lynch et al., 2024; Lermen et al., 2024). This suggests that in modern LLMs, knowing a piece of information and being willing to express it are partly separable, and disentangling the two remains an open direction.

Evaluation has converged on benchmarks such as WMDP (Li et al., 2024d), TOFU (Maini et al., 2024), and MUSE (Shi et al., 2024), and typically examines three aspects. Forget quality asks whether the target content is actually removed, measured, for example, by membership inference or by comparison against a retrained reference model. Retain utility asks whether performance on held-out data and downstream tasks is preserved. Robustness asks whether the forgetting survives relearning or jailbreak attacks (Shumailov et al., 2024).

6.3 Side Effects of Editing and Unlearning

Knowledge editing and unlearning change the factual, behavioral, or conceptual content of a pre-trained language model after training. These interventions are intended to be local, but in practice they often affect a broader set of behaviors. An intervention that succeeds on its target fact may still damage neighboring knowledge, weaken retained capabilities, or leave recoverable traces of the original content. We organize these side effects along three axes. Completeness (§6.3.1) asks whether the intended change reaches the whole semantic region it should reach. Containment (§6.3.2) asks whether the change leaves unrelated knowledge intact. Durability (§6.3.3) asks whether the change survives stronger evaluation and adversarial access. Finally, we discuss the mitigation strategies (§6.3.4).

6.3.1 COMPLETENESS

Finding

Success on the target prompt may overstate how far an edit or deletion has propagated, because knowledge is encoded through multiple pathways that a single intervention does not cover.

Completeness concerns whether an intervention propagates far enough. An update can succeed on its target query and still fail on logically related queries. Typical signs are poor generalization to paraphrases and failure on multi-hop questions. Multi-hop and relational editing benchmarks such as MQuAKE and RippleEdits report this pattern clearly (Zhong et al., 2023; Cohen et al., 2024a). MULFE extends this observation to free-text editing, where probes that range from literal recall to deeper reasoning expose inconsistent behaviors across generalization levels (Wang et al., 2024a).

The gap takes different forms across editing and unlearning. In knowledge editing, an edited fact can be correct for the canonical prompt and wrong under paraphrases, multi-hop queries, or multilingual transfer. In multimodal models, the gap widens further because knowledge at different granularities is stored across distinct components that must be edited jointly (Zhou et al., 2025a). The difficulty grows for free-text and commonsense knowledge, which is dispersed across both MLP and attention layers rather than localized in a single module, so an edit applied at one site cannot cover the whole region (Huang et al., 2024c). When the update is introduced through context, retrieval, or side memory, the new fact may also remain outside the model’s internal reasoning (Mitchell et al., 2022b; Hartvigsen et al., 2023). The model recalls it when the memory is active but may not use it in compositional reasoning.

In unlearning, completeness requires removing every expression of the target. A forgotten fact may disappear under the exact deletion prompt while paraphrases or related facts still reveal it, because the content is encoded through several alternative pathways, and removing one observable expression does not remove the rest (Li et al., 2024d; Wang et al., 2025c).

6.3.2 CONTAINMENT

Finding

Edited and retained knowledge compete within shared parameter subspaces, so parametric overwrites can disturb neighboring facts and accumulate into catastrophic forgetting over repeated edits.

Containment concerns whether an intervention leaves unrelated knowledge intact. At the fine end, an edit changes a specific neighboring fact that should have stayed fixed. At the coarse end, an edit degrades broad capabilities such as factual recall, reasoning accuracy, and fluency, producing symptoms such as higher perplexity, catastrophic forgetting, and collapse after repeated edits.

For knowledge editing, containment risk depends on how the update is applied. A parametric overwrite is the most destructive case, because the old and new facts compete within the same parameter subspace, and the local change can disturb other associations that share it. Locate-and-edit methods such as ROME and MEMIT (§6.1) show this tension, as repeated edits accumulate in shared layers and disturb general behavior (Meng et al., 2022, 2023; Yang et al., 2024c). Later work links this degradation to edit overfitting and norm growth. Edit overfitting means the model learns a narrow link between the edit prompt and the target answer rather than a stable semantic change. Norm growth means repeated edits amplify certain parameter directions until they interfere with broader abilities (Zhang et al., 2025c; Gupta et al., 2025; Huang et al., 2024b). When the update is stored in external memory or a lightweight adapter instead, it leaves the base parameters mostly intact and carries lower containment risk. Its main risk is then poor integration rather than global degradation.

Unlearning faces a sharper version of the same conflict. The objective must reduce the model’s ability to reproduce the forget set while preserving performance on the retain set. Naive gradient ascent on the forget set tends to break this balance. It can erase more than the intended content or push the model toward unstable generations (Zhang et al., 2024b).

6.3.3 DURABILITY

Finding

Interventions often merely suppress knowledge rather than remove it. Unlearned knowledge resurfaces under adversarial prompts or light relearning, and edited models can revert to original answers that remain encoded in the network.

Durability concerns whether an intervention stays valid under stronger evaluation. An intervention can look successful under a narrow test and fail under a harder one. This axis is central to unlearning. A model may stop producing a forgotten answer under standard prompts while the same knowledge remains recoverable through paraphrases, adversarial suffixes, stochastic decoding, probing, or small amounts of relearning (Li et al., 2024d; Wang et al., 2025c; Xu et al., 2026; Reisizadeh et al., 2026). Benchmarks built around this concern, such as RWKU, therefore pair standard recall probes with adversarial-attack probes and neighbor-knowledge tests under a strict setting where neither the forget corpus nor the retain corpus is available (Jin et al., 2024b).

Refusal-style training can prevent direct disclosure without removing the underlying representation. Concept-erasure methods show the same limitation from the representation side. LEACE gives a closed-form guarantee for removing linearly recoverable information from representations, yet the guarantee holds only for the specified probe class, and non-linear pathways can still preserve recoverable information (Belrose et al., 2025).

Durability also matters for knowledge editing. When the update is stored in side memory, it is available only when that memory is retrieved, so its validity depends on routing (Mitchell et al., 2022b; Hartvigsen et al., 2023). Parametric edits can also be superficial. Models that score near-perfectly on conventional editing metrics still regenerate the original answer, and mechanistic analysis traces this reversion to the residual stream in earlier layers and to specific later-layer attention heads that still encode the old knowledge (Xie et al., 2025). Failures of completeness, such as paraphrase and multi-hop errors, are also durability failures (Zhong et al., 2023; Cohen et al., 2024a), which is why evaluation must test more than target accuracy.

6.3.4 MITIGATION STRATEGIES

Finding

Effective mitigation separates the update from retained knowledge, either outside the base parameters or within a protected subspace, and reliability must be verified by audits beyond output-level metrics.

Mitigation strategies try to control the scope, magnitude, and auditability of post-training knowledge changes. For knowledge editing, two mitigation families dominate. The first keeps the new knowledge explicit and traceable. In-context editing, retrieval-based editing, SERAC, and GRACE store new facts outside the base model or in lightweight adapters, which reduces parameter drift (Zheng et al., 2023a; Mitchell et al., 2022b; Hartvigsen et al., 2023). This design improves locality and reversibility, and its main requirement is accurate routing, since the system must activate the stored knowledge when it is relevant and leave it inactive otherwise. MeLLO and WISE suggest that side memory supports more stable lifelong editing when it is combined with explicit reasoning or memory partitioning (Zhong et al., 2023; Wang et al., 2024e).

The second family constrains a parametric update so that it interferes less with retained knowledge. AlphaEdit projects edits into a null space tied to preserved knowledge, which reduces interference with unrelated facts (Fang et al., 2025), and WilKE reduces interference across sequential edits by selecting which layer to edit for each fact (Hu et al., 2024a). StableKE expands the edit target with paraphrases and contextual descriptions, so the update covers a semantic neighborhood rather than a single prompt form (Wei et al., 2024c). EvoEdit extends this idea to lifelong free-text editing, combining latent perturbation augmentation with knowledge-driven parameter fusion to keep sequential edits both general and stable (Cao et al., 2025). ENCORE targets long edit sequences by controlling edit overfitting and norm growth (Gupta et al., 2025), and spectral-preservation methods aim to protect the dominant directions that support general ability (Zhang et al., 2026a). Both families follow the same principle of separation. The update should be kept apart

from what supports retained knowledge, either outside the base parameters entirely or in a subspace that retained knowledge does not depend on.

For unlearning, effective mitigation needs both stable objectives and stronger audits. NPO replaces unconstrained gradient ascent with a preference-style objective that limits divergence from the reference model (Zhang et al., 2024b). Robust unlearning methods add adversarial prompts, paraphrases, and related facts during training or evaluation (Li et al., 2024d; Wang et al., 2025c). Safety-oriented unlearning shows that removing harmful knowledge can be more robust than training the model to refuse direct requests (Zhang et al., 2025d). Output-level metrics remain insufficient on their own. Reliable evaluation should also include retain-set utility, paraphrase recovery, adversarial attacks, relearning tests, and representation-level probes.

6.4 Discussion

Editing and unlearning are usually studied as separate tasks, but the side effects reviewed above trace back to a shared cause. Both interventions are specified at the level of an individual fact or a bounded corpus, yet their parametric implementations must act on parameters that many pieces of knowledge share, because knowledge is stored in a distributed rather than record-like form (§4). Completeness, containment, and durability describe three ways this mismatch surfaces. The change misses some of the pathways that express the target, or touches pathways that also support unrelated knowledge, or alters only the output behavior while the underlying representation survives. The mitigation strategies reviewed above follow the same logic from two directions, either keeping the change outside the base parameters in explicit and traceable form, or confining the parametric update to a subspace that retained knowledge does not depend on.

Two problems remain open. First, most current evaluation still measures the expression of knowledge rather than knowledge itself, so a suppressed answer and a removed representation can receive the same score, and claims of successful editing or unlearning need representation-level and adversarial audits before they can be trusted. Second, most methods are validated on isolated interventions, whereas deployment requires long sequences of edits and deletions applied to the same model, and how side effects accumulate and interact over such sequences is only beginning to be understood (Gupta et al., 2025; Cao et al., 2025). Progress on both fronts will likely depend on the localization and representation analyses of §4, since knowing where and how a fact is encoded is what would turn editing and unlearning from behavioral patching into controlled modification of model knowledge.

7 Knowledge Transfer

Knowledge transfer turns the focus outward, asking how knowledge moves between models of different scales and architectures, between modalities, and between languages. Unlike knowledge editing, which modifies existing knowledge in situ, knowledge transfer involves replicating or projecting knowledge from a source to a target, typically with the goal of endowing the target with capabilities it could not efficiently learn on its own.

We organize the literature according to the nature of the boundary being crossed. Cross-model transfer (§7.1) encompasses knowledge distillation and model merging. Cross-modal transfer (§7.2) moves knowledge across modality boundaries. Cross-lingual transfer (§7.3)

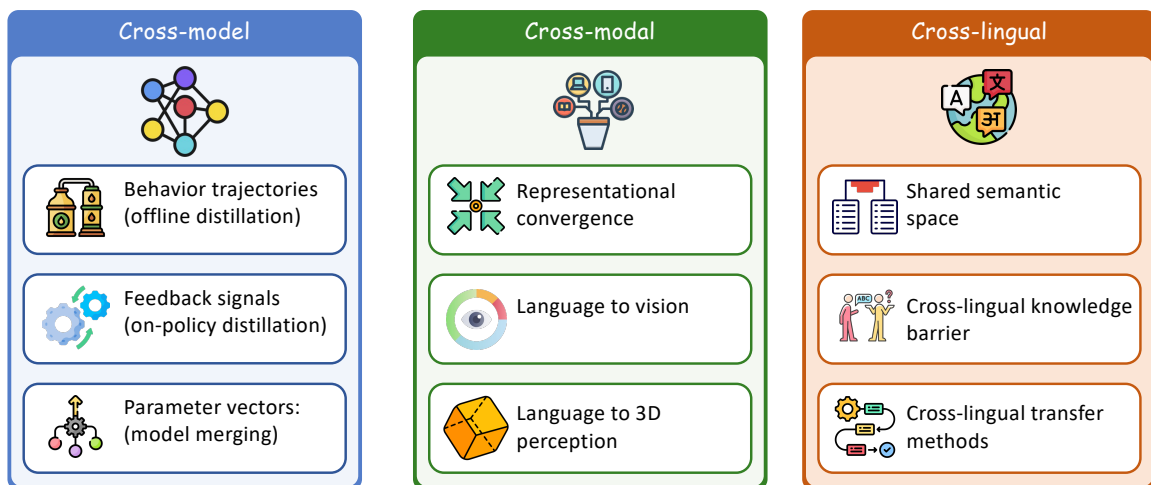


Figure 8: Taxonomy of knowledge transfer in LLMs, organized by the boundary being crossed. Cross-model transfer is categorized by the knowledge carrier (behavior trajectories, feedback signals, and parameter vectors). Cross-modal transfer proceeds from the theoretical foundation of representational convergence to its applications in vision and 3D perception. Cross-lingual transfer is structured around the shared semantic space that enables transfer, the knowledge barrier that limits it, and the methods developed to bridge the gap.

moves knowledge across languages. Each category addresses a distinct type of boundary that knowledge must traverse, and the methods and challenges differ accordingly. The overall landscape is depicted in Figure 8.

7.1 Cross-Model Knowledge Transfer

Cross-model transfer moves knowledge between neural models that may differ in scale, architecture, or training recipe. We organize these methods by what serves as the carrier of knowledge during transfer. In offline distillation, knowledge is encoded in behavior trajectories, specifically the teacher’s reasoning traces that the student learns to reproduce. In on-policy distillation, knowledge takes the form of feedback signals, namely the teacher’s evaluative responses to the student’s own outputs. In model merging, knowledge is carried directly by the parameters themselves as directions in weight space that can be composed through arithmetic.

7.1.1 TRANSFER VIA BEHAVIOR TRAJECTORIES

Finding

Distillation has shifted from aligning teacher-student output distributions to transferring the reasoning process itself. However, static traces may favor superficial imitation over genuine reasoning transfer and amplify teacher biases in the student.

Classical knowledge distillation (KD) (Hinton et al., 2015) treats the teacher’s softened logits as the primary supervision signal, transferring “dark knowledge” about inter-class relationships. This formulation assumes that the teacher’s value lies in its output distribution, and the student’s goal is to reproduce it at lower cost. The emergence of reasoning-capable LLMs, notably o1 (OpenAI, 2024b) and DeepSeek-R1 (Guo et al., 2025), has reshaped this assumption. The teacher’s primary value now lies not in what it predicts but in how it reasons, through chain-of-thought trajectories, problem decomposition strategies, and multi-step inference chains that underlie its answers. Accordingly, the dominant form of LLM distillation has become offline (off-policy), in which the teacher generates reasoning traces and the student trains on them as static supervised data (Guo et al., 2025; Hsieh et al., 2023; Huang et al., 2024d).

This shift from behavior imitation to reasoning transplant introduces qualitatively new challenges. First, not all correct reasoning trajectories are equally useful. The choice of which traces to distill and from which teacher affects student performance (Tian et al., 2025; Zhang et al., 2026b), making trajectory curation a design decision. Second, whether reasoning genuinely transfers or is merely superficially imitated remains an open concern. Work on teacher hacking (Tiapkin et al., 2025) shows that students can exploit systematic weaknesses in teacher outputs rather than internalizing the underlying reasoning strategies. Third, distillation can propagate and amplify undesirable properties. Chaudhari et al. (2025) demonstrate that adversarial biases injected into a teacher cascade through the distillation pipeline, sometimes emerging stronger in the student than in the teacher. Beyond these conceptual challenges, offline distillation also faces a training-inference mismatch in which the student trains on flawless teacher prefixes but must generate its own at inference, causing errors to compound autoregressively with severity that scales roughly quadratically with sequence length (Song and Zheng, 2026). This exposure bias, inherent to using static behavior trajectories as the knowledge carrier, motivates a shift to a different carrier altogether.

Another practical barrier is tokenizer mismatch. When teacher and student use different vocabularies, logit-level supervision becomes impossible. Solutions remain nascent, including byte-level interfaces that operate at a sub-token granularity shared across all tokenizers (Singh et al., 2026) and statistics-based token alignment across vocabularies (Wan et al., 2024).

7.1.2 TRANSFER AS FEEDBACK SIGNALS

Finding

On-policy distillation and RL provide complementary training signals and are increasingly combined within a single optimization. However, moving supervision onto student-generated trajectories can cause distribution drift, in which teacher feedback becomes unreliable once the student’s outputs leave the teacher’s reliable supervision range.

On-policy distillation (OPD) addresses exposure bias by changing the knowledge carrier from static trajectories to dynamic feedback. The student generates its own trajectories and receives teacher evaluations on these self-generated outputs, grounding distillation in inter-

active imitation learning (Agarwal et al., 2024; Gu et al., 2024). OPD has rapidly expanded into a dedicated subfield with surveys (Song and Zheng, 2026) and diverse instantiations spanning divergence minimization, reward-guided learning, and self-distillation.

A central theoretical result underpinning this growth is that standard reverse-KL distillation is mathematically equivalent to policy-gradient RL with the teacher-student log-probability ratio as a dense per-token reward (Yang et al., 2026c). This equivalence reveals that distillation and RL are not independent pipeline stages but complementary signals within the same optimization. Distillation supplies dense gradients that stabilize training, while task-level rewards enable the student to discover solutions the teacher itself would not produce. Methods such as RLAD (Zhang et al., 2026c) operationalize this insight through joint optimization rather than sequential execution, and industrial systems including Qwen3 (Yang et al., 2025a) have converged on multi-stage pipelines that tightly couple distillation with RL.

However, OPD introduces failure modes absent from offline distillation. When the student generates severely out-of-distribution prefixes, the teacher’s conditional distribution on those prefixes becomes unreliable, a phenomenon known as the flawed prefix trap (Fu et al., 2026; Song and Zheng, 2026). Additional instabilities such as optimization collapse and sensitivity to loss formulation have been documented (Zhu et al., 2026; Li et al., 2026b). A parallel line of work on self-distillation removes the external teacher entirely. The model exploits information asymmetries within itself (for instance, conditioning on ground-truth answers at training time) to serve as its own teacher (Zhao et al., 2026b), though this approach is bounded by the knowledge already encoded in the model’s parameters (Kim and Lee, 2026). These findings suggest that offline and on-policy distillation face complementary bottlenecks. The former suffers from exposure bias as errors compound on the student’s own distribution, while the latter suffers from distribution drift as the evolving student moves beyond the teacher’s reliable supervision range.

7.1.3 TRANSFER VIA PARAMETER VECTORS

Finding

Task vectors, the parameter-space displacements from pre-training to fine-tuning, serve as portable units of knowledge that can be arithmetically composed when models share a pre-trained initialization.

While distillation relies on training signals, whether trajectories or feedback, model merging operates directly in parameter space, treating the parameters themselves as the carrier of knowledge. The simplest instantiation is weighted parameter averaging, which averages the weights of several fine-tuned models (Wortsman et al., 2022a) and already yields measurable gains in out-of-distribution robustness. However, naive averaging often produces unacceptably poor results when the constituent models encode diverse capabilities. The introduction of task vectors (Ilharco et al., 2023), defined as the parameter-space difference between a fine-tuned model and its pre-trained ancestor, transformed model merging from a blunt averaging heuristic into a structured arithmetic framework, and catalyzed a rapidly growing body of methods and applications.

Task Arithmetic (Ilharco et al., 2023) demonstrated that these vectors can be composed through elementary operations. Adding multiple task vectors to a pre-trained model achieves multi-task learning, negating a task vector induces targeted forgetting, and vector analogies enable zero-shot transfer to new task combinations. This arithmetic extends naturally to parameter-efficient modules. PEM (Zhang et al., 2023) shows that LoRA and (IA)³ modules trained on different tasks can be composed through the same additive operations. Accumulating evidence suggests that the capabilities encoded by task vectors occupy identifiable, low-dimensional directions. Zbeeb et al. (2025) demonstrate this concretely by extracting a reasoning vector, the parameter difference between an RL-trained model and its supervised fine-tuned predecessor, that consistently improves reasoning benchmarks when added to compatible models. Balasubramanian et al. (2026) generalize this in the Master Key Hypothesis, positing that capabilities correspond to transferable directions in a low-dimensional latent subspace. From the NTK perspective, Ortiz-Jimenez et al. (2023) establish that when task vectors are approximately orthogonal in the NTK feature space, arithmetic composition preserves each task’s performance. Together, these findings suggest that different models may converge on a shared geometry of capabilities, echoing the Platonic Representation Hypothesis (Huh et al., 2024) at the level of behavioral directions rather than individual representations.

The effectiveness of parameter-space composition rests on the geometry of the loss landscape. Linear mode connectivity (LMC) (Frankle et al., 2020) holds when linear interpolation between two sets of weights traverses no high-loss barrier, so that every intermediate model performs about as well as the two endpoints. Multiple studies (Entezari et al., 2022; Frankle et al., 2020) have shown that models fine-tuned from a shared pre-trained checkpoint reliably satisfy LMC, placing them in the same loss basin and making weight interpolation a low-risk operation. This property underpins the empirical success of Model Soups (Wortsman et al., 2022a), which averages models fine-tuned with different hyperparameters, and WiSE-FT (Wortsman et al., 2022b), which interpolates between pre-trained and fine-tuned weights to improve robustness under distribution shift. Multimodal experiments reinforce this principle. Sung et al. (2023) find that merging vision and language models succeeds only when both start from the same pre-trained backbone. While shared initialization is the most reliable enabler, it is not strictly necessary. Weight alignment methods such as Git Re-basin (Ainsworth et al., 2023) can permute the neurons of independently trained models to place them in a common basin before merging, and architecture transformation techniques (Wan et al., 2024) use distillation to unify heterogeneous model structures, though both approaches incur additional cost.

When task vectors from different objectives point in opposing directions at the same parameter positions, their composition produces destructive interference. TIES-Merging (Yadav et al., 2023) first systematically identified this sign conflict problem and proposed retaining only a small fraction of the largest-magnitude parameters while resolving sign disagreements. Following the taxonomy of Yang et al. (2026a), we organize subsequent methods by the granularity at which merging coefficients are determined. At the task level, DARE (Yu et al., 2024b) randomly sparsifies and rescales task vectors, requiring no extra data. At the layer level, AdaMerging (Yang et al., 2024b) learns per-layer merging coefficients via entropy minimization on unlabeled test data. At the parameter level, Fisher-Merging (Matena and Raffel, 2022) weights each parameter by its Fisher informa-

tion, requiring labeled validation data. At the sample level, WEMoE (Tang et al., 2024) dynamically computes input-dependent merging coefficients through routing networks at inference time. Empirically, finer-grained methods tend to perform better, approaching the performance of independent task-specific models (Yang et al., 2026a). However, each step demands more data, computation, or inference overhead. This trade-off between conflict resolution quality and practical cost remains the central tension driving methodological innovation in model merging.

7.2 Cross-Modal Knowledge Transfer

Cross-modal knowledge transfer investigates how knowledge encoded in one modality, typically language, can be projected onto other modalities such as vision and 3D spatial understanding. A notable shift in the field is the reversal of knowledge flow. Where early multimodal work such as CLIP (Radford et al., 2021) learned shared representations from paired multimodal data, the dominant paradigm now treats pre-trained text LLMs as rich knowledge sources and projects their capabilities outward onto other modalities.

7.2.1 REPRESENTATIONAL CONVERGENCE

Finding

Models trained independently on different modalities and architectures develop increasingly similar representational structures.

A growing body of evidence shows that large-scale models, even when trained on entirely separate modalities, develop internally similar representation geometries. Huh et al. (2024) articulate this as the Platonic Representation Hypothesis, arguing that as vision models and language models grow larger, they measure distances between data points in increasingly similar ways and converge toward a shared statistical model of the underlying reality. This convergence is not limited to models that share training data or objectives. Merullo et al. (2023) demonstrate that a single learned linear projection suffices to map frozen vision-transformer embeddings into the token-embedding space of a frozen language model, enabling the stitched system to caption images and answer visual questions even though neither model was trained on multimodal data, a result that would be difficult to explain without pre-existing structural alignment. Rahmazadehgervi et al. (2025) further confirm that independently trained unimodal vision and language encoders encode semantically similar structures, enabling zero-shot latent-space communication without explicit alignment training. At a finer grain, Lee et al. (2025b) reveal that cross-modal representations between text and speech converge progressively over model depth. Shallow layers remain modality-specific and specialized for raw input processing, while deeper layers develop modality-agnostic abstractions. This layer-wise convergence pattern is consistent across multiple multimodal foundation models.

These findings collectively suggest that cross-modal transfer succeeds not mainly because alignment is engineered between modalities, but because the underlying representations are already converging. Much of the practical challenge thus lies in exposing and exploiting this latent alignment. Gu et al. (2025) pursue this direction through universal embedding learning that unifies text, image, and other modalities within a single representation space, while

Boizard et al. (2026) show in BidirLM that causal LLMs can be adapted into omnimodal bidirectional encoders by composing modality-specific components onto a shared backbone. The existence of spontaneous representational convergence provides a theoretical basis for the empirical successes reviewed in the following sections.

7.2.2 FROM LANGUAGE TO VISION

Finding

Language-to-vision transfer can work with little architectural machinery, from a single linear projection between frozen models to directly reusing frozen LLM blocks as visual encoder layers.

The most mature application of cross-modal transfer projects language knowledge into visual understanding. A foundational result is the discovery that the representational barrier between vision and language can be bridged by surprisingly simple mappings. Merullo et al. (2023) show that training only a single linear layer between a frozen image encoder and a frozen generative language model enables image captioning and visual question answering, without any modification to either model. That such a simple mapping suffices suggests that the conceptual representations learned by vision and language models are largely alignable through a linear transformation. The LLaVA framework (Liu et al., 2023) operationalizes this insight at scale. By projecting CLIP visual tokens into the LLM input space through a lightweight MLP and freezing both the vision encoder and the LLM, it achieves strong multimodal performance with minimal trainable parameters.

Pang et al. (2024) further reveal that frozen LLM transformer blocks, trained only on text, can directly process visual tokens as effective visual encoder layers, consistently improving performance across 2D image classification, 3D point cloud recognition, and action recognition tasks. They propose the *information filtering hypothesis*, in which the pre-trained LLM blocks distinguish informative visual tokens and amplify their effect, a general capability that transfers across modalities. Takishita et al. (2025) corroborate this from a complementary angle, showing that when the upper layers of a vision encoder’s self-attention are progressively ablated, the LLM can fully compensate for the missing high-level semantic features. The recognition accuracy remains nearly unchanged even after removing the top six layers, suggesting that the LLM internally reconstructs visual semantics that the vision encoder would otherwise have provided.

Beyond architectural transfer, a growing body of work distills specific reasoning capabilities from text LLMs into vision systems. Carbune et al. (2024) and He et al. (2025) transfer chart reasoning from LLMs to VLMs, enabling structured visual reasoning that VLMs could not acquire from image-text data alone. Du et al. (2025) explore reproducing o1-like slow-thinking reasoning in multimodal LLMs, while Huang et al. (2026a) develop directional reasoning injection that selectively transfers reasoning chains from LLMs to enhance multimodal inference. A parallel thread uses language knowledge to improve visual representations themselves. Hu et al. (2024c) equip diffusion models with LLM-derived semantic understanding to improve text-to-image alignment, and Jeon et al. (2026) enhance visual features through language-guided spatial reasoning. Across both threads, the converging conclusion is that text LLMs, through training on large corpora, have developed

semantic structures and reasoning capabilities that complement and can improve purely visual systems, and that accessing these capabilities requires little architectural machinery.

7.2.3 FROM LANGUAGE TO 3D PERCEPTION

Finding

LLMs’ computational mechanisms generalize to 3D tokens, but text corpora provide few spatial-structural priors. Direct projection alone therefore cannot supply these priors, and practical systems route 3D input through dedicated encoders or intermediate 2D vision instead.

An emerging frontier extends cross-modal transfer from 2D visual understanding to 3D spatial reasoning. A key observation distinguishes this setting from the language-to-vision case. While the computational machinery of LLMs generalizes to 3D data, the knowledge content does not transfer as readily. Pang et al. (2024) demonstrate that frozen LLM transformer blocks can directly process point cloud tokens and improve 3D classification. Notably, the same blocks that enhance 2D tasks also benefit 3D recognition, confirming that the attention mechanism itself is modality-agnostic and provides a useful inductive bias for 3D data. However, this architectural transferability does not resolve the deeper challenge that language pre-training corpora provide little coverage of the spatial-structural priors that are essential for 3D understanding.

This knowledge gap is reflected in the architectures that have emerged for language-to-3D transfer. Rather than the linear projections that suffice for 2D vision, practical 3D systems increasingly rely on multi-step intermediary strategies. Xu et al. (2024b) introduce PointLLM, which encodes colored point clouds through a dedicated point cloud encoder before projecting into the LLM’s latent space, fusing geometric, appearance, and linguistic information. Hong et al. (2023) inject 3D scene representations into LLMs through specialized encoders. More recent systems go further, adopting indirect transfer pathways. Deng et al. (2025) and hybrid methods (Fu et al., 2024) reconstruct 3D scenes from multi-view images using neural radiance fields or Gaussian splatting, then summarize the resulting feature fields for processing by vision-language models. This two-hop design, in which language knowledge reaches 3D through 2D vision-language models, has become a prominent paradigm precisely because a direct language-to-3D projection cannot supply the missing spatial priors. At the scene level, the representational gap widens further. Yang et al. (2024e) transfer 2D segmentation knowledge to 3D objects, while systems such as Chat-Scene (Huang et al., 2024a) and LL3DA (Chen et al., 2024a) ground dialogue and reasoning in reconstructed 3D scenes using object identifiers and position-aware features, all of which constitute additional structural scaffolding that has no counterpart in the 2D vision pipeline.

These findings point to a clear asymmetry in language-centric cross-modal transfer. The further a target modality lies from the statistical structure captured in text corpora, the more architectural scaffolding and domain-specific inductive biases are required to compensate for the missing knowledge. For 3D perception, a promising direction is to incorporate spatial-structural supervision directly into the language model’s pre-training or intermediate training stages, rather than relying solely on downstream alignment to bridge the gap.

7.3 Cross-Lingual Knowledge Transfer

Cross-lingual knowledge transfer concerns whether knowledge acquired predominantly through one language during pre-training can be retrieved and utilized in other languages. This question is practically urgent because the vast majority of high-quality training data is concentrated in a handful of high-resource languages. Recent research has converged on three key findings. LLMs develop an internal shared semantic space that underpins transfer (§7.3.1). Despite this, a systematic knowledge barrier limits deep transfer in practice (§7.3.2). Among existing transfer methods spanning a wide cost spectrum, mixed-language fine-tuning has emerged as the most consistently effective strategy, though not a universal one (§7.3.3).

7.3.1 SHARED SEMANTIC SPACE

Finding

LLMs develop a shared semantic space, and cross-lingual transfer is most effective when computation stays within this space.

Interpretability studies have revealed that multilingual LLMs do not process each language in isolation but develop an internal shared representation space through which knowledge can, in principle, flow across languages. [Lim et al. \(2025\)](#) apply the logit lens to Gemma 2 and Qwen 2.5 models and find that when the model’s intermediate hidden states operate within a shared semantic subspace, one that is closest to English representations, cross-lingual knowledge transfer is most effective. Conversely, when hidden states drift into language-specific subspaces, predictions become inconsistent across languages. Complementing this view from the perspective of knowledge boundaries, [Xiao et al. \(2025\)](#) probe LLM internal representations when processing known and unknown questions across multiple languages, discovering that knowledge boundary perception is encoded in the middle-to-upper layers and that the differences between languages follow a *linear* structure. As a result, simple mean-shifting or linear projection suffices to transfer boundary perception across languages without any training. At a coarser architectural level, [Bandarkar et al. \(2025\)](#) provide a striking demonstration by swapping layers between an English math-expert model and a target-language model. Upper layers encode task-specific knowledge while lower layers encode language-specific representations, and recombining them yields zero-shot cross-lingual math performance. [Arnett et al. \(2025\)](#) show that bilingual models spontaneously develop shared grammatical abstractions, further suggesting that language-universal structure emerges as an intrinsic property of large-scale training.

A notable paradox accompanies these findings. [Lim et al. \(2025\)](#) observe that *larger* models, while more accurate overall, tend to dissociate their hidden states from the shared space and reason more in language-specific subspaces. These larger models compensate through sheer capacity, encoding knowledge redundantly across languages rather than utilizing the shared space more fully. This finding challenges the assumption that scaling alone will resolve cross-lingual transfer limitations.

7.3.2 CROSS-LINGUAL KNOWLEDGE BARRIER

Finding

Multilinguality does not entail crosslinguality. Models translate well when source text is in context, yet struggle to retrieve parametric knowledge learned in one language when queried in another.

Despite the existence of a shared semantic space, mounting evidence shows that current LLMs fail to fully exploit it. [Chua et al. \(2025\)](#) provide the most systematic demonstration of this *cross-lingual knowledge barrier*. By constructing mixed-language versions of MMLU, where the question appears in one language and the answer options in other languages, they show that all 15 evaluated models, including GPT-4, suffer accuracy drops under mixed-language conditions, even though these models perform competitively on standard machine translation. The key distinction is between *explicit* cross-lingual tasks, where the source text is provided in context (e.g., translation), and *implicit* cross-lingual tasks, where the model must retrieve parametric knowledge learned in one language to answer a query posed in another. The barrier is pronounced for the latter.

[Goldman et al. \(2025\)](#) operationalize this distinction through the ECLeKTic benchmark, which tests knowledge sourced from Wikipedia articles that exist in only a single language. For example, the benchmark asks whether a fact recorded only on Indonesian Wikipedia can be retrieved when the question is posed in German. Even the best-performing model (Gemini 2.5 Pro) achieves only modest overall success, confirming that knowledge does not freely cross language boundaries. [Beibitkhan \(2026\)](#) extend this finding to genuinely low-resource languages (Kazakh and Mongolian), reporting a consistent gap between English and low-resource conditions, with models maintaining surface-level fluency while producing less accurate content. At the behavioral level, [Zhao et al. \(2024a\)](#) observe that reasoning abilities transfer cross-lingually far more readily than factual knowledge, which tends to remain bound to the language in which it was acquired. These results collectively establish that multilinguality, the ability to generate fluent text in many languages, does not entail crosslinguality, the ability to transfer knowledge across them.

7.3.3 CROSS-LINGUAL TRANSFER METHODS

Finding

Mixed-language fine-tuning most consistently narrows the cross-lingual gap and generalizes to unseen languages, but no transfer strategy is universal. Which strategy helps, and how much, depends on model architecture and language similarity.

Existing approaches to bridging the cross-lingual knowledge gap span a wide cost spectrum. At the heavy end, continual pre-training pipelines extend the model’s vocabulary and then train on large-scale target-language corpora ([Csaki et al., 2024](#); [Zhao et al., 2024a](#); [Han et al., 2025](#)). These are effective but expensive and risk catastrophic forgetting of the original capabilities. A lighter alternative is tokenizer and embedding transfer. [Remy et al. \(2024\)](#) initialize target-language embeddings as weighted averages of semantically similar source-language embeddings, enabling zero-shot cross-lingual generation by replacing only the surface-level input/output interface. At the lightest end, [Shaham et al. \(2024\)](#) show

that a small amount of multilingual data during instruction tuning suffices to activate broad cross-lingual generalization, and Yong et al. (2025) achieve improved cross-lingual reasoning purely through test-time compute scaling with no language-specific training at all.

Among these approaches, mixed-language fine-tuning stands out for its consistency. Chua et al. (2025) show that fine-tuning on mixed-language data, even out-of-domain corpora such as WikiText, narrows the cross-lingual knowledge gap, and that the benefits generalize to languages not seen during fine-tuning. Xiao et al. (2025) find that fine-tuning on bilingual question-translation pairs further enhances cross-lingual knowledge boundary perception, with the improvement transferring to unseen languages.

However, no single transfer strategy is universally effective. Beibitkhan (2026) demonstrate that cross-lingual transfer prompting, instructing the model to reason in English before translating back, yields meaningful gains for bilingual architectures but provides no benefit to English-dominant models, showing that effective strategies are *architecture-dependent*. Language similarity also matters. Lim et al. (2025) find that languages sharing typological features or writing systems exhibit markedly higher mutual transfer rates. These constraints suggest that closing the cross-lingual knowledge gap will require a portfolio of approaches tailored to specific model architectures and language pairs.

7.4 Discussion

Transfer across all three boundaries rests on a shared foundation. Linear mode connectivity enables parameter-space composition across models (Frankle et al., 2020), the Platonic Representation Hypothesis suggests that vision and language models converge toward shared statistical structures (Huh et al., 2024), and multilingual LLMs develop internal shared semantic spaces that support cross-lingual knowledge flow (Bandarkar et al., 2025; Arnett et al., 2025). In each case, convergence, whether of weights within a shared loss basin or of representations across models and languages, is what makes transfer possible.

However, the barriers that limit transfer are distinct across categories. In cross-model distillation, the primary obstacles are training-inference distribution mismatch (exposure bias in offline distillation, distribution drift in on-policy distillation) and the risk of superficial imitation rather than genuine capability internalization. In model merging, the bottleneck is inter-task parameter interference when task vectors from different objectives conflict in the same parameter positions. In cross-modal transfer, the barrier depends on the target modality. 2D vision transfer can succeed with surprisingly simple linear projections, but 3D transfer faces a deeper knowledge gap where the LLM’s pre-training corpus lacks spatial-structural priors that no amount of alignment can supply. In cross-lingual transfer, the challenge is the cross-lingual knowledge barrier, where models may generate fluent text in many languages while failing to access knowledge originally learned in another language, a limitation that goes beyond surface-level tokenizer mismatch. These diverse failure modes underscore that no single intervention will solve knowledge transfer across all boundaries.

This shared-foundation, distinct-barrier picture leaves several questions open. First, **the theoretical foundations for representational convergence remain fragmented**. The Platonic Representation Hypothesis (Huh et al., 2024), language-independent abstractions in multilingual models (Arnett et al., 2025), and linear mode connectivity in model merging (Frankle et al., 2020) all point in the same direction, namely that independently

trained models converge toward compatible structures, whether in weight space or in representation space, but no unified framework yet explains why this convergence occurs, under what conditions it holds, or where its boundaries lie. Developing such a framework would transform knowledge transfer from empirical recipe-tuning into a theoretically grounded discipline. Second, **the right abstraction level for transferable knowledge is an open question.** The methods surveyed here transfer knowledge through behavior trajectories and feedback signals (distillation), parameters (merging), or representations (embedding transfer, alignment). Yet an emerging body of work in agentic settings suggests that higher-level abstractions, including reusable skills (Zheng et al., 2025; Ni et al., 2026), episodic memories (Kim et al., 2026), and structured knowledge bases (Wang et al., 2026b; Tang et al., 2025), may be more natural and effective units of transfer. Whether the field will converge on parameters, representations, skills, or natural language as the primary transfer medium remains to be seen. Third, **the relationship between knowledge transfer and alignment is underexplored.** Current distillation and merging pipelines focus almost exclusively on capability transfer, with little systematic study of whether safety properties, value alignment, and refusal behaviors transfer reliably or degrade silently. As knowledge transfer becomes increasingly central to the LLM development pipeline, understanding and guaranteeing alignment preservation across transfer steps is a pressing practical concern.

8 Knowledge Evaluation

Evaluating the knowledge of large language models is complicated by the fact that knowledge itself takes two qualitatively distinct forms. *Parametric knowledge* is encoded in model weights during training and can only be accessed indirectly through model behavior. *Contextual knowledge* is supplied at inference time, via retrieved documents, few-shot demonstrations, or tool outputs, and is directly inspectable, though not guaranteed to be correct. Beyond this source distinction, we must also decide what properties to measure. Existing work highlights four recurring dimensions, namely the factual correctness of stored knowledge (Hendrycks et al., 2020), the ability to compose knowledge across multi-step reasoning (Yang et al., 2018), the temporal currency of knowledge relative to the current world (Vu et al., 2024), and the model’s capacity to recognize the boundaries of what it knows (Kadavath et al., 2022). Together, knowledge source and knowledge property form a two-by-four evaluation matrix (Figure 9). Our goal is not only to catalog benchmarks, but also to show that the same property manifests differently across the two sources. For instance, currency in parametric knowledge concerns the staleness of pre-training data, while currency in contextual knowledge concerns whether retrieved information can actually override outdated beliefs.

8.1 Evaluating Parametric Knowledge

Parametric knowledge is stored implicitly in model weights, cannot be read out directly, and must instead be inferred from model behavior on carefully constructed probes. This indirection introduces a persistent methodological challenge. A model that answers a question correctly may be retrieving memorized facts, exploiting spurious correlations, or relying on surface-level patterns in the prompt.

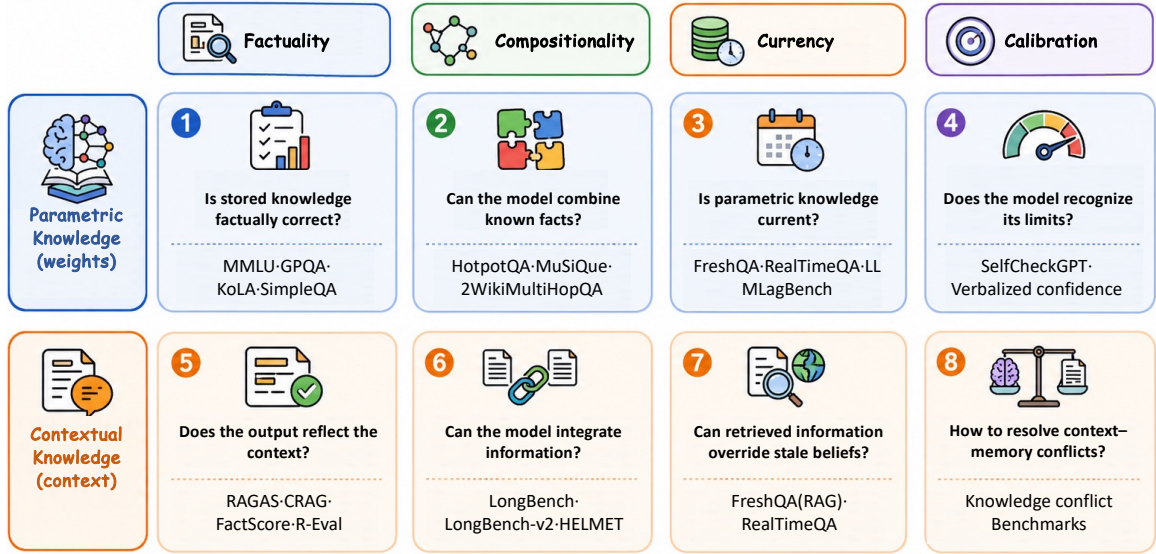


Figure 9: A two-by-four framework for evaluating knowledge in LLMs, organized by knowledge source (parametric vs. contextual) and knowledge property (factuality, compositionality, currency, and calibration). Each cell states the core evaluation question and lists representative benchmarks.

8.1.1 FACTUALITY

Finding

High accuracy on static factual benchmarks does not imply robust factual knowledge. Models can achieve strong scores through surface-level pattern matching, and benchmark saturation combined with data contamination makes it increasingly difficult to distinguish genuine fact retrieval from statistical shortcut learning.

The most natural way to assess parametric knowledge is to ask factual questions and measure accuracy. MMLU (Hendrycks et al., 2020) establishes this paradigm at scale, covering 57 academic subjects. Subsequent work has raised the difficulty ceiling with MMLU-Pro (Wang et al., 2024g) and GPQA (Rein et al., 2023), which require graduate-level reasoning and resist shallow heuristics. KoLA (Yu et al., 2023) takes a complementary approach, stratifying tasks along a cognitive hierarchy from simple fact recall to application and creation, and reveals that models saturate lower tiers while struggling at higher ones. SimpleQA (Wei et al., 2024a) focuses instead on calibration-aware accuracy. Questions are designed to have unambiguous short answers, making it easier to distinguish genuine knowledge from hedged or hallucinated responses.

Two challenges have grown increasingly acute as model capabilities improve. The first is benchmark saturation. Frontier models now approach or exceed human performance on MMLU, which reduces its diagnostic value. The second is data contamination. Static benchmarks risk being absorbed into pre-training corpora, inflating reported scores. DyVal (Zhu et al., 2023) addresses contamination by generating evaluation instances dynamically, while

LatestEval (Li et al., 2024g) constructs questions from texts published after a model’s training cutoff, providing a contamination-resistant probe of knowledge boundaries.

8.1.2 COMPOSITIONALITY

Finding

Compositional evaluation reveals a dissociation between reasoning output and knowledge use. Models can produce plausible reasoning chains that do not reflect the computation actually producing the answer, so evaluating verbalized reasoning does not guarantee evaluation of compositional knowledge use.

Factual recall is a necessary but not sufficient condition for knowledge competence. A model that knows individual facts may still fail to combine them across reasoning steps. HotpotQA (Yang et al., 2018) introduces multi-hop question answering as a testbed for this capability. 2WikiMultiHopQA (Ho et al., 2020) draws on structured knowledge from two Wikipedia articles per question, providing a richer source of cross-document compositional inference. MuSiQue (Trivedi et al., 2022) subsequently shows that models can exploit spurious single-hop shortcuts in HotpotQA, and constructs questions that are unsolvable without genuine multi-step reasoning.

A deeper question has emerged around chain-of-thought prompting. Even when models produce correct reasoning chains, those chains may not faithfully reflect the underlying computation. Sprague et al. (2024) find that models often arrive at correct answers through paths inconsistent with their stated reasoning. Yee et al. (2024) demonstrate a systematic dissociation between the expressed reasoning trace and the model’s actual knowledge retrieval process. FaithCoT-Bench (Shen et al., 2025) operationalizes this concern as a benchmark, measuring whether the intermediate steps in a chain-of-thought are each individually supported by the model’s parametric knowledge.

8.1.3 CURRENCY

Finding

Parametric knowledge becomes stale after the training cutoff, and the lag can be measured, but models provide no explicit signal of where their knowledge ends.

Parametric knowledge is frozen at training time, yet the world continues to change. FreshQA (Vu et al., 2024) provides a systematic evaluation of this gap, pairing time-sensitive questions with answers that are updated as ground truth changes. In its no-retrieval setting, it measures how quickly model accuracy degrades for questions whose answers have changed since the training cutoff. RealTimeQA (Kasai et al., 2023) adopts a streaming evaluation protocol, releasing new questions weekly alongside current news, and thus provides a continuously refreshed benchmark that resists contamination. LLMlagBench (Pkezik et al., 2025) quantifies the lag between real-world events and model knowledge by measuring the temporal gap at which accuracy drops below a threshold.

8.1.4 CALIBRATION

Finding

Calibration in LRMs is not simply a property of the final output distribution but also involves the allocation of reasoning resources. The over-thinking phenomenon, in which models expend heavy computation on easy queries yet remain underconfident, shows that reasoning depth must be matched to query difficulty.

A well-calibrated model should express high confidence when it is likely to be correct and low confidence otherwise. Kadavath et al. (2022) show that large models can verbalize meaningful confidence estimates when prompted explicitly, and that these estimates correlate with empirical accuracy, though the correlation degrades on harder tasks. Self-CheckGPT (Manakul et al., 2023) takes a consistency-based approach. By sampling multiple responses to the same question and measuring their agreement, it estimates factual reliability without requiring external ground truth. Yang et al. (2024a) provide a broader analysis of verbalized confidence, finding systematic overconfidence in factual domains and underconfidence in reasoning-heavy ones.

The emergence of large reasoning models (LRMs) has introduced a new calibration phenomenon. Models such as DeepSeek-R1 (Guo et al., 2025) generate extended reasoning chains before producing answers. This extra computation can improve accuracy but also introduces over-thinking, where the model expends large amounts of computation on straightforward queries while remaining underconfident about the output. Sui et al. (2025) characterize this behavior empirically and propose adaptive mechanisms that match reasoning depth to query difficulty. AdaptThink (Zhang et al., 2025a) formalizes this as an optimization objective, training models to allocate deliberation selectively.

8.2 Evaluating Contextual Knowledge

When knowledge is supplied at inference time rather than retrieved from weights, the evaluation problem shifts. The question is no longer whether a model *has* the right knowledge, but whether it correctly *uses* the knowledge it has been given. This seemingly simpler setting introduces its own difficulties. Injected knowledge may be scattered across multiple passages, may conflict with parametric beliefs, and even when correct may fail to dislodge the beliefs the model has already formed.

8.2.1 FACTUALITY

Finding

Faithfulness failures concentrate in domains where parametric knowledge is strongest. This pattern suggests that knowledge interference, not retrieval quality alone, limits contextual factuality, because models may over-rely on parametric knowledge precisely where it is most reliable.

In the contextual setting, factuality is typically evaluated as *faithfulness*. Does the model’s output accurately reflect the provided context, without importing ungrounded claims from parametric memory? This reframing presumes that the provided context is itself trustworthy; the case where it is not is taken up under calibration below. RAGAS (Es

et al., 2024) operationalizes this with a suite of reference-free metrics that decompose RAG quality into retrieval relevance, answer faithfulness, and answer relevance, enabling automated evaluation without human annotation. R-Eval (Yu et al., 2024a) extends this to domain-specific RAG pipelines, revealing that faithfulness failures are more common in specialized domains where parametric knowledge is strong and may override retrieved evidence. CRAG (Yang et al., 2024d) introduces a benchmark covering five domains and three question types, and finds that even strong systems hallucinate on many queries where retrieved documents provide sufficient information. FActScore (Min et al., 2023) approaches faithfulness at the granularity of individual atomic claims, decomposing a generated response and verifying each claim against a reference corpus. This methodology has been widely adopted for long-form generation evaluation.

8.2.2 COMPOSITIONALITY

Finding

The lost-in-the-middle effect reveals that knowledge integration in long contexts is limited by evidence position, not by context window size alone. Models handle evidence near the beginning and end of the context well but systematically underuse evidence in the middle.

Multi-document reasoning requires a model to locate, integrate, and synthesize information scattered across multiple retrieved passages. This capability goes well beyond single-passage reading comprehension. LongBench (Bai et al., 2024) provides a bilingual benchmark that covers multi-document QA, summarization, and code completion at context lengths up to tens of thousands of tokens, and reveals that models perform consistently worse on multi-document tasks than on single-document ones. LongBench-v2 (Bai et al., 2025) extends this with harder instances requiring deeper cross-document reasoning, showing that the gap persists even for models with large context windows.

A central challenge identified across this line of work is the *lost-in-the-middle* effect (Liu et al., 2024a). Models consistently underuse information located in the middle of long contexts, favoring evidence near the beginning and end. HELMET (Yen et al., 2024) provides a holistic evaluation across diverse long-context tasks and confirms that recall of middle-positioned evidence is a strong predictor of overall compositional performance. DetectiveQA (Xu et al., 2024c) constructs questions that require evidence chains spanning multiple documents, providing a targeted probe of genuine multi-hop integration rather than positional recall.

8.2.3 CURRENCY

Finding

Providing current information through retrieval is necessary but not sufficient for knowledge updating. Models can exhibit an anchoring effect in which parametric priors override retrieved evidence even when that evidence is both correct and relevant.

One of the original motivations for retrieval-augmented generation was to allow models to answer questions about events that occurred after their training cutoff, bypassing the

currency limitation of parametric knowledge. FreshQA (Vu et al., 2024) provides a direct test of this premise by pairing time-sensitive questions with ground truth that is updated as the world changes. In its RAG configuration, it measures whether retrieved current information can substitute for absent parametric knowledge. RealTimeQA (Kasai et al., 2023) evaluates the same capability in a streaming setting, releasing questions alongside weekly news to ensure freshness.

These evaluations reveal a subtler problem. Even when a model is provided with retrieved documents containing the correct current answer, it may still report an outdated response, anchoring on parametric beliefs that conflict with the retrieved evidence. Currency evaluation in the contextual setting therefore overlaps with calibration. The model must not only access fresh information but also actively override stale parametric priors.

8.2.4 CALIBRATION

Finding

The resolution of context-memory conflict still lacks an agreed normative framework. Model behavior shows no consistent strategy for when to trust context over memory and is highly sensitive to prompt features such as source framing and perceived authority.

The most distinctive challenge in contextual knowledge evaluation is the resolution of *context-memory conflict*. When retrieved evidence contradicts parametric knowledge, which source should the model trust? Xu et al. (2024a) survey this literature systematically, identifying three conflict types (context-memory, inter-context, and intra-memory) and cataloging the conditions under which each type degrades model reliability.

Subsequent work has probed specific aspects of this resolution problem. Sun et al. (2025c) show that the optimal resolution strategy is task-dependent. For knowledge-intensive QA, trusting context is usually preferable, but for reasoning tasks, parametric knowledge is often more reliable. Zhao et al. (2026c) explore how framing and source attribution in the prompt shift model preferences between context and memory. Lin et al. (2025) examine the complementary failure mode, where models are too easily swayed by injected misinformation, and propose training interventions that improve resistance to contextual interference without sacrificing appropriate context-following behavior.

8.3 Discussion

The preceding sections reveal an evaluation literature that is maturing but still fragmented. We highlight three challenges that cut across both parametric and contextual settings.

The first challenge is to align behavioral evaluation with mechanistic understanding. Almost all existing benchmarks evaluate knowledge through model outputs, including correct or incorrect answers, calibrated or miscalibrated confidence, and faithful or hallucinated text. This behavioral perspective cannot distinguish between a model that genuinely encodes a fact and one that exploits surface-level correlations to produce the right answer. Mechanistic interpretability has begun to offer tools for probing internal representations directly (Chang et al., 2024b), but connecting these internal probes to standard behavioral benchmarks remains an open methodological problem. Until this gap is bridged, it is dif-

difficult to know whether high benchmark scores reflect genuine knowledge or sophisticated pattern matching.

The second challenge concerns the reliability of LLM-as-Judge evaluation. As human annotation becomes a bottleneck, the field has increasingly turned to LLMs themselves as evaluators (Zheng et al., 2023b). Wang et al. (2024d) show that LLM judges are systematically biased. They favor longer responses, prefer outputs that share their own stylistic tendencies, and exhibit position bias when comparing answer pairs. TruthfulQA (Lin et al., 2022) demonstrated early on that models can generate fluent, confident, and incorrect answers that fool both humans and automated evaluators. These biases are particularly problematic for knowledge evaluation, where the ground truth may be subtle and the incorrect answer may be superficially plausible. Developing evaluation protocols that are robust to judge bias, whether through calibration, ensemble approaches, or hybrid human-LLM pipelines, remains an active area.

The third challenge is the limited lifespan of static benchmarks, which face two compounding pressures. Model capabilities improve faster than benchmarks are updated, and training data contamination erodes the validity of fixed test sets over time. DyVal (Zhu et al., 2023) and LatestEval (Li et al., 2024g) address contamination through dynamic generation and temporal filtering, respectively, but neither fully solves the capability saturation problem. A natural direction is to develop continuously updated evaluation frameworks that release new instances on a rolling basis and track performance trajectories rather than point-in-time scores. This shift also demands that evaluation infrastructure scale beyond academic lab settings, a practical challenge that the community is only beginning to address.

9 Conclusion

This survey has reviewed knowledge mechanisms in large language models through a single lifecycle, following knowledge as it is acquired, represented, utilized, and evolved within a model and then transferred beyond it, and closing with how knowledge is evaluated at every stage. Most existing work studies these stages in isolation. By treating them instead as successive states of one underlying object, knowledge itself, we bring into view the dependencies that connect them, where what a model acquires sets the limits on how knowledge can be represented, how it is represented governs whether it can be recalled and used, and a single edit propagates along the same pathways that serve recall, reaching or failing to reach the knowledge it was meant to change.

One observation recurs across the entire lifecycle. A model’s behavior is bounded not only by what it has learned but by whether that knowledge can be reached and expressed. Failures that look like missing knowledge are often failures of access, arbitration, or propagation instead, and this gap between what a model encodes and what it expresses surfaces again and again, in how facts are recalled, in how conflicting sources are arbitrated, in how far an edit actually spreads, and in how knowledge learned in one language can fail to surface in another. Strong benchmark scores show the same mismatch in reverse, masking shallow pattern matching that succeeds without the relevant knowledge ever being encoded. No single stage reveals the full extent of this gap on its own, and bringing it into one frame is much of what the lifecycle view contributes.

We hope this lifecycle perspective offers more than a way to organize a fast-growing literature. It provides a vantage point for asking how the knowledge-related failures of current models arise and how an intervention at one stage can propagate to the others, and it marks a step from describing what models do toward explaining why they do it. Closing the distance between behavioral observation and mechanistic understanding is, in our view, the central challenge ahead, and progress on it is what will let us build models whose knowledge we can understand and steer.

10 Author

Core Contributors: Hao Xiang¹, Xiusheng Huang, Shangqing Tu, Jiasheng Zheng, Jiaquan Xie, Chenhui Hu, JiaXiang Liu, Qiao Liang

Instructors: Boxi Cao, Pengfei Cao, Lei Hou, Hongyu Lin, Jian Luan, Yaojie Lu, Jun Zhao, Xianpei Han, Juanzi Li, Kang Liu, Bin Wang, Le Sun

References

- Marah Abdin, Jyoti Aneja, Harkirat Behl, Sébastien Bubeck, Ronen Eldan, Suriya Gunasekar, Michael Harrison, Russell J. Hewett, Mojan Javaheripi, Piero Kauffmann, James R. Lee, Yin Tat Lee, Yuanzhi Li, Weishung Liu, Caio C. T. Mendes, Anh Nguyen, Eric Price, Gustavo de Rosa, Olli Saarikivi, Adil Salim, Shital Shah, Xin Wang, Rachel Ward, Yue Wu, Dingli Yu, Cyril Zhang, and Yi Zhang. Phi-4 technical report, 2024. URL <https://arxiv.org/abs/2412.08905>.
- Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, Piotr Stanczyk, Sabela Ramos, Matthieu Geist, and Olivier Bachem. On-policy distillation of language models: Learning from self-generated mistakes, 2024. URL <https://arxiv.org/abs/2306.13649>.
- Samuel K. Ainsworth, Jonathan Hayase, and Siddhartha Srinivasa. Git re-basin: Merging models modulo permutation symmetries, 2023. URL <https://arxiv.org/abs/2209.04836>.
- Zeyuan Allen-Zhu and Yuanzhi Li. Physics of language models: Part 3.1, knowledge storage and extraction. In Ruslan Salakhutdinov, Zico Kolter, Katherine Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp, editors, *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 1067–1077. PMLR, 21–27 Jul 2024. URL <https://proceedings.mlr.press/v235/allen-zhu24a.html>.
- Zeyuan Allen-Zhu and Yuanzhi Li. Physics of language models: Part 3.3, knowledge capacity scaling laws. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/26d3c9a66836ded8f34a944f2bfe868e-Abstract-Conference.html.
- Emmanuel Ameisen, Jack Lindsey, Adam Pearce, Wes Gurnee, Nicholas L Turner, Brian Chen, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, et al. Circuit tracing:

1. Leading Contributor

- Revealing computational graphs in language models. *Transformer Circuits Thread*, 6: 16318–16352, 2025.
- Catherine Arnett, Tyler A. Chang, James A. Michaelov, and Benjamin K. Bergen. On the acquisition of shared grammatical representations in bilingual language models, 2025. URL <https://arxiv.org/abs/2503.03962>.
- Mikel Artetxe, Shruti Bhosale, Naman Goyal, Todor Mihaylov, Myle Ott, Sam Shleifer, Xi Victoria Lin, Jingfei Du, Srinivasan Iyer, Ramakanth Pasunuru, Giridharan Anantharaman, Xian Li, Shuohui Chen, Halil Akin, Mandeep Baines, Louis Martin, Xing Zhou, Punit Singh Koura, Brian O’Horo, Jeffrey Wang, Luke Zettlemoyer, Mona Diab, Zornitsa Kozareva, and Veselin Stoyanov. Efficient large scale language modeling with mixtures of experts. In Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang, editors, *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 11699–11732, Abu Dhabi, United Arab Emirates, December 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-main.804. URL <https://aclanthology.org/2022.emnlp-main.804/>.
- Jinheon Baek, Sun Jae Lee, Prakhar Gupta, Geunseob, Oh, Siddharth Dalmia, and Prateek Kolhar. Revisiting in-context learning with long context language models, 2024. URL <https://arxiv.org/abs/2412.16926>.
- Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. LongBench: A bilingual, multitask benchmark for long context understanding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3119–3137, Bangkok, Thailand, August 2024. doi: 10.18653/v1/2024.acl-long.172. URL <https://aclanthology.org/2024.acl-long.172>.
- Yushi Bai, Shangqing Tu, Jiajie Zhang, Hao Peng, Xiaozhi Wang, Xin Lv, Shulin Cao, Jiazheng Xu, Lei Hou, Yuxiao Dong, et al. Longbench v2: Towards deeper understanding and reasoning on realistic long-context multitasks. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3639–3664, 2025.
- Rishab Balasubramanian, Pin-Jie Lin, Rituraj Sharma, Anjie Fang, Fardin Abdi, Viktor Rozgic, Zheng Du, Mohit Bansal, and Tu Vu. The master key hypothesis: Unlocking cross-model capability transfer via linear subspace alignment, 2026. URL <https://arxiv.org/abs/2604.06377>.
- Lucas Bandarkar, Benjamin Muller, Pritish Yuvraj, Rui Hou, Nayan Singhal, Hongjiang Lv, and Bing Liu. Layer swapping for zero-shot cross-lingual transfer in large language models, 2025. URL <https://arxiv.org/abs/2410.01335>.
- Lucas Bandarkar, Alan Ansell, and Trevor Cohn. Knowledge localization in mixture-of-experts llms using cross-lingual inconsistency, 2026. URL <https://arxiv.org/abs/2603.17102>.

- Logan Barnhart, Reza Akbarian Bafghi, Stephen Becker, and Maziar Raissi. Aligning to what? limits to rlhf based alignment, 2025. URL <https://arxiv.org/abs/2503.09025>.
- Joshua Barron and Devin White. Too big to think: Capacity, memorization, and generalization in pre-trained transformers. In *Tiny Titans: The Next Wave of On-Device Learning for Foundation Models Workshop at ICML*, 2025. URL <https://icml.cc/virtual/2025/47826>.
- Abdul-Salem Beibitkhan. Left behind: Cross-lingual transfer as a bridge for low-resource languages in large language models, 2026. URL <https://arxiv.org/abs/2603.21036>.
- Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. Leace: Perfect linear concept erasure in closed form, 2025. URL <https://arxiv.org/abs/2306.03819>.
- Gagan Bhatia, Somayajulu G Sripada, Kevin Allan, and Jacobo Azcona. Distributional semantics tracing: A framework for explaining hallucinations in large language models, 2025. URL <https://arxiv.org/abs/2510.06107>.
- Sid Black, Lee Sharkey, Leo Grinsztajn, Eric Winsor, Dan Braun, Jacob Merizian, Kip Parker, Carlos Ramón Guevara, Beren Millidge, Gabriel Alfour, et al. Interpreting neural networks through the polytope lens. *arXiv preprint arXiv:2211.12312*, 2022.
- Nicolas Boizard, Théo Deschamps-Berger, Hippolyte Gisserot-Boukhlef, Céline Hudelot, and Pierre Colombo. Bidirlm: From text to omnimodal bidirectional encoders by adapting and composing causal llms, 2026. URL <https://arxiv.org/abs/2604.02045>.
- Sebastian Borgeaud, Arthur Mensch, Jordan Hoffmann, Trevor Cai, Eliza Rutherford, Katie Millican, George van den Driessche, Jean-Baptiste Lespiau, Bogdan Damoc, Aidan Clark, Diego de Las Casas, Aurelia Guy, Jacob Menick, Roman Ring, Tom Hennigan, Saffron Huang, Loren Maggiore, Chris Jones, Albin Cassirer, Andy Brock, Michela Paganini, Geoffrey Irving, Oriol Vinyals, Simon Osindero, Karen Simonyan, Jack W. Rae, Erich Elsen, and Laurent Sifre. Improving language models by retrieving from trillions of tokens, 2022. URL <https://arxiv.org/abs/2112.04426>.
- Lucas Bourtole, Varun Chandrasekaran, Christopher A. Choquette-Choo, Hengrui Jia, Adelin Travers, Baiwu Zhang, David Lie, and Nicolas Papernot. Machine unlearning. In *IEEE Symposium on Security and Privacy (S&P)*, 2021.
- Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermy, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Christopher Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. <https://transformer-circuits.pub/2023/monosemantic-features/index.html>.
- Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prfulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell,

- Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. Language models are few-shot learners. In *Advances in Neural Information Processing Systems*, volume 33, pages 1877–1901. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/hash/1457c0d6bfc4967418bfb8ac142f64a-Abstract.html.
- Weilin Cai, Juyong Jiang, Fan Wang, Jing Tang, Sunghun Kim, and Jiayi Huang. A survey on mixture of experts in large language models. *IEEE Transactions on Knowledge and Data Engineering*, 37:3896–3915, 2025. doi: 10.1109/TKDE.2025.3554028.
- Nitay Calderon, Eyal Ben-David, Zorik Gekhman, Eran Ofek, and Gal Yona. Empty shelves or lost keys? recall is the bottleneck for parametric factuality, 2026. URL <https://arxiv.org/abs/2602.14080>.
- Boxi Cao, Hongyu Lin, Xianpei Han, and Le Sun. The life cycle of knowledge in big language models: A survey. *arXiv preprint arXiv:2303.07616*, 2023.
- Pengfei Cao, Zeao Ji, Daojian Zeng, Jun Zhao, and Kang Liu. Evoedit: Lifelong free-text knowledge editing through latent perturbation augmentation and knowledge-driven parameter fusion, 2025. URL <https://arxiv.org/abs/2512.04545>.
- Pengfei Cao, Yuheng Chen, Zhuoran Jin, Yubo Chen, Kang Liu, and Jun Zhao. One mind, many tongues: A deep dive into language-agnostic knowledge neurons in large language models. *Artificial Intelligence*, 353:104490, 2026. ISSN 0004-3702. doi: <https://doi.org/10.1016/j.artint.2026.104490>. URL <https://www.sciencedirect.com/science/article/pii/S0004370226000160>.
- Victor Carbune, Hassan Mansoor, Fangyu Liu, Rahul Aralikkatte, Gilles Baechler, Jindong Chen, and Abhanshu Sharma. Chart-based reasoning: Transferring capabilities from llms to vlms, 2024. URL <https://arxiv.org/abs/2403.12596>.
- Arie Cattan, Alon Jacovi, Ori Ram, Jonathan Herzig, Roei Aharoni, Sasha Goldshtein, Eran Ofek, Idan Szpektor, and Avi Caciularu. Dragged into conflicts: Detecting and addressing conflicting sources in search-augmented llms, 2025. URL <https://arxiv.org/abs/2506.08500>.
- Hoyeon Chang, Jinho Park, Seonghyeon Ye, Sohee Yang, Youngkyung Seo, Du-Seong Chang, and Minjoon Seo. How do large language models acquire factual knowledge during pretraining? In *Advances in Neural Information Processing Systems*, volume 37. Curran Associates, Inc., 2024a. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/6fdf57c71bc1f1ee29014b8dc52e723f-Abstract-Conference.html.
- Yupeng Chang, Xu Wang, Jindong Wang, Yuan Wu, Linyi Yang, Kaijie Zhu, Hao Chen, Xiaoyuan Yi, Cunxiang Wang, Yidong Wang, et al. A survey on evaluation of large language models. *ACM Transactions on Intelligent Systems and Technology*, 15(3):1–45, 2024b.

- Harsh Chaudhari, Jamie Hayes, Matthew Jagielski, Ilia Shumailov, Milad Nasr, and Alina Oprea. Cascading adversarial bias from injection to distillation in language models, 2025. URL <https://arxiv.org/abs/2505.24842>.
- Marmik Chaudhari, Nishkal Hundia, and Idhant Gulati. Sparse crosscoders for diffing moes and dense models. *arXiv preprint arXiv:2603.05805*, 2026.
- Shouyuan Chen, Sherman Wong, Liangjian Chen, and Yuandong Tian. Extending context window of large language models via positional interpolation, 2023. URL <https://arxiv.org/abs/2306.15595>.
- Sijin Chen, Xin Chen, Chi Zhang, Mingsheng Li, Gang Yu, Hao Fei, Hongyuan Zhu, Jiayuan Fan, and Tao Chen. Ll3da: Visual interactive instruction tuning for omni-3d understanding reasoning and planning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 26428–26438, 2024a.
- Yuheng Chen, Pengfei Cao, Yubo Chen, Kang Liu, and Jun Zhao. Journey to the center of the knowledge neurons: Discoveries of language-independent knowledge neurons and degenerate knowledge neurons. *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(16):17817–17825, Mar. 2024b. doi: 10.1609/aaai.v38i16.29735. URL <https://ojs.aaai.org/index.php/AAAI/article/view/29735>.
- Yuheng Chen, Pengfei Cao, Yubo Chen, Kang Liu, and Jun Zhao. Knowledge localization: Mission not accomplished? enter query localization! In *International Conference on Learning Representations*, volume 2025, pages 29589–29613, 2025a.
- Yuheng Chen, Pengfei Cao, Yubo Chen, Yining Wang, Shengping Liu, Kang Liu, and Jun Zhao. Cracking factual knowledge: A comprehensive analysis of degenerate knowledge neurons in large language models. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10240–10261, Vienna, Austria, July 2025b. Association for Computational Linguistics. ISBN 979-8-89176-251-0. doi: 10.18653/v1/2025.acl-long.505. URL <https://aclanthology.org/2025.acl-long.505/>.
- Yuheng Chen, Pengfei Cao, Kang Liu, and Jun Zhao. The knowledge microscope: Features as better analytical lenses than neurons. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 10493–10515, 2025c.
- Daixuan Cheng, Shaohan Huang, and Furu Wei. Adapting large language models via reading comprehension. In *International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=y886UXPEZ0>.
- Minyeong Choe, Haehyun Cho, Changho Seo, and Hyunil Kim. Do all autoregressive transformers remember facts the same way? a cross-architecture analysis of recall mechanisms, 2025. URL <https://arxiv.org/abs/2509.08778>.

- Lynn Chua, Badih Ghazi, Yangsibo Huang, Pritish Kamath, Ravi Kumar, Pasin Manurangsi, Amer Sinha, Chulin Xie, and Chiyuan Zhang. Crosslingual capabilities and knowledge barriers in multilingual large language models, 2025. URL <https://arxiv.org/abs/2406.16135>.
- Yung-Sung Chuang, Yujia Xie, Hongyin Luo, Yoon Kim, James Glass, and Pengcheng He. Dola: Decoding by contrasting layers improves factuality in large language models, 2024. URL <https://arxiv.org/abs/2309.03883>.
- Yung-Sung Chuang, Benjamin Cohen-Wang, Shannon Zejiang Shen, Zhaofeng Wu, Hu Xu, Xi Victoria Lin, James Glass, Shang-Wen Li, and Wen tau Yih. Selfcite: Self-supervised alignment for context attribution in large language models, 2025. URL <https://arxiv.org/abs/2502.09604>.
- Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Yunxuan Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, Albert Webson, Shixiang Shane Gu, Zhuyun Dai, Mirac Suzgun, Xinyun Chen, Aakanksha Chowdhery, Alex Castro-Ros, Marie Pellat, Kevin Robinson, Dasha Valter, Sharan Narang, Gaurav Mishra, Adams Yu, Vincent Zhao, Yanping Huang, Andrew Dai, Hongkun Yu, Slav Petrov, Ed H. Chi, Jeff Dean, Jacob Devlin, Adam Roberts, Denny Zhou, Quoc V. Le, and Jason Wei. Scaling instruction-finetuned language models. *Journal of Machine Learning Research*, 25(70):1–53, 2024. URL <https://www.jmlr.org/papers/v25/23-0870.html>.
- Roi Cohen, Eden Biran, Ori Yoran, Amir Globerson, and Mor Geva. Evaluating the ripple effects of knowledge editing in language models. *Transactions of the Association for Computational Linguistics (TACL)*, 2024a.
- Roi Cohen, Konstantin Dobler, Eden Biran, and Gerard de Melo. I don’t know: Explicit modeling of uncertainty with an [idk] token, 2024b. URL <https://arxiv.org/abs/2412.06676>.
- Arthur Conmy, Augustine N. Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. In *Thirty-seventh Conference on Neural Information Processing Systems*, 2023. URL <https://openreview.net/forum?id=89ia77nZ8u>.
- Zoltan Csaki, Bo Li, Jonathan Li, Qiantong Xu, Pian Pawakapan, Leon Zhang, Yun Du, Hengyu Zhao, Changran Hu, and Urmish Thakker. Sambalingo: Teaching large language models new languages, 2024. URL <https://arxiv.org/abs/2404.05829>.
- Hoagy Cunningham, Aidan Ewart, Logan Riggs, Robert Huben, and Lee Sharkey. Sparse autoencoders find highly interpretable features in language models. *arXiv preprint arXiv:2309.08600*, 2023.
- Damai Dai, Li Dong, Yaru Hao, Zhifang Sui, Baobao Chang, and Furu Wei. Knowledge neurons in pretrained transformers. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022.

- Damai Dai, Chengqi Deng, Chenggang Zhao, R. X. Xu, Huazuo Gao, Deli Chen, Jiashi Li, Wangding Zeng, Xingkai Yu, Y. Wu, Zhenda Xie, Y. K. Li, Panpan Huang, Fuli Luo, Chong Ruan, Zhifang Sui, and Wenfeng Liang. DeepSeekMoE: Towards ultimate expert specialization in mixture-of-experts language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1280–1297, Bangkok, Thailand, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.70. URL <https://aclanthology.org/2024.acl-long.70/>.
- Yijia Dai, Zhaolin Gao, Yahya Satter, Sarah Dean, and Jennifer J. Sun. Pre-trained large language models learn hidden markov models in-context, 2025. URL <https://arxiv.org/abs/2506.07298>.
- Nicola De Cao, Wilker Aziz, and Ivan Titov. Editing factual knowledge in language models. In *Proceedings of the 2021 conference on empirical methods in natural language processing*, pages 6491–6506, 2021.
- DeepSeek-AI. Deepseek-v4: Towards highly efficient million-token context intelligence, 2026. URL https://huggingface.co/deepseek-ai/DeepSeek-V4-Pro/blob/main/DeepSeek_V4.pdf.
- Jiajun Deng, Tianyu He, Li Jiang, Tianyu Wang, Feras Dayoub, and Ian Reid. 3d-llava: Towards generalist 3d lmms with omni superpoint transformer, 2025. URL <https://arxiv.org/abs/2501.01163>.
- Shizhe Diao, Yu Yang, Yonggan Fu, Xin Dong, Dan Su, Markus Kliegl, Zijia Chen, Peter Belcak, Yoshi Suhara, Hongxu Yin, Mostofa Patwary, Yingyan Lin, Jan Kautz, and Pavlo Molchanov. Nemotron-climb: Clustering-based iterative data mixture bootstrapping for language model pre-training, 2025. URL <https://arxiv.org/abs/2504.13161>.
- Nan Du, Yanping Huang, Andrew M. Dai, Simon Tong, Dmitry Lepikhin, Yuanzhong Xu, Maxim Krikun, Yanqi Zhou, Adams Wei Yu, Orhan Firat, Barret Zoph, Liam Fedus, Maarten Bosma, Zongwei Zhou, Tao Wang, Yu Emma Wang, Kellie Webster, Marie Pellat, Kevin Robinson, Kathleen Meier-Hellstern, Toju Duke, Lucas Dixon, Kun Zhang, Quoc V Le, Yonghui Wu, Zhifeng Chen, and Claire Cui. Glam: Efficient scaling of language models with mixture-of-experts, 2022. URL <https://arxiv.org/abs/2112.06905>.
- Yifan Du, Zikang Liu, Yifan Li, Wayne Xin Zhao, Yuqi Huo, Bingning Wang, Weipeng Chen, Zheng Liu, Zhongyuan Wang, and Ji-Rong Wen. Virgo: A preliminary exploration on reproducing o1-like mllm, 2025. URL <https://arxiv.org/abs/2501.01904>.
- Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- MohammadReza Ebrahimi, Sunny Panchal, and Roland Memisevic. Your context is not an array: Unveiling random access limitations in transformers, 2024. URL <https://arxiv.org/abs/2408.05506>.

- Ronen Eldan and Mark Russinovich. Who’s harry potter? approximate unlearning in LLMs. *arXiv preprint arXiv:2310.02238*, 2023.
- Mohamed Elgaar and Hadi Amiri. Curriculum learning for llm pretraining: An analysis of learning dynamics, 2026. URL <https://arxiv.org/abs/2601.21698>.
- Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, et al. Toy models of superposition. *arXiv preprint arXiv:2209.10652*, 2022.
- Eric Elmoznino, Tom Marty, Tejas Kasetty, Leo Gagnon, Sarthak Mittal, Mahan Fathi, Dhanya Sridhar, and Guillaume Lajoie. In-context learning and occam’s razor, 2024. URL <https://arxiv.org/abs/2410.14086>.
- Josh Engels, Eric Michaud, Isaac Liao, Wes Gurnee, and Max Tegmark. Not all language model features are one-dimensionally linear. In *International Conference on Learning Representations*, volume 2025, pages 84591–84622, 2025.
- Joshua Engels, Logan Riggs, and Max Tegmark. Decomposing the dark matter of sparse autoencoders. *arXiv preprint arXiv:2410.14670*, 2024.
- Rahim Entezari, Hanie Sedghi, Olga Saukh, and Behnam Neyshabur. The role of permutation invariance in linear mode connectivity of neural networks, 2022. URL <https://arxiv.org/abs/2110.06296>.
- Shahul Es, Jithin James, Luis Espinosa Anke, and Steven Schockaert. Ragas: Automated evaluation of retrieval augmented generation. In *Proceedings of the 18th conference of the european chapter of the association for computational linguistics: system demonstrations*, pages 150–158, 2024.
- Esakkivel Esakkiraja, Sai Rajeswar, Denis Akhiyarov, and Rajagopal Venkatesaramani. Therefore i am. i think, 2026. URL <https://arxiv.org/abs/2604.01202>.
- Chenrui Fan, Ming Li, Lichao Sun, and Tianyi Zhou. Missing premise exacerbates overthinking: Are reasoning models losing critical thinking skill?, 2025. URL <https://arxiv.org/abs/2504.06514>.
- Simin Fan and Martin Jaggi. Irreducible curriculum for language model pretraining. In *Attributing Model Behavior at Scale*, 2023. URL <https://openreview.net/forum?id=zsNbN3NkzR>.
- Junfeng Fang, Houcheng Jiang, Kun Wang, Yunshan Ma, Jie Shi, Xiang Wang, Xiangnan He, and Tat-Seng Chua. Alphaedit: Null-space constrained model editing for language models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=HvSyvtvg3Jh>.
- Eoin Farrell, Yeu-Tong Lau, and Arthur Conmy. Applying sparse autoencoders to unlearn knowledge in language models. *arXiv preprint arXiv:2410.19278*, 2024.

- William Fedus, Barret Zoph, and Noam Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity, 2022. URL <https://arxiv.org/abs/2101.03961>.
- Javier Ferrando, Oscar Obeso, Senthooan Rajamanoharan, and Neel Nanda. Do i know this entity? knowledge awareness and hallucinations in language models, 2024. URL <https://arxiv.org/abs/2411.14257>.
- Jonathan Frankle, Gintare Karolina Dziugaite, Daniel M. Roy, and Michael Carbin. Linear mode connectivity and the lottery ticket hypothesis, 2020. URL <https://arxiv.org/abs/1912.05671>.
- Rao Fu, Jingyu Liu, Xilun Chen, Yixin Nie, and Wenhan Xiong. Scene-llm: Extending language model for 3d visual understanding and reasoning, 2024. URL <https://arxiv.org/abs/2403.11401>.
- Yuqian Fu, Haohuan Huang, Kaiwen Jiang, Jiakai Liu, Zhuo Jiang, Yuanheng Zhu, and Dongbin Zhao. Revisiting on-policy distillation: Empirical failure modes and simple fixes, 2026. URL <https://arxiv.org/abs/2603.25562>.
- Leo Gao, Tom Dupre la Tour, Henk Tillman, Gabriel Goh, Rajan Troll, Alec Radford, Ilya Sutskever, Jan Leike, and Jeffrey Wu. Scaling and evaluating sparse autoencoders. In *International Conference on Learning Representations*, volume 2025, pages 26721–26754, 2025a.
- Tianyu Gao, Alexander Wettig, Howard Yen, and Danqi Chen. How to train long-context language models (effectively). In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7376–7399, Vienna, Austria, July 2025b. Association for Computational Linguistics. doi: 10.18653/v1/2025.acl-long.366. URL <https://aclanthology.org/2025.acl-long.366/>.
- Zorik Gekhman, Eyal Ben David, Hadas Orgad, Eran Ofek, Yonatan Belinkov, Idan Spector, Jonathan Herzig, and Roi Reichart. Inside-out: Hidden factual knowledge in llms, 2025. URL <https://arxiv.org/abs/2503.15299>.
- Zorik Gekhman, Roei Aharoni, Eran Ofek, Mor Geva, Roi Reichart, and Jonathan Herzig. Thinking to recall: How reasoning unlocks parametric knowledge in llms, 2026. URL <https://arxiv.org/abs/2603.09906>.
- Aryo Pradipta Gema, Chen Jin, Ahmed Abdulaal, Tom Diethe, Philip Teare, Beatrice Alex, Pasquale Minervini, and Amrutha Saseendran. Decore: Decoding by contrasting retrieval heads to mitigate hallucinations, 2024. URL <https://arxiv.org/abs/2410.18860>.
- Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. Transformer feed-forward layers are key-value memories. In Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih, editors, *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 5484–5495, Online and Punta Cana, Dominican

- Republic, November 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.emnlp-main.446. URL <https://aclanthology.org/2021.emnlp-main.446/>.
- Mor Geva, Avi Caciularu, Kevin Wang, and Yoav Goldberg. Transformer feed-forward layers build predictions by promoting concepts in the vocabulary space. In Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang, editors, *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 30–45, Abu Dhabi, United Arab Emirates, December 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-main.3. URL <https://aclanthology.org/2022.emnlp-main.3/>.
- Mor Geva, Jasmijn Bastings, Katja Filippova, and Amir Globerson. Dissecting recall of factual associations in auto-regressive language models. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12216–12235, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.751. URL <https://aclanthology.org/2023.emnlp-main.751/>.
- Amirata Ghorbani and James Zou. Neuron shapley: Discovering the responsible neurons. In H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 5922–5932. Curran Associates, Inc., 2020. URL https://proceedings.neurips.cc/paper_files/paper/2020/file/41c542dfe6e4fc3deb251d64cf6ed2e4-Paper.pdf.
- Omer Goldman, Uri Shaham, Dan Malkin, Sivan Eiger, Avinatan Hassidim, Yossi Matias, Joshua Maynez, Adi Mayrav Gilady, Jason Riesa, Shruti Rijhwani, Laura Rimell, Idan Szpektor, Reut Tsarfaty, and Matan Eyal. Eclectic: a novel challenge set for evaluation of cross-lingual knowledge transfer, 2025. URL <https://arxiv.org/abs/2502.21228>.
- Boyu Gou, Zanming Huang, Yuting Ning, Yu Gu, Michael Lin, Weijian Qi, Andrei Kopanav, Botao Yu, Bernal Jimenez Gutierrez, Yiheng Shu, Chan Hee Song, Jiaman Wu, Shijie Chen, Hanane Nour Moussa, Tianshu Zhang, Jian Xie, Yifei Li, Tianci Xue, Zeyi Liao, Kai Zhang, Boyuan Zheng, Zhaowei Cai, Viktor Rozgic, Morteza Ziyadi, Huan Sun, and Yu Su. Mind2web 2: Evaluating agentic search with agent-as-a-judge. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2026. URL <https://openreview.net/forum?id=AUaW6DS9si>.
- Andrei Gromov, Kushal Tirumala, Hassan Shapourian, Paolo Glorioso, and Daniel A. Roberts. The unreasonable ineffectiveness of the deeper layers, 2024. URL <https://arxiv.org/abs/2403.17887>.
- Tiancheng Gu, Kaicheng Yang, Ziyong Feng, Xingjun Wang, Yanzhao Zhang, Dingkun Long, Yingda Chen, Weidong Cai, and Jiankang Deng. Breaking the modality barrier: Universal embedding learning with multimodal llms, 2025. URL <https://arxiv.org/abs/2504.17432>.
- Yuxian Gu, Li Dong, Furu Wei, and Minlie Huang. Minillm: Knowledge distillation of large language models. In *International Conference on Learning Representations*, volume 2024, pages 32694–32717, 2024.

- Xinyan Guan, Jiali Zeng, Fandong Meng, Chunlei Xin, Yaojie Lu, Hongyu Lin, Xianpei Han, Le Sun, and Jie Zhou. Deeprag: Thinking to retrieval step by step for large language models, 2025. URL <https://arxiv.org/abs/2502.01142>.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, et al. Deepseek-r1 incentivizes reasoning in llms through reinforcement learning. *Nature*, 645(8081):633–638, 2025.
- Akshat Gupta, Anurag Rao, and Gopala Anumanchipalli. Model editing at scale leads to gradual and catastrophic forgetting. In *Findings of the Association for Computational Linguistics: ACL*, 2024.
- Akshat Gupta, Phudish Prateepamornkul, Maochuan Lu, Ahmed Alaa, Thomas Hartvigsen, and Gopala Anumanchipalli. Lifelong knowledge editing requires better regularization, 2025. URL <https://arxiv.org/abs/2502.01636>.
- Wes Gurnee and Max Tegmark. Language models represent space and time. In *International Conference on Learning Representations*, volume 2024, pages 2483–2503, 2024.
- Kelvin Guu, Kenton Lee, Zora Tung, Panupong Pasupat, and Ming-Wei Chang. Realm: Retrieval-augmented language model pre-training, 2020. URL <https://arxiv.org/abs/2002.08909>.
- Seungwook Han, Jinyeop Song, Jeff Gore, and Pulkit Agrawal. Emergence of abstractions: Concept encoding and decoding mechanism for in-context learning in transformers, 2024. URL <https://arxiv.org/abs/2412.12276>.
- Sungjun Han, Juyoung Suk, Suyeong An, Hyungguk Kim, Kyuseok Kim, Wonsuk Yang, Seungtaek Choi, and Jamin Shin. Trillion 7b technical report, 2025. URL <https://arxiv.org/abs/2504.15431>.
- Thomas Hartvigsen, Swami Sankaranarayanan, Hamid Palangi, Yoon Kim, and Marzyeh Ghassemi. Aging with GRACE: Lifelong model editing with discrete key-value adaptors. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Peter Hase, Mohit Bansal, Been Kim, and Asma Ghandeharioun. Does localization inform editing? surprising differences in causality-based localization vs. knowledge editing in language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Wei He, Zhiheng Xi, Wanxu Zhao, Xiaoran Fan, Yiwen Ding, Zifei Shan, Tao Gui, Qi Zhang, and Xuanjing Huang. Distill visual chart reasoning ability from llms to mllms, 2025. URL <https://arxiv.org/abs/2410.18798>.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*, 2020.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network, 2015. URL <https://arxiv.org/abs/1503.02531>.

- Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. Constructing a multi-hop qa dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6609–6625, 2020.
- Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, Tom Hennigan, Eric Noland, Katie Millican, George van den Driessche, Bogdan Damoc, Aurelia Guy, Simon Osindero, Karen Simonyan, Erich Elsen, Jack W. Rae, Oriol Vinyals, and Laurent Sifre. Training compute-optimal large language models. In *Advances in Neural Information Processing Systems*, volume 35. Curran Associates, Inc., 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/c1e2faff6f588870935f114ebe04a3e5-Abstract-Conference.html.
- Yining Hong, Haoyu Zhen, Peihao Chen, Shuhong Zheng, Yilun Du, Zhenfang Chen, and Chuang Gan. 3d-llm: Injecting the 3d world into large language models. *Advances in Neural Information Processing Systems*, 36:20482–20494, 2023.
- Cheng-Yu Hsieh, Chun-Liang Li, Chih-Kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alex Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 8003–8017, 2023.
- Chenhui Hu, Pengfei Cao, Yubo Chen, Kang Liu, and Jun Zhao. WilKE: Wise-layer knowledge editor for lifelong knowledge editing. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Findings of the Association for Computational Linguistics: ACL 2024*, pages 3476–3503, Bangkok, Thailand, August 2024a. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.207. URL <https://aclanthology.org/2024.findings-acl.207/>.
- Chenhui Hu, Pengfei Cao, Yubo Chen, Kang Liu, and Jun Zhao. Knowledge in superposition: Unveiling the failures of lifelong knowledge editing for large language models. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pages 24086–24094, 2025a.
- Chenhui Hu, Pengfei Cao, Yubo Chen, Kang Liu, and Jun Zhao. Towards atoms of large language models. *arXiv preprint arXiv:2509.20784*, 2025b.
- Shengding Hu, Yuge Tu, Xu Han, Chaoqun He, Ganqu Cui, Xiang Long, Zhi Zheng, Yewei Fang, Yuxiang Huang, Weilin Zhao, Xinrong Zhang, Zheng Leng Thai, Kaihuo Zhang, Chongyi Wang, Yuan Yao, Chenyang Zhao, Jie Zhou, Jie Cai, Zhongwu Zhai, Ning Ding, Chao Jia, Guoyang Zeng, Dahai Li, Zhiyuan Liu, and Maosong Sun. MiniCPM: Unveiling the potential of small language models with scalable training strategies. In *First Conference on Language Modeling*, 2024b. URL <https://openreview.net/forum?id=3X2L2TFr0f>.
- Xiwei Hu, Rui Wang, Yixiao Fang, Bin Fu, Pei Cheng, and Gang Yu. Ella: Equip diffusion models with llm for enhanced semantic alignment, 2024c. URL <https://arxiv.org/abs/2403.05135>.

- Yunhai Hu, Yilun Zhao, Chen Zhao, and Arman Cohan. Mcts-rag: Enhancing retrieval-augmented generation with monte carlo tree search, 2025c. URL <https://arxiv.org/abs/2503.20757>.
- Chao Huang, Zeliang Zhang, Jiang Liu, Ximeng Sun, Jialian Wu, Xiaodong Yu, Ze Wang, Chenliang Xu, Emad Barsoum, and Zicheng Liu. Drift: Transferring reasoning priors for efficient mllm fine-tuning, 2026a. URL <https://arxiv.org/abs/2510.15050>.
- Haifeng Huang, Yilun Chen, Zehan Wang, Rongjie Huang, Runsen Xu, Tai Wang, Luping Liu, Xize Cheng, Yang Zhao, Jiangmiao Pang, and Zhou Zhao. Chat-scene: Bridging 3d scene and large language models with object identifiers, 2024a. URL <https://arxiv.org/abs/2312.08168>.
- Junjie Huang, Jiarui Qin, Di Yin, Weiwen Liu, Yong Yu, Xing Sun, and Weinan Zhang. ReMiT: RL-guided mid-training for iterative LLM evolution, 2026b. URL <https://arxiv.org/abs/2602.03075>.
- Xiusheng Huang, Jiayang Liu, Yequan Wang, and Kang Liu. Reasons and solutions for the decline in model performance after editing. In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024b. URL <https://openreview.net/forum?id=xjXYgdFM5M>.
- Xiusheng Huang, Yequan Wang, Jun Zhao, and Kang Liu. Commonsense knowledge editing based on free-text in LLMs. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen, editors, *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 14870–14880, Miami, Florida, USA, November 2024c. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.826. URL <https://aclanthology.org/2024.emnlp-main.826/>.
- Zhen Huang, Haoyang Zou, Xuefeng Li, Yixiu Liu, Yuxiang Zheng, Ethan Chern, Shijie Xia, Yiwei Qin, Weizhe Yuan, and Pengfei Liu. O1 replication journey – part 2: Surpassing o1-preview through simple distillation, big progress or bitter lesson?, 2024d. URL <https://arxiv.org/abs/2411.16489>.
- Ziyang Huang, Xiaowei Yuan, Yiming Ju, Jun Zhao, and Kang Liu. Reinforced internal-external knowledge synergistic reasoning for efficient adaptive search agent, 2025. URL <https://arxiv.org/abs/2505.07596>.
- Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. The platonic representation hypothesis, 2024. URL <https://arxiv.org/abs/2405.07987>.
- Nan Huo, Jinyang Li, Bowen Qin, Ge Qu, Xiaolong Li, Xiaodong Li, Chenhao Ma, and Reynold Cheng. Micro-act: Mitigate knowledge conflict in question answering via actionable self-reasoning, 2025. URL <https://arxiv.org/abs/2506.05278>.
- Adam Ibrahim, Benjamin Thérien, Kshitij Gupta, Mats L. Richter, Quentin Anthony, Timothée Lesort, Eugene Belilovsky, and Irina Rish. Simple and scalable strategies to continually pre-train large language models, 2024. URL <https://arxiv.org/abs/2403.08763>.

- Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. Editing models with task arithmetic. In *International Conference on Learning Representations (ICLR)*, 2023.
- Sekh Mainul Islam, Pepa Atanasova, and Isabelle Augenstein. Multi-step knowledge interaction analysis via rank-2 subspace disentanglement, 2025. URL <https://arxiv.org/abs/2511.01706>.
- Gautier Izacard and Edouard Grave. Leveraging passage retrieval with generative models for open domain question answering, 2021. URL <https://arxiv.org/abs/2007.01282>.
- Joel Jang, Dongkeun Yoon, Sohee Yang, Sungmin Cha, Moontae Lee, Lajanugen Logeswaran, and Minjoon Seo. Knowledge unlearning for mitigating privacy risks in language models. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.
- Byungwoo Jeon, Dongyoung Kim, Huiwon Jang, Insoo Kim, and Jinwoo Shin. Spatialboost: Enhancing visual representation through language-guided reasoning, 2026. URL <https://arxiv.org/abs/2603.22057>.
- Haonian Ji, Kaiwen Xiong, Siwei Han, Peng Xia, Shi Qiu, Yiyang Zhou, Jiaqi Liu, Jinlong Li, Bingzhou Li, Zeyu Zheng, Cihang Xie, and Huaxiu Yao. Clawarena: Benchmarking ai agents in evolving information environments, 2026. URL <https://arxiv.org/abs/2604.04202>.
- jiajie Zhang, Yushi Bai, Xin Lv, Wanjuan Gu, Danqing Liu, Minhao Zou, Shulin Cao, Lei Hou, Yuxiao Dong, Ling Feng, and Juanzi Li. Longcite: Enabling llms to generate fine-grained citations in long-context qa, 2024. URL <https://arxiv.org/abs/2409.02897>.
- Albert Q. Jiang, Alexandre Sablayrolles, Antoine Roux, Arthur Mensch, Blanche Savary, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Emma Bou Hanna, Florian Bressand, Gianna Lengyel, Guillaume Bour, Guillaume Lample, L  lio Renard Lavaud, Lucile Saulnier, Marie-Anne Lachaux, Pierre Stock, Sandeep Subramanian, Sophia Yang, Szymon Antoniak, Teven Le Scao, Th  ophile Gervet, Thibaut Lavril, Thomas Wang, Timoth  e Lacroix, and William El Sayed. Mixtral of experts, 2024a. URL <https://arxiv.org/abs/2401.04088>.
- Dongwei Jiang, Alvin Zhang, Andrew Wang, Nicholas Andrews, and Daniel Khashabi. Feedback friction: Llms struggle to fully incorporate external feedback, 2025. URL <https://arxiv.org/abs/2506.11930>.
- Haitao Jiang, Wenbo Zhang, Jiarui Yao, Hengrui Cai, Sheng Wang, and Rui Song. Supervised fine-tuning versus reinforcement learning: A study of post-training methods for large language models, 2026. URL <https://arxiv.org/abs/2603.13985>.
- Ziyan Jiang, Xueguang Ma, and Wenhui Chen. Longrag: Enhancing retrieval-augmented generation with long-context llms, 2024b. URL <https://arxiv.org/abs/2406.15319>.

- Bowen Jin, Hansi Zeng, Zhenrui Yue, Dong Wang, Hamed Zamani, and Jiawei Han. Search-rl: Training llms to reason and leverage search engines with reinforcement learning, 2025a. URL <https://arxiv.org/abs/2503.09516>.
- Jiajie Jin, Xiaoxi Li, Guanting Dong, Yuyao Zhang, Yutao Zhu, Yang Zhao, Hongjin Qian, and Zhicheng Dou. Decoupled planning and execution: A hierarchical reasoning framework for deep search, 2025b. URL <https://arxiv.org/abs/2507.02652>.
- Mingyu Jin, Qinkai Yu, Jingyuan Huang, Qingcheng Zeng, Zhenting Wang, Wenyue Hua, Haiyan Zhao, Kai Mei, Yanda Meng, Kaize Ding, Fan Yang, Mengnan Du, and Yongfeng Zhang. Exploring concept depth: How large language models acquire knowledge at different layers?, 2024a. URL <https://arxiv.org/abs/2404.07066>.
- Zhuoran Jin, Pengfei Cao, Chenhao Wang, Zhitao He, Hongbang Yuan, Jiachun Li, Yubo Chen, Kang Liu, and Jun Zhao. RWKU: Benchmarking real-world knowledge unlearning for large language models. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2024b. URL <https://openreview.net/forum?id=wOmtZ5FgMH>.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- Feiyang Kang, Newsha Ardalani, Michael Kuchnik, Youssef Emad, Mostafa Elhoushi, Shubhabrata Sengupta, Shang-Wen Li, Ramya Raghavendra, Ruoxi Jia, and Carole-Jean Wu. Demystifying synthetic data in llm pre-training: A systematic study of scaling laws, benefits, and pitfalls, 2025a. URL <https://arxiv.org/abs/2510.01631>.
- Feiyang Kang, Yifan Sun, Bingbing Wen, Si Chen, Dawn Song, Rafid Mahmood, and Ruoxi Jia. AutoScale: Scale-aware data mixing for pre-training LLMs. In *Conference on Language Modeling*, 2025b. URL <https://openreview.net/forum?id=rujwIvjooA>.
- Minki Kang, Jongwon Jeong, and Jaewoong Cho. T1: Tool-integrated verification for test-time compute scaling in small language models, 2026. URL <https://arxiv.org/abs/2504.04718>.
- Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models, 2020. URL <https://arxiv.org/abs/2001.08361>.
- Jungo Kasai, Keisuke Sakaguchi, Ronan Le Bras, Akari Asai, Xinyan Yu, Dragomir Radev, Noah A Smith, Yejin Choi, Kentaro Inui, et al. Realtime qa: What’s the answer right now? *Advances in neural information processing systems*, 36:49025–49043, 2023.
- Sourena Khanzadeh. Project ariadne: A structural causal framework for auditing faithfulness in llm agents, 2026. URL <https://arxiv.org/abs/2601.02314>.
- Jaehoon Kim and Dongha Lee. Opsd compresses what rlvr teaches: A post-rl compaction stage for reasoning models, 2026. URL <https://arxiv.org/abs/2605.06188>.

- Kangsan Kim, Minki Kang, Taeil Kim, Yanlai Yang, Mengye Ren, and Sung Ju Hwang. Memory transfer learning: How memories are transferred across domains in coding agents, 2026. URL <https://arxiv.org/abs/2604.14004>.
- Chuyi Kong, Gao Wei, Jing Ma, Hongzhan Lin, and Yaxin Fan. Reflex: Self-refining explainable fact-checking via disentangling truth into style and substance, 2025. URL <https://arxiv.org/abs/2511.20233>.
- János Kramár, Tom Lieberum, Rohin Shah, and Neel Nanda. Atp*: An efficient and scalable method for localizing llm behaviour to components, 2024. URL <https://arxiv.org/abs/2403.00745>.
- Meghdad Kurmanji, Peter Triantafillou, Jamie Hayes, and Eleni Triantafillou. Towards unbounded machine unlearning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Philippe Laban, Alexander R. Fabbri, Caiming Xiong, and Chien-Sheng Wu. Summary of a haystack: A challenge to long-context llms and rag systems, 2024. URL <https://arxiv.org/abs/2407.01370>.
- Patrick Leask, Bart Bussmann, Michael Pearce, Joseph Bloom, Curt Tigges, Noura Al Moubayed, Lee Sharkey, and Neel Nanda. Sparse autoencoders do not find canonical units of analysis. In *International Conference on Learning Representations*, volume 2025, pages 53617–53642, 2025.
- Hyunji Lee, Franck Dernoncourt, Trung Bui, and Seunghyun Yoon. Corg: Generating answers from complex, interrelated contexts, 2025a. URL <https://arxiv.org/abs/2505.00023>.
- Hyunji Lee, Danni Liu, Supriti Sinhamahapatra, and Jan Niehues. How do multimodal foundation models encode text and speech? an analysis of cross-lingual and cross-modal representations, 2025b. URL <https://arxiv.org/abs/2411.17666>.
- Jaeho Lee and Atharv Chowdhary. Assertbench: A benchmark for evaluating self-assertion in large language models, 2025. URL <https://arxiv.org/abs/2506.11110>.
- Katherine Lee, Daphne Ippolito, Andrew Nystrom, Chiyuan Zhang, Douglas Eck, Chris Callison-Burch, and Nicholas Carlini. Deduplicating training data makes language models better. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio, editors, *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8424–8445, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-long.577. URL <https://aclanthology.org/2022.acl-long.577/>.
- Simon Lermen, Charlie Rogers-Smith, and Jeffrey Ladish. LoRA fine-tuning efficiently undoes safety training in Llama 2-Chat 70B. *arXiv preprint arXiv:2310.20624*, 2024.
- Mosh Levy, Zohar Elyoseph, Shauli Ravfogel, and Yoav Goldberg. State over tokens: Characterizing the role of reasoning tokens, 2025. URL <https://arxiv.org/abs/2512.12777>.

- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, 33:9459–9474, 2020.
- Bryan Li, Samar Haider, Fiona Luo, Adwait Agashe, and Chris Callison-Burch. Bordirlines: A dataset for evaluating cross-lingual retrieval-augmented generation, 2024a. URL <https://arxiv.org/abs/2410.01171>.
- Chengpeng Li, Mingfeng Xue, Zhenru Zhang, Jiayi Yang, Beichen Zhang, Xiang Wang, Bowen Yu, Binyuan Hui, Junyang Lin, and Dayiheng Liu. Start: Self-taught reasoner with tools, 2025a. URL <https://arxiv.org/abs/2503.04625>.
- Junjie Li, Xinrui Guo, Yuhao Wu, Roy Ka-Wei Lee, Hongzhi Li, and Yutao Xie. Lost in stories: Consistency bugs in long story generation by llms, 2026a. URL <https://arxiv.org/abs/2603.05890>.
- Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model, 2024b. URL <https://arxiv.org/abs/2306.03341>.
- Kuan Li, Zhongwang Zhang, Huifeng Yin, Liwen Zhang, Litu Ou, Jialong Wu, Wenbiao Yin, Baixuan Li, Zhengwei Tao, Xinyu Wang, Weizhou Shen, Junkai Zhang, Dingchu Zhang, Xixi Wu, Yong Jiang, Ming Yan, Pengjun Xie, Fei Huang, and Jingren Zhou. Websailor: Navigating super-human reasoning for web agent, 2025b. URL <https://arxiv.org/abs/2507.02592>.
- Mo Li, Songyang Zhang, Yunxin Liu, and Kai Chen. Needlebench: Can llms do retrieval and reasoning in 1 million context window?, 2024c. URL <https://arxiv.org/abs/2407.11963>.
- Nathaniel Li, Alexander Pan, Anjali Gopal, Summer Yue, Daniel Berrios, Alice Gatti, Justin D. Li, Ann-Kathrin Dombrowski, Shashwat Goel, Long Phan, Gabriel Mukobi, Nathan Helm-Burger, Rassin Lababidi, Lennart Justen, Andrew B. Liu, Michael Chen, Isabelle Barrass, Oliver Zhang, Xiaoyuan Zhu, Rishub Tamirisa, Bhurugu Bharathi, Adam Khoja, Zhenqi Zhao, Ariel Herbert-Voss, Cort B. Breuer, Samuel Marks, Oam Patel, Andy Zou, Mantas Mazeika, Zifan Wang, Palash Oswal, Weiran Lin, Adam A. Hunt, Justin Tienken-Harder, Kevin Y. Shih, Kemper Talley, John Guan, Russell Kaplan, Ian Steneker, David Campbell, Brad Jokubaitis, Alex Levinson, Jean Wang, William Qian, Kallol Krishna Karmakar, Steven Basart, Stephen Fitz, Mindy Levine, Ponnurangam Kumaraguru, Uday Tupakula, Vijay Varadharajan, Ruoyu Wang, Yan Shoshitaishvili, Jimmy Ba, Kevin M. Esvelt, Alexandr Wang, and Dan Hendrycks. The wmdp benchmark: Measuring and reducing malicious use with unlearning, 2024d. URL <https://arxiv.org/abs/2403.03218>.
- Tianle Li, Ge Zhang, Quy Duc Do, Xiang Yue, and Wenhui Chen. Long-context llms struggle with long in-context learning, 2024e. URL <https://arxiv.org/abs/2404.02060>.

- Xiaopeng Li, Shasha Li, Shezheng Song, Jing Yang, Jun Ma, and Jie Yu. PMET: Precise model editing in a transformer. In *AAAI Conference on Artificial Intelligence*, 2024f.
- Xiaoxi Li, Jiajie Jin, Guanting Dong, Hongjin Qian, Yutao Zhu, Yongkang Wu, Ji-Rong Wen, and Zhicheng Dou. Webthinker: Empowering large reasoning models with deep research capability, 2025c. URL <https://arxiv.org/abs/2504.21776>.
- Yangning Li, Weizhi Zhang, Yuyao Yang, Wei-Chieh Huang, Yaozu Wu, Junyu Luo, Yuanchen Bei, Henry Peng Zou, Xiao Luo, Yusheng Zhao, Chunkit Chan, Yankai Chen, Zhongfen Deng, Yinghui Li, Hai-Tao Zheng, Dongyuan Li, Renhe Jiang, Ming Zhang, Yangqiu Song, and Philip S. Yu. Towards agentic rag with deep reasoning: A survey of rag-reasoning systems in llms, 2025d. URL <https://arxiv.org/abs/2507.09477>.
- Yaxuan Li, Yuxin Zuo, Bingxiang He, Jinqian Zhang, Chaojun Xiao, Cheng Qian, Tianyu Yu, Huan ang Gao, Wenkai Yang, Zhiyuan Liu, and Ning Ding. Rethinking on-policy distillation of large language models: Phenomenology, mechanism, and recipe, 2026b. URL <https://arxiv.org/abs/2604.13016>.
- Yucheng Li, Frank Guerin, and Chenghua Lin. Latesteval: Addressing data contamination in language model evaluation through dynamic and time-sensitive test construction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 18600–18607, 2024g.
- Kaiqu Liang, Haimin Hu, Xuandong Zhao, Dawn Song, Thomas L. Griffiths, and Jaime Fernández Fisac. Machine bullshit: Characterizing the emergent disregard for truth in large language models, 2025a. URL <https://arxiv.org/abs/2507.07484>.
- Xun Liang, Shichao Song, Zifan Zheng, Hanyu Wang, Qingchen Yu, Xunkai Li, Rong-Hua Li, Feiyu Xiong, and Zhiyu Li. Internal consistency and self-feedback in large language models: A survey, 2024. URL <https://arxiv.org/abs/2407.14507>.
- Xun Liang, Simin Niu, Zhiyu Li, Sensen Zhang, Hanyu Wang, Feiyu Xiong, Jason Zhaoxin Fan, Bo Tang, Shichao Song, Mengwei Wang, and Jiawei Yang. Saferag: Benchmarking security in retrieval-augmented generation of large language model, 2025b. URL <https://arxiv.org/abs/2501.18636>.
- Tom Lieberum, Senthoooran Rajamanoharan, Arthur Conmy, Lewis Smith, Nicolas Sonnerat, Vikrant Varma, János Kramár, Anca Dragan, Rohin Shah, and Neel Nanda. Gemma scope: Open sparse autoencoders everywhere all at once on gemma 2. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 278–300, 2024.
- Zheng Wei Lim, Alham Fikri Aji, and Trevor Cohn. Language-specific latent process hinders cross-lingual performance, 2025. URL <https://arxiv.org/abs/2505.13141>.
- Chenyu Lin, Yilin Wen, Du Su, Hexiang Tan, Fei Sun, Muhan Chen, Chenfu Bao, and Zhonghou Lyu. Resisting contextual interference in rag via parametric-knowledge reinforcement. *arXiv preprint arXiv:2506.05154*, 2025.

- Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. In *Proceedings of the 60th annual meeting of the association for computational linguistics (volume 1: long papers)*, pages 3214–3252, 2022.
- Yong Lin, Hangyu Lin, Wei Xiong, Shizhe Diao, Jianmeng Liu, Jipeng Zhang, Rui Pan, Haoxiang Wang, Wenbin Hu, Hanning Zhang, Hanze Dong, Renjie Pi, Han Zhao, Nan Jiang, Heng Ji, Yuan Yao, and Tong Zhang. Mitigating the alignment tax of RLHF. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen, editors, *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 580–606, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.35. URL <https://aclanthology.org/2024.emnlp-main.35/>.
- Jack Lindsey, Wes Gurnee, Emmanuel Ameisen, Brian Chen, Adam Pearce, Nicholas L. Turner, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam Jermyn, Andy Jones, Andrew Persic, Zhenyi Qi, T. Ben Thompson, Sam Zimmerman, Kelley Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson. On the biology of a large language model. *Transformer Circuits Thread*, 2025. URL <https://transformer-circuits.pub/2025/attribution-graphs/biology.html>.
- Emmy Liu, Graham Neubig, and Chenyan Xiong. Midtraining bridges pretraining and posttraining distributions. In *Methods and Opportunities at Small Scale*, 2026a. URL <https://openreview.net/forum?id=tu0VUHXmWU>.
- Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning, 2023. URL <https://arxiv.org/abs/2304.08485>.
- Jiaxiang Liu, Boxuan Xing, Chenhao Yuan, ChenxiangZhang ChenxiangZhang, Di Wu, Xiusheng Huang, Haida Yu, Chuhan Lang, Pengfei Cao, Jun Zhao, et al. Know-mri: A knowledge mechanisms revealer&interpreter for large language models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, pages 199–210, 2025a.
- Nelson F Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the association for computational linguistics*, 12:157–173, 2024a.
- Ruibo Liu, Jerry Wei, Fangyu Liu, Chenglei Si, Yanzhe Zhang, Jinmeng Rao, Steven Zheng, Daiyi Peng, Diyi Yang, Denny Zhou, and Andrew M. Dai. Best practices and lessons learned on synthetic data. In *Conference on Language Modeling*, 2024b. URL <https://openreview.net/forum?id=0JaWBhh61C>.
- Sijia Liu, Yuanshun Yao, Jinghan Jia, Stephen Casper, Nathalie Baracaldo, Peter Hase, Yuguang Yao, Chris Yuhao Liu, Xiaojun Xu, Hang Li, Kush R. Varshney, Mohit Bansal, Sanmi Koyejo, and Yang Liu. Rethinking machine unlearning for large language models. *Nature Machine Intelligence*, 2024c.

- Yajiao Liu, Congliang Chen, Junchi Yang, and Ruoyu Sun. Rethinking data mixture for large language models: A comprehensive survey and new perspectives, 2025b. URL <https://arxiv.org/abs/2505.21598>.
- Yizhou Liu, Ziming Liu, and Jeff Gore. Superposition yields robust neural scaling. *Advances in Neural Information Processing Systems*, 38:159269–159305, 2026b.
- Ka Man Lo, Zeyu Huang, Zihan Qiu, Zili Wang, and Jie Fu. A closer look into mixture-of-experts in large language models. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 4427–4447, Albuquerque, New Mexico, April 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-naacl.251. URL <https://aclanthology.org/2025.findings-naacl.251/>.
- Xingyu Lu, Xiaonan Li, Qinyuan Cheng, Kai Ding, Xuanjing Huang, and Xipeng Qiu. Scaling laws for fact memorization of large language models. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen, editors, *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 11263–11282, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-emnlp.658. URL <https://aclanthology.org/2024.findings-emnlp.658/>.
- Ziyang Luo, Zhiqi Shen, Wenzhuo Yang, Zirui Zhao, Prathyusha Jwalapuram, Amrita Saha, Doyen Sahoo, Silvio Savarese, Caiming Xiong, and Junnan Li. Mcp-universe: Benchmarking large language models with real-world model context protocol servers, 2025. URL <https://arxiv.org/abs/2508.14704>.
- Ang Lv, Yuhan Chen, Kaiyi Zhang, Yulong Wang, Lifeng Liu, Ji-Rong Wen, Jian Xie, and Rui Yan. Interpreting key mechanisms of factual recall in transformer-based language models, 2024. URL <https://arxiv.org/abs/2403.19521>.
- Aengus Lynch, Phillip Guo, Aidan Ewart, Stephen Casper, and Dylan Hadfield-Menell. Eight methods to evaluate robust unlearning in LLMs. *arXiv preprint arXiv:2402.16835*, 2024.
- Shengjie Ma, Xuhui Jiang, Chengjin Xu, Cehao Yang, Liyu Zhang, Jiaxin Mao, and Jian Guo. Synthesize-on-graph: Knowledgeable synthetic data generation for continued pre-training of large language models. In *Learning on Graphs Conference*, 2025. URL <https://openreview.net/forum?id=mkx3UyFPL1>.
- Pratyush Maini, Zhili Feng, Avi Schwarzschild, Zachary C. Lipton, and J. Zico Kolter. Tofu: A task of fictitious unlearning for llms, 2024. URL <https://arxiv.org/abs/2401.06121>.
- Ankit Maloo. Kwbench: Measuring unprompted problem recognition in knowledge work, 2026. URL <https://arxiv.org/abs/2604.15760>.
- Potsawee Manakul, Adian Liusie, and Mark Gales. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 conference on empirical methods in natural language processing*, pages 9004–9017, 2023.

- Sara Vera Marjanović, Arkil Patel, Vaibhav Adlakha, Milad Aghajohari, Parishad BehnamGhader, Mehar Bhatia, Aditi Khandelwal, Austin Kraft, Benno Krojer, Xing Han Lù, Nicholas Meade, Dongchan Shin, Amirhossein Kazemnejad, Gaurav Kamath, Marius Mosbach, Karolina Stańczak, and Siva Reddy. Deepseek-r1 thoughtology: Let’s jthinkj about llm reasoning, 2025. URL <https://arxiv.org/abs/2504.07128>.
- Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *arXiv preprint arXiv:2310.06824*, 2023.
- Samuel Marks, Can Rager, Eric J Michaud, Yonatan Belinkov, David Bau, and Aaron Mueller. Sparse feature circuits: Discovering and editing interpretable causal graphs in language models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=I4e82CIDxv>.
- Michael S Matena and Colin Raffel. Merging models with fisher-weighted averaging. *Advances in Neural Information Processing Systems*, 35:17703–17716, 2022.
- Nick Mecklenburg, Yiyu Lin, Xiaoxiao Li, Daniel Holstein, Leonardo Nunes, Sara Malvar, Bruno Silva, Ranveer Chandra, Vijay Aski, Pavan Kumar Reddy Yannam, Tolga Aktas, and Todd Hendry. Injecting new knowledge into large language models via supervised fine-tuning, 2024. URL <https://arxiv.org/abs/2404.00213>.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. Mass-editing memory in a transformer. In *International Conference on Learning Representations (ICLR)*, 2023.
- Jack Merullo, Louis Castricato, Carsten Eickhoff, and Ellie Pavlick. Linearly mapping from image to text space, 2023. URL <https://arxiv.org/abs/2209.15162>.
- Jack Merullo, Carsten Eickhoff, and Ellie Pavlick. Language models implement simple word2vec-style vector arithmetic. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 5030–5047, 2024.
- Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 12076–12100, 2023.
- Julian Minder, Clément Dumas, Caden Juang, Bilal Chughtai, and Neel Nanda. Overcoming sparsity artifacts in crosscoders to interpret chat-tuning. *Advances in Neural Information Processing Systems*, 38:106423–106474, 2026.

- Adrian Mirza, Nawaf Alampara, Sreekanth Kunchapu, Benedict Emoekabu, Aswanth Krishnan, Mara Wilhelmi, Macjonathan Okereke, Juliane Eberhardt, Amir Mohammad Elahi, Maximilian Greiner, Caroline T. Holick, Tanya Gupta, Mehrdad Asgari, Christina Glaubitz, Lea C. Klepsch, Yannik Köster, Jakob Meyer, Santiago Miret, Tim Hoffmann, Fabian Alexander Kreth, Michael Ringleb, Nicole Roesner, Ulrich S. Schubert, Leanne M. Stafast, Dinga Wonanke, Michael Pieler, Philippe Schwaller, and Kevin Maik Jablonka. Are large language models superhuman chemists?, 2024. URL <https://arxiv.org/abs/2404.01475>.
- Eric Mitchell, Charles Lin, Antoine Bosselut, Chelsea Finn, and Christopher D. Manning. Fast model editing at scale. In *International Conference on Learning Representations (ICLR)*, 2022a.
- Eric Mitchell, Charles Lin, Antoine Bosselut, Christopher D. Manning, and Chelsea Finn. Memory-based model editing at scale. In *International Conference on Machine Learning (ICML)*, 2022b.
- Atsuyuki Miyai, Zaiying Zhao, Kazuki Egashira, Atsuki Sato, Tatsumi Sunada, Shota Onohara, Hiromasa Yamanishi, Mashiro Toyooka, Kunato Nishina, Ryoma Maeda, Kiyoharu Aizawa, and Toshihiko Yamasaki. Webchorearena: Evaluating web browsing agents on realistic tedious web tasks, 2025. URL <https://arxiv.org/abs/2506.01952>.
- Kaixiang Mo, Yuxin Shi, Weiwei Weng, Zhiqiang Zhou, Shuman Liu, Haibo Zhang, and Anxiang Zeng. Mid-training of large language models: A survey, 2025. URL <https://arxiv.org/abs/2510.06826>.
- Ali Modarressi, Hanieh Deilamsalehy, Franck Dernoncourt, Trung Bui, Ryan A. Rossi, Seunghyun Yoon, and Hinrich Schütze. Nolima: Long-context evaluation beyond literal matching, 2025. URL <https://arxiv.org/abs/2502.05167>.
- Niklas Muennighoff, Alexander M. Rush, Boaz Barak, Teven Le Scao, Nouamane Tazi, Aleksandra Piktus, Sampo Pyysalo, Thomas Wolf, and Colin A. Raffel. Scaling data-constrained language models. In *Advances in Neural Information Processing Systems*, volume 36, pages 50358–50376. Curran Associates, Inc., 2023. doi: 10.52202/075280-2191. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/9d89448b63ce1e2e8dc7af72c984c196-Abstract-Conference.html.
- Jingwei Ni, Yihao Liu, Xinpeng Liu, Yutao Sun, Mengyu Zhou, Pengyu Cheng, Dexin Wang, Erchao Zhao, Xiaoxi Jiang, and Guanjin Jiang. Trace2skill: Distill trajectory-local lessons into transferable agent skills, 2026. URL <https://arxiv.org/abs/2603.25158>.
- Jingcheng Niu, Andrew Liu, Zining Zhu, and Gerald Penn. What does the knowledge neuron thesis have to do with knowledge? In *International Conference on Learning Representations*, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/hash/013d743db3c684957305d32017f13339-Abstract-Conference.html.
- Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Dawn

- Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Scott Johnston, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. In-context learning and induction heads. *arXiv preprint arXiv:2209.11895*, 2022.
- OpenAI. Learning to reason with LLMs. OpenAI Blog, September 2024a. URL <https://openai.com/index/learning-to-reason-with-llms/>.
- OpenAI. Openai o1 system card, 2024b. URL <https://arxiv.org/abs/2412.16720>.
- Guillermo Ortiz-Jimenez, Alessandro Favero, and Pascal Frossard. Task arithmetic in the tangent space: Improved editing of pre-trained models. *Advances in Neural Information Processing Systems*, 36:66727–66754, 2023.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul F. Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744. Curran Associates, Inc., 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/b1efde53be364a73914f58805a001731-Abstract-Conference.html.
- Ziqi Pang, Ziyang Xie, Yunze Man, and Yu-Xiong Wang. Frozen transformers in language models are effective visual encoder layers, 2024. URL <https://arxiv.org/abs/2310.12973>.
- Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. *arXiv preprint arXiv:2311.03658*, 2023.
- Kiho Park, Yo Joong Choe, Yibo Jiang, and Victor Veitch. The geometry of categorical and hierarchical concepts in large language models. In *International Conference on Learning Representations*, volume 2025, pages 76441–76463, 2025.
- Gonçalo Paulo and Nora Belrose. Sparse autoencoders trained on the same data learn different features. *arXiv preprint arXiv:2501.16615*, 2025.
- Guilherme Penedo, Hynek Kydlíček, Loubna Ben Allal, Anton Lozhkov, Margaret Mitchell, Colin A. Raffel, Leandro von Werra, and Thomas Wolf. The FineWeb datasets: Decanting the web for the finest text data at scale. In *Advances in Neural Information Processing Systems*, volume 37, pages 30811–30849. Curran Associates, Inc., 2024. doi: 10.52202/079017-0970. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/370df50ccfdf8bde18f8f9c2d9151bda-Abstract-Datasets_and_Benchmarks_Track.html.
- Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. YaRN: Efficient context window extension of large language models. In *International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=wHBfxhZu1u>.

- Letian Peng, Chenyang An, Shibo Hao, Chengyu Dong, and Jingbo Shang. Linear correlation in lm’s compositional generalization and hallucination, 2025. URL <https://arxiv.org/abs/2502.04520>.
- Fabio Petroni, Tim Rocktäschel, Sebastian Riedel, Patrick Lewis, Anton Bakhtin, Yuxiang Wu, and Alexander Miller. Language Models as Knowledge Bases? In *Proceedings of EMNLP*, pages 2463–2473, 2019. doi: 10.18653/v1/D19-1250. URL <https://aclanthology.org/D19-1250>.
- Thinh Pham, Nguyen Nguyen, Pratibha Zunjare, Weiyuan Chen, Yu-Min Tseng, and Tu Vu. Sealqa: Raising the bar for reasoning in search-augmented language models, 2025. URL <https://arxiv.org/abs/2506.01062>.
- Piotr Pkezik, Konrad Kaczynski, Maria Szymanska, Filip Zarnecki, Zuzanna Deckert, Jakub Kwiatkowski, and Wojciech Janowski. Llm1agbench: Identifying temporal training boundaries in large language models. *arXiv preprint arXiv:2511.12116*, 2025.
- Neha Prakriya, Jui-Nan Yen, Cho-Jui Hsieh, and Jason Cong. Accelerating large language model pretraining via LFR pedagogy: Learn, focus, and review. In Gemma Boleda and Michael Roth, editors, *Proceedings of the 29th Conference on Computational Natural Language Learning*, pages 268–290, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-271-8. doi: 10.18653/v1/2025.conll-1.18. URL <https://aclanthology.org/2025.conll-1.18/>.
- Jirui Qi, Gabriele Sarti, Raquel Fernández, and Arianna Bisazza. Model internals-based answer attribution for trustworthy retrieval-augmented generation, 2024. URL <https://arxiv.org/abs/2406.13663>.
- Xiaoye Qu, Yafu Li, Zhaochen Su, Weigao Sun, Jianhao Yan, Dongrui Liu, Ganqu Cui, Daizong Liu, Shuxian Liang, Junxian He, Peng Li, Wei Wei, Jing Shao, Chaochao Lu, Yue Zhang, Xian-Sheng Hua, Bowen Zhou, and Yu Cheng. A survey of efficient reasoning for large reasoning models: Language, multimodality, and beyond, 2025. URL <https://arxiv.org/abs/2503.21614>.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PmLR, 2021.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, volume 36, pages 53728–53741. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/a85b405ed65c6477a4fe8302b5e06ce7-Abstract-Conference.html.
- Pooyan Rahmanzadehgervi, Logan Bolton, Mohammad Reza Taesiri, and Anh Totti Nguyen. Vision language models are blind: Failing to translate detailed visual features into words, 2025. URL <https://arxiv.org/abs/2407.06581>.

- Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null it out: Guarding protected attributes by iterative nullspace projection. In *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.
- Shauli Ravfogel, Gilad Yehudai, Joan Bruna, and Alberto Bietti. Geometric factual recall in transformers, 2026. URL <https://arxiv.org/abs/2605.12426>.
- Abhilasha Ravichander, Shruti Ghela, David Wadden, and Yejin Choi. Halogen: Fantastic llm hallucinations and where to find them, 2025. URL <https://arxiv.org/abs/2501.08292>.
- David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R Bowman. Gpqa: A graduate-level google-proof q&a benchmark. *arXiv preprint arXiv:2311.12022*, 2023.
- Hadi Reisizadeh, Jiajun Ruan, Yiwei Chen, Soumyadeep Pal, Sijia Liu, and Mingyi Hong. Leak@k: Unlearning does not make llms forget under probabilistic decoding, 2026. URL <https://arxiv.org/abs/2511.04934>.
- François Remy, Pieter Delobelle, Hayastan Avetisyan, Alfiya Khabibullina, Miryam de Lhoneux, and Thomas Demeester. Trans-tokenization and cross-lingual vocabulary transfers: Language adaptation of llms for low-resource nlp, 2024. URL <https://arxiv.org/abs/2408.04303>.
- Adam Roberts, Colin Raffel, and Noam Shazeer. How much knowledge can you pack into the parameters of a language model? In Bonnie Webber, Trevor Cohn, Yulan He, and Yang Liu, editors, *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 5418–5426, Online, November 2020a. Association for Computational Linguistics. doi: 10.18653/v1/2020.emnlp-main.437. URL <https://aclanthology.org/2020.emnlp-main.437/>.
- Adam Roberts, Colin Raffel, and Noam Shazeer. How much knowledge can you pack into the parameters of a language model? In *Proceedings of the 2020 conference on empirical methods in natural language processing (EMNLP)*, pages 5418–5426, 2020b.
- Laura Ruis, Maximilian Mozes, Juhan Bae, Siddhartha Rao Kamalakara, Dwaraknath Gnaneshwar, Acyr Locatelli, Robert Kirk, Tim Rocktäschel, Edward Grefenstette, and Max Bartolo. Procedural knowledge in pretraining drives reasoning in large language models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=1hQKHHUsMx>.
- Bharat Runwal, Ashish Agrawal, Anurag Roy, and Rameswar Panda. PRISM: A systematic study of the retention-interaction tradeoff in mid-training language models, 2026. URL <https://arxiv.org/abs/2603.17074>.
- Raphaël Sarfati, Eric Bigelow, Daniel Wurgaft, Siddharth Boppana, Jack Merullo, Atticus Geiger, Owen Lewis, Tom McGrath, and Ekdeep Singh Lubana. The shape of beliefs: Geometry, dynamics, and interventions along representation manifolds of language models’ posteriors. *arXiv preprint arXiv:2602.02315*, 2026.

- Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools, 2023. URL <https://arxiv.org/abs/2302.04761>.
- Sina J. Semnani, Jirayu Burapachee, Arpandeeep Khatua, Thanawan Atchariyachanvanit, Zheng Wang, and Monica S. Lam. Detecting corpus-level knowledge inconsistencies in wikipedia with large language models, 2025. URL <https://arxiv.org/abs/2509.23233>.
- Or Shafran, Shaked Ronen, Omri Fahn, Shauli Ravfogel, Atticus Geiger, and Mor Geva. From directions to regions: Decomposing activations in language models via local geometry. *arXiv preprint arXiv:2602.02464*, 2026.
- Uri Shaham, Jonathan Herzig, Roei Aharoni, Idan Szpektor, Reut Tsarfaty, and Matan Eyal. Multilingual instruction tuning with just a pinch of multilinguality, 2024. URL <https://arxiv.org/abs/2401.01854>.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024. URL <https://arxiv.org/abs/2402.03300>.
- Asankhaya Sharma. Eliciting fine-tuned transformer capabilities via inference-time techniques, 2025a. URL <https://arxiv.org/abs/2506.08060>.
- Chaitanya Sharma. Retrieval-augmented generation: A comprehensive survey of architectures, enhancements, and robustness frontiers, 2025b. URL <https://arxiv.org/abs/2506.00054>.
- Noam Shazeer, Azalia Mirhoseini, Krzysztof Maziarczyk, Andy Davis, Quoc Le, Geoffrey Hinton, and Jeff Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations*, 2017. URL <https://openreview.net/forum?id=B1ckMDqlg>.
- Xu Shen, Song Wang, Zhen Tan, Laura Yao, Xinyu Zhao, Kaidi Xu, Xin Wang, and Tianlong Chen. Faithcot-bench: Benchmarking instance-level faithfulness of chain-of-thought reasoning. *arXiv preprint arXiv:2510.04040*, 2025.
- WeiJia Shi, Jaechan Lee, Yangsibo Huang, Sadhika Malladi, Jieyu Zhao, Ari Holtzman, Daogao Liu, Luke Zettlemoyer, Noah A. Smith, and Chiyuan Zhang. Muse: Machine unlearning six-way evaluation for language models, 2024. URL <https://arxiv.org/abs/2407.06460>.
- Ilya Shumailov, Jamie Hayes, Eleni Triantafyllou, Guillermo Ortiz-Jimenez, Nicolas Papernot, Milad Nasr, Matthew Jagielski, Itay Yona, Ronen Eldan, and Alexey Kurakin. Un-unlearning: Unlearning is not sufficient for content regulation in advanced generative AI. *arXiv preprint arXiv:2407.00106*, 2024.
- Shuzheng Si, Haozhe Zhao, Cheng Gao, Yuzhuo Bai, Zhitong Wang, Bofei Gao, Kangyang Luo, Wenhao Li, Yufei Huang, Gang Chen, Fanchao Qi, Minjia Zhang, Baobao Chang,

- and Maosong Sun. Teaching large language models to maintain contextual faithfulness via synthetic tasks and reinforcement learning, 2025. URL <https://arxiv.org/abs/2505.16483>.
- Avyav Kumar Singh, Yen-Chen Wu, Alexandru Cioba, Alberto Bernacchia, and Davide Buffelli. Cross-tokenizer llm distillation through a byte-level interface, 2026. URL <https://arxiv.org/abs/2604.07466>.
- Linxin Song, Xuwei Ding, Jieyu Zhang, Taiwei Shi, Ryotaro Shimizu, Rahul Gupta, Yang Liu, Jian Kang, and Jieyu Zhao. Discovering knowledge deficiencies of language models on massive knowledge base, 2025a. URL <https://arxiv.org/abs/2503.23361>.
- Mingyang Song and Mao Zheng. A survey of on-policy distillation for large language models, 2026. URL <https://arxiv.org/abs/2604.00626>.
- Zirui Song, Bin Yan, Yuhan Liu, Miao Fang, Mingzhe Li, Rui Yan, and Xiuying Chen. Injecting domain-specific knowledge into large language models: A comprehensive survey. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 25297–25311, Suzhou, China, November 2025b. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-emnlp.1379. URL <https://aclanthology.org/2025.findings-emnlp.1379/>.
- Zayne Sprague, Fangcong Yin, Juan Diego Rodriguez, Dongwei Jiang, Manya Wadhwa, Prasann Singhal, Xinyu Zhao, Xi Ye, Kyle Mahowald, and Greg Durrett. To cot or not to cot? chain-of-thought helps mainly on math and symbolic reasoning. *arXiv preprint arXiv:2409.12183*, 2024.
- Zhaochen Su, Jun Zhang, Xiaoye Qu, Tong Zhu, Yanshu Li, Jiashuo Sun, Juntao Li, Min Zhang, and Yu Cheng. Conflictbank: A benchmark for evaluating the influence of knowledge conflicts in llm, 2024. URL <https://arxiv.org/abs/2408.12076>.
- Yang Sui, Yu-Neng Chuang, Guanchu Wang, Jiamu Zhang, Tianyi Zhang, Jiayi Yuan, Hongyi Liu, Andrew Wen, Shaochen Zhong, Na Zou, et al. Stop overthinking: A survey on efficient reasoning for large language models. *arXiv preprint arXiv:2503.16419*, 2025.
- Sainbayar Sukhbaatar, Edouard Grave, Guillaume Lample, Herve Jegou, and Armand Joulin. Augmenting self-attention with persistent memory, 2019. URL <https://arxiv.org/abs/1907.01470>.
- Chen Sun, Renat Aksitov, Andrey Zhmoginov, Nolan Andrew Miller, Max Vladymyrov, Ulrich Rueckert, Been Kim, and Mark Sandler. How new data permeates LLM knowledge and how to dilute it. In *The Thirteenth International Conference on Learning Representations*, 2025a. URL <https://openreview.net/forum?id=NGKQoaqLpo>.
- Jingyi Sun, Greta Warren, Irina Shklovski, and Isabelle Augenstein. Explaining sources of uncertainty in automated fact-checking, 2025b. URL <https://arxiv.org/abs/2505.17855>.

- Kaiser Sun, Fan Bai, and Mark Dredze. Task matters: Knowledge requirements shape llm responses to context-memory conflict. *arXiv preprint arXiv:2506.06485*, 2025c.
- Mukund Sundararajan, Ankur Taly, and Qiqi Yan. Axiomatic attribution for deep networks. In Doina Precup and Yee Whye Teh, editors, *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 3319–3328. PMLR, 06–11 Aug 2017. URL <https://proceedings.mlr.press/v70/sundararajan17a.html>.
- Yi-Lin Sung, Linjie Li, Kevin Lin, Zhe Gan, Mohit Bansal, and Lijuan Wang. An empirical study of multimodal model merging. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 1563–1575, 2023.
- Aaquib Syed, Can Rager, and Arthur Conmy. Attribution patching outperforms automated circuit discovery. In Yonatan Belinkov, Najoung Kim, Jaap Jumelet, Hosein Mohebbi, Aaron Mueller, and Hanjie Chen, editors, *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 407–416, Miami, Florida, US, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.blackboxnlp-1.25. URL <https://aclanthology.org/2024.blackboxnlp-1.25/>.
- Jihoon Tack, Jack Lanchantin, Jane Yu, Andrew Cohen, Ilia Kulikov, Janice Lan, Shibo Hao, Yuandong Tian, Jason E. Weston, and Xian Li. LLM pretraining with continuous concepts. In *The Fourteenth International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=wTGcb3Dx0n>.
- Sho Takishita, Jay Gala, Abdelrahman Mohamed, Kentaro Inui, and Yova Kementchedzhieva. Llms can compensate for deficiencies in visual representations, 2025. URL <https://arxiv.org/abs/2506.05439>.
- Chenmien Tan, Ge Zhang, and Jie Fu. Massive editing for large language models via meta learning. In *International Conference on Learning Representations (ICLR)*, 2024.
- Anke Tang, Li Shen, Yong Luo, Nan Yin, Lefei Zhang, and Dacheng Tao. Merging multi-task models via weight-ensembling mixture of experts. In *Proceedings of the 41st International Conference on Machine Learning*, ICML’24. JMLR.org, 2024.
- Xiangru Tang, Tianrui Qin, Tianhao Peng, Ziyang Zhou, Daniel Shao, Tingting Du, Xinming Wei, Peng Xia, Fang Wu, He Zhu, Ge Zhang, Jiaheng Liu, Xingyao Wang, Sirui Hong, Chenglin Wu, Hao Cheng, Chi Wang, and Wangchunshu Zhou. Agent kb: Leveraging cross-domain experience for agentic problem solving, 2025. URL <https://arxiv.org/abs/2507.06229>.
- Gemma Team, Morgane Riviere, Shreya Pathak, Pier Giuseppe Sessa, Cassidy Hardin, Surya Bhupatiraju, Léonard Hussenot, Thomas Mesnard, Bobak Shahriari, Alexandre Ramé, Johan Ferret, Peter Liu, Pouya Tafti, Abe Friesen, Michelle Casbon, Sabela Ramos, Ravin Kumar, Charline Le Lan, Sammy Jerome, Anton Tsitsulin, Nino Vieillard, Piotr Stanczyk, Sertan Girgin, Nikola Momchev, Matt Hoffman, Shantanu Thakoor, Jean-Bastien Grill, Behnam Neyshabur, Olivier Bachem, Alanna Walton, Aliaksei Severyn, Alicia Parrish, Aliya Ahmad, Allen Hutchison, Alvin Abdagic, Amanda Carl, Amy

Shen, Andy Brock, Andy Coenen, Anthony Laforge, Antonia Paterson, Ben Bastian, Bilal Piot, Bo Wu, Brandon Royal, Charlie Chen, Chintu Kumar, Chris Perry, Chris Welty, Christopher A. Choquette-Choo, Danila Sinopalnikov, David Weinberger, Dimple Vijaykumar, Dominika Rogozińska, Dustin Herbison, Elisa Bandy, Emma Wang, Eric Noland, Erica Moreira, Evan Senter, Evgenii Eltyshev, Francesco Visin, Gabriel Rasskin, Gary Wei, Glenn Cameron, Gus Martins, Hadi Hashemi, Hanna Klimczak-Plucińska, Harleen Batra, Harsh Dhand, Ivan Nardini, Jacinda Mein, Jack Zhou, James Svensson, Jeff Stanway, Jetha Chan, Jin Peng Zhou, Joana Carrasqueira, Joana Iljazi, Jocelyn Becker, Joe Fernandez, Joost van Amersfoort, Josh Gordon, Josh Lipschultz, Josh Newlan, Ju yeong Ji, Kareem Mohamed, Kartikeya Badola, Kat Black, Katie Millican, Keelin McDonell, Kelvin Nguyen, Kiranbir Sodhia, Kish Greene, Lars Lowe Sjoesund, Lauren Usui, Laurent Sifre, Lena Heuermann, Leticia Lago, Lilly McNealus, Livio Baldini Soares, Logan Kilpatrick, Lucas Dixon, Luciano Martins, Machel Reid, Manvinder Singh, Mark Iverson, Martin Görner, Mat Velloso, Mateo Wirth, Matt Davidow, Matt Miller, Matthew Rahtz, Matthew Watson, Meg Risdal, Mehran Kazemi, Michael Moynihan, Ming Zhang, Minsuk Kahng, Minwoo Park, Mofi Rahman, Mohit Khatwani, Natalie Dao, Nenshad Bardoliwalla, Nesh Devanathan, Neta Dumai, Nilay Chauhan, Oscar Wahltinez, Pankil Botarda, Parker Barnes, Paul Barham, Paul Michel, Pengchong Jin, Petko Georgiev, Phil Culliton, Pradeep Kuppala, Ramona Comanescu, Ramona Merhej, Reena Jana, Reza Ardeshir Rokni, Rishabh Agarwal, Ryan Mullins, Samaneh Saadat, Sara Mc Carthy, Sarah Cogan, Sarah Perrin, Sébastien M. R. Arnold, Sebastian Krause, Shengyang Dai, Shruti Garg, Shruti Sheth, Sue Ronstrom, Susan Chan, Timothy Jordan, Ting Yu, Tom Eccles, Tom Hennigan, Tomas Kocisky, Tulsee Doshi, Vihan Jain, Vikas Yadav, Vilobh Meshram, Vishal Dharmadhikari, Warren Barkley, Wei Wei, Wenming Ye, Woohyun Han, Woosuk Kwon, Xiang Xu, Zhe Shen, Zhitao Gong, Zichuan Wei, Victor Cotruta, Phoebe Kirk, Anand Rao, Minh Giang, Ludovic Peran, Tris Warkentin, Eli Collins, Joelle Barral, Zoubin Ghahramani, Raia Hadsell, D. Sculley, Jeanine Banks, Anca Dragan, Slav Petrov, Oriol Vinyals, Jeff Dean, Demis Hassabis, Koray Kavukcuoglu, Clement Farabet, Elena Buchatskaya, Sebastian Borgeaud, Noah Fiedel, Armand Joulin, Kathleen Kenealy, Robert Dadashi, and Alek Andreev. Gemma 2: Improving open language models at a practical size, 2024. URL <https://arxiv.org/abs/2408.00118>.

Qwen Team. Qwen3.5: Towards native multimodal agents, 2026. URL <https://qwen.ai/blog?id=qwen3.5>.

Adly Templeton. *Scaling monosemanticity: Extracting interpretable features from claude 3 sonnet*. Anthropic, 2024.

Bozhong Tian, Siyuan Liang, Siyuan Cheng, Zhiwei Xu, Ningyu Zhang, and Huajun Chen. InstructEdit: Instruction-based knowledge editing for large language models. In *International Joint Conference on Artificial Intelligence (IJCAI)*, 2024.

Xiaoyu Tian, Yunjie Ji, Haotian Wang, Shuaiting Chen, Sitong Zhao, Yiping Peng, Han Zhao, and Xiangang Li. Not all correct answers are equal: Why your distillation source matters, 2025. URL <https://arxiv.org/abs/2505.14464>.

- Daniil Tiapkin, Daniele Calandriello, Johan Ferret, Sarah Perrin, Nino Vieillard, Alexandre Ramé, and Mathieu Blondel. On teacher hacking in language model distillation, 2025. URL <https://arxiv.org/abs/2502.02671>.
- Kushal Tirumala, Daniel Simig, Armen Aghajanyan, and Ari S. Morcos. D4: Improving LLM pretraining via document de-duplication and diversification. In *Advances in Neural Information Processing Systems*, volume 36. Curran Associates, Inc., 2023. URL <https://openreview.net/forum?id=CGOL2PFrb1>.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. Llama: Open and efficient foundation language models, 2023a. URL <https://arxiv.org/abs/2302.13971>.
- Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, Dan Bikel, Lukas Blecher, Cristian Canton Ferrer, Moya Chen, Guillem Cucurull, David Esiobu, Jude Fernandes, Jeremy Fu, Wenyin Fu, Brian Fuller, Cynthia Gao, Vedanuj Goswami, Naman Goyal, Anthony Hartshorn, Saghar Hosseini, Rui Hou, Hakan Inan, Marcin Kardas, Viktor Kerkez, Madian Khabsa, Isabel Kloumann, Artem Korenev, Punit Singh Koura, Marie-Anne Lachaux, Thibaut Lavril, Jenya Lee, Diana Liskovich, Yinghai Lu, Yuning Mao, Xavier Martinet, Todor Mihaylov, Pushkar Mishra, Igor Molybog, Yixin Nie, Andrew Poulton, Jeremy Reizenstein, Rashi Rungta, Kalyan Saladi, Alan Schelten, Ruan Silva, Eric Michael Smith, Ranjan Subramanian, Xiaoqing Ellen Tan, Binh Tang, Ross Taylor, Adina Williams, Jian Xiang Kuan, Puxin Xu, Zheng Yan, Iliyan Zarov, Yuchen Zhang, Angela Fan, Melanie Kambadur, Sharan Narang, Aurelien Rodriguez, Robert Stojnic, Sergey Edunov, and Thomas Scialom. Llama 2: Open foundation and fine-tuned chat models, 2023b. URL <https://arxiv.org/abs/2307.09288>.
- Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. MuSiQue: Multihop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554, 2022. doi: 10.1162/tacl_a.00475. URL <https://aclanthology.org/2022.tacl-1.31>.
- Chengying Tu, Xuemiao Zhang, Rongxiang Weng, Rumei Li, Chen Zhang, Yang Bai, Hongfei Yan, Jingang Wang, and Xunliang Cai. A survey on LLM mid-training, 2025. URL <https://arxiv.org/abs/2510.23081>.
- Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc., 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.
- Max Vladymyrov, Johannes von Oswald, Mark Sandler, and Rong Ge. Linear transformers are versatile in-context learners, 2024. URL <https://arxiv.org/abs/2402.14180>.

- Tu Vu, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Jason Wei, Chris Tar, Yun-Hsuan Sung, Denny Zhou, Quoc Le, et al. Freshllms: Refreshing large language models with search engine augmentation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 13697–13720, 2024.
- Hitesh Wadhwa, Rahul Seetharaman, Somyaa Aggarwal, Reshmi Ghosh, Samyadeep Basu, Soundararajan Srinivasan, Wenlong Zhao, Shreyas Chaudhari, and Ehsan Aghazadeh. From rags to rich parameters: Probing how language models utilize external knowledge over parametric information for factual queries, 2024. URL <https://arxiv.org/abs/2406.12824>.
- Fanqi Wan, Longguang Zhong, Ziyi Yang, Ruijun Chen, and Xiaojun Quan. Fusechat: Knowledge fusion of chat models, 2024. URL <https://arxiv.org/abs/2408.07990>.
- Bo Wang, Junzhuo Li, Hong Chen, Yuanlin Chu, Yuxuan Fan, and Xuming Hu. Deconstructing pre-training: Knowledge attribution analysis in moe and dense models, 2026a. URL <https://arxiv.org/abs/2601.08383>.
- Chenhao Wang, Pengfei Cao, Zhuoran Jin, Yubo Chen, Daojian Zeng, Kang Liu, and Jun Zhao. MULFE: A multi-level benchmark for free text model editing. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13570–13587, Bangkok, Thailand, August 2024a. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.732. URL <https://aclanthology.org/2024.acl-long.732/>.
- Chenxi Wang, Zhuoyun Yu, Xin Xie, Wuguannan Yao, Runnan Fang, Shuofei Qiao, Kexin Cao, Guozhou Zheng, Xiang Qi, Peng Zhang, and Shumin Deng. Skillx: Automatically constructing skill knowledge bases for agents, 2026b. URL <https://arxiv.org/abs/2604.04804>.
- Fei Wang, Xingchen Wan, Ruoxi Sun, Jiefeng Chen, and Serkan Ö. Arık. Astute rag: Overcoming imperfect retrieval augmentation and knowledge conflicts for large language models, 2025a. URL <https://arxiv.org/abs/2410.07176>.
- Han Wang, Archiki Prasad, Elias Stengel-Eskin, and Mohit Bansal. Retrieval-augmented generation with conflicting evidence, 2025b. URL <https://arxiv.org/abs/2504.13079>.
- Huazheng Wang, Yongcheng Jing, Haifeng Sun, Yingjie Wang, Jingyu Wang, Jianxin Liao, and Dacheng Tao. Erasing without remembering: Implicit knowledge forgetting in large language models, 2025c. URL <https://arxiv.org/abs/2502.19982>.
- Jize Wang, Zerun Ma, Yining Li, Songyang Zhang, Cailian Chen, Kai Chen, and Xinyi Le. Gta: A benchmark for general tool agents, 2024b. URL <https://arxiv.org/abs/2407.08713>.
- Kevin Ro Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *The Eleventh International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=NpsVSN6o4ul>.

- Mengru Wang, Yunzhi Yao, Ziwen Xu, Shuofei Qiao, Shumin Deng, Peng Wang, Xiang Chen, Jia-Chen Gu, Yong Jiang, Pengjun Xie, Fei Huang, Huajun Chen, and Ningyu Zhang. Knowledge mechanisms in large language models: A survey and perspective, 2024c. URL <https://arxiv.org/abs/2407.15017>.
- Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Lingpeng Kong, Qi Liu, Tianyu Liu, and Zhifang Sui. Large language models are not fair evaluators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 9440–9450, Bangkok, Thailand, August 2024d. doi: 10.18653/v1/2024.acl-long.511. URL <https://aclanthology.org/2024.acl-long.511/>.
- Peng Wang, Zexi Li, Ningyu Zhang, Ziwen Xu, Yunzhi Yao, Yong Jiang, Pengjun Xie, Fei Huang, and Huajun Chen. Wise: Rethinking the knowledge memory for lifelong model editing of large language models, 2024e. URL <https://arxiv.org/abs/2405.14768>.
- Shuhe Wang, Shengyu Zhang, Jie Zhang, Runyi Hu, Xiaoya Li, Tianwei Zhang, Jiwei Li, Fei Wu, Guoyin Wang, and Eduard Hovy. Reinforcement learning enhanced llms: A survey, 2024f. URL <https://arxiv.org/abs/2412.10400>.
- Wenxiao Wang, Parsa Hosseini, and Soheil Feizi. Chain-of-defensive-thought: Structured reasoning elicits robustness in large language models against reference corruption, 2025d. URL <https://arxiv.org/abs/2504.20769>.
- Yanting Wang, Wei Zou, Runpeng Geng, and Jinyuan Jia. Tracllm: A generic framework for attributing long context llms, 2025e. URL <https://arxiv.org/abs/2506.04202>.
- Yanting Wang, Runpeng Geng, Ying Chen, and Jinyuan Jia. Attntrace: Contextual attribution of prompt injection and knowledge corruption, 2026c. URL <https://arxiv.org/abs/2508.03793>.
- Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, et al. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. *Advances in Neural Information Processing Systems*, 37:95266–95290, 2024g.
- Jason Wei, Maarten Bosma, Vincent Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M. Dai, and Quoc V. Le. Finetuned language models are zero-shot learners. In *International Conference on Learning Representations*, 2022a. URL <https://openreview.net/forum?id=gEZrGCozdqR>.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, volume 35, pages 24824–24837. Curran Associates, Inc., 2022b. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/9d5609613524ecf4f15af0f7b31abca4-Abstract-Conference.html.

- Jason Wei, Nguyen Karina, Hyung Won Chung, Yunxin Joy Jiao, Spencer Papay, Amelia Glaese, John Schulman, and William Fedus. Measuring short-form factuality in large language models. *arXiv preprint arXiv:2411.04368*, 2024a.
- Yifan Wei, Xiaoyan Yu, Yixuan Weng, Huanhuan Ma, Yuanzhe Zhang, Jun Zhao, and Kang Liu. Does knowledge localization hold true? surprising differences between entity and relation perspectives in language models. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management, CIKM '24*, page 4118–4122, New York, NY, USA, 2024b. Association for Computing Machinery. ISBN 9798400704369. doi: 10.1145/3627673.3679900. URL <https://doi.org/10.1145/3627673.3679900>.
- Zhepei Wei, Xiao Yang, Kai Sun, Jiaqi Wang, Rulin Shao, Sean Chen, Mohammad Kachuee, Teja Gollapudi, Tony Liao, Nicolas Scheffer, Rakesh Wanga, Anuj Kumar, Yu Meng, Wen tau Yih, and Xin Luna Dong. Truthrl: Incentivizing truthful llms via reinforcement learning, 2025. URL <https://arxiv.org/abs/2509.25760>.
- Zihao Wei, Liang Pang, Hanxing Ding, Jingcheng Deng, Huawei Shen, and Xueqi Cheng. Stable knowledge editing in large language models, 2024c. URL <https://arxiv.org/abs/2402.13048>.
- Liang Ze Wong. Paying attention to facts: Quantifying the knowledge capacity of attention layers, 2025. URL <https://arxiv.org/abs/2502.05076>.
- Mitchell Wortsman, Gabriel Ilharco, Samir Ya Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, et al. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *International conference on machine learning*, pages 23965–23998. PMLR, 2022a.
- Mitchell Wortsman, Gabriel Ilharco, Jong Wook Kim, Mike Li, Simon Kornblith, Rebecca Roelofs, Raphael Gontijo Lopes, Hannaneh Hajishirzi, Ali Farhadi, Hongseok Namkoong, et al. Robust fine-tuning of zero-shot models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 7959–7971, 2022b.
- Xinwei Wu, Junzhuo Li, Minghui Xu, Weilong Dong, Shuangzhi Wu, Chao Bian, and Deyi Xiong. DEPN: Detecting and editing privacy neurons in pretrained language models. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2875–2886, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.174. URL <https://aclanthology.org/2023.emnlp-main.174/>.
- Mengzhou Xia, Sathika Malladi, Suchin Gururangan, Sanjeev Arora, and Danqi Chen. LESS: Selecting influential data for targeted instruction tuning. In Ruslan Salakhutdinov, Zico Kolter, Katherine Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp, editors, *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 54104–54132. PMLR, 21–27 Jul 2024. URL <https://proceedings.mlr.press/v235/xia24c.html>.

- Chenghao Xiao, Hou Pong Chan, Hao Zhang, Mahani Aljunied, Lidong Bing, Noura Al Moubayed, and Yu Rong. Analyzing llms’ knowledge boundary cognition across languages through the lens of internal representations, 2025. URL <https://arxiv.org/abs/2504.13816>.
- Jiakuan Xie, Pengfei Cao, Yubo Chen, Kang Liu, and Jun Zhao. Revealing the deceptiveness of knowledge editing: A mechanistic analysis of superficial editing. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 17756–17780, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-251-0. doi: 10.18653/v1/2025.acl-long.868. URL <https://aclanthology.org/2025.acl-long.868/>.
- Sang Michael Xie, Hieu Pham, Xuanyi Dong, Nan Du, Hanxiao Liu, Yifeng Lu, Percy Liang, Quoc V. Le, Tengyu Ma, and Adams Wei Yu. DoReMi: Optimizing data mixtures speeds up language model pretraining. In *Advances in Neural Information Processing Systems*, volume 36. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/dcba6be91359358c2355cd920da3fcbd-Abstract-Conference.html.
- Rongwu Xu, Zehan Qi, Zhijiang Guo, Cunxiang Wang, Hongru Wang, Yue Zhang, and Wei Xu. Knowledge conflicts for llms: A survey. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 8541–8565, 2024a.
- Runsen Xu, Xiaolong Wang, Tai Wang, Yilun Chen, Jiangmiao Pang, and Dahua Lin. Pointllm: Empowering large language models to understand point clouds. In *European Conference on Computer Vision*, pages 131–147. Springer, 2024b.
- Xiaoyu Xu, Xiang Yue, Yang Liu, Qingqing Ye, Huadi Zheng, Peizhao Hu, Minxin Du, and Haibo Hu. Unlearning isn’t deletion: Investigating reversibility of machine unlearning in llms, 2026. URL <https://arxiv.org/abs/2505.16831>.
- Zhe Xu, Jiasheng Ye, Xiangyang Liu, Tianxiang Sun, Xiaoran Liu, Qipeng Guo, Linlin Li, Qun Liu, Xuanjing Huang, and Xipeng Qiu. Detectiveqa: Evaluating long-context reasoning on detective novels. *arXiv preprint arXiv:2409.02465*, 2024c.
- Fuzhao Xue, Yao Fu, Wangchunshu Zhou, Zangwei Zheng, and Yang You. To repeat or not to repeat: Insights from scaling LLM under token-crisis. In *Advances in Neural Information Processing Systems*, volume 36. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/b9e472cd579c83e2f6aa3459f46aac28-Abstract-Conference.html.
- Prateek Yadav, Derek Tam, Leshem Choshen, Colin A Raffel, and Mohit Bansal. Ties-merging: Resolving interference when merging models. *Advances in neural information processing systems*, 36:7093–7115, 2023.
- Kai Yan, Yufei Xu, Zhengyin Du, Xuesong Yao, Zheyu Wang, Xiaowen Guo, and Jiecao Chen. Recitation over reasoning: How cutting-edge language models can fail on elementary school-level reasoning problems?, 2025. URL <https://arxiv.org/abs/2504.00509>.

- An Yang et al. Qwen3 technical report, 2025a. URL <https://arxiv.org/abs/2505.09388>.
- Daniel Yang, Yao-Hung Hubert Tsai, and Makoto Yamada. On verbalized confidence scores for llms. *arXiv preprint arXiv:2412.14737*, 2024a.
- Enneng Yang, Zhenyi Wang, Li Shen, Shiwei Liu, Guibing Guo, Xingwei Wang, and Dacheng Tao. Adamerging: Adaptive model merging for multi-task learning. In *International Conference on Learning Representations*, volume 2024, pages 22743–22763, 2024b.
- Enneng Yang, Li Shen, Guibing Guo, Xingwei Wang, Xiaochun Cao, Jie Zhang, and Dacheng Tao. Model merging in llms, mllms, and beyond: Methods, theories, applications, and opportunities. *ACM Comput. Surv.*, 58(8), February 2026a. ISSN 0360-0300. doi: 10.1145/3787849. URL <https://doi.org/10.1145/3787849>.
- Junxiao Yang, Jinzhe Tu, Haoran Liu, Xiaoce Wang, Chujie Zheng, Zhixin Zhang, Shiyao Cui, Caishun Chen, Tiantian He, Hongning Wang, Yew-Soon Ong, and Minlie Huang. Barrel: Boundary-aware reasoning for factual and reliable llms, 2025b. URL <https://arxiv.org/abs/2505.13529>.
- Sohee Yang, Elena Gribovskaya, Nora Kassner, Mor Geva, and Sebastian Riedel. Do large language models latently perform multi-hop reasoning?, 2025c. URL <https://arxiv.org/abs/2402.16837>.
- Wanli Yang, Fei Sun, Xinyu Ma, Xun Liu, Dawei Yin, and Xueqi Cheng. The butterfly effect of model editing: Few edits can trigger large language models collapse. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Findings of the Association for Computational Linguistics: ACL 2024*, pages 5419–5437, Bangkok, Thailand, August 2024c. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.322. URL <https://aclanthology.org/2024.findings-acl.322/>.
- Wanli Yang, Hongyu Zang, Junwei Zhang, Wenjie Shi, Du Su, Jingang Wang, Xueqi Cheng, and Fei Sun. Beyond reasoning: Reinforcement learning unlocks parametric knowledge in llms, 2026b. URL <https://arxiv.org/abs/2605.07153>.
- Wenkai Yang, Weijie Liu, Ruobing Xie, Kai Yang, Saiyong Yang, and Yankai Lin. Learning beyond teacher: Generalized on-policy distillation with reward extrapolation, 2026c. URL <https://arxiv.org/abs/2602.12125>.
- Xiao Yang, Kai Sun, Hao Xin, Yushi Sun, Nikita Bhalla, Xiangsen Chen, Sajal Choudhary, Rongze Daniel Gui, Ziran Will Jiang, Ziyu Jiang, Lingkun Kong, Brian Moran, Jiaqi Wang, Yifan Ethan Xu, An Yan, Chenyu Yang, Eting Yuan, Hanwen Zha, Nan Tang, Lei Chen, Nicolas Scheffer, Yue Liu, Nirav Shah, Rakesh Wanga, Anuj Kumar, Wentao Yih, and Xin Luna Dong. Crag – comprehensive rag benchmark, 2024d. URL <https://arxiv.org/abs/2406.04744>.
- Yunhan Yang, Yukun Huang, Yuan-Chen Guo, Liangjun Lu, Xiaoyang Wu, Edmund Y. Lam, Yan-Pei Cao, and Xihui Liu. Sampart3d: Segment any part in 3d objects, 2024e. URL <https://arxiv.org/abs/2411.07184>.

- Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of EMNLP*, pages 2369–2380, 2018. doi: 10.18653/v1/D18-1259. URL <https://aclanthology.org/D18-1259>.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. React: Synergizing reasoning and acting in language models, 2023a. URL <https://arxiv.org/abs/2210.03629>.
- Yunzhi Yao, Peng Wang, Bozhong Tian, Siyuan Cheng, Zhoubo Li, Shumin Deng, Huajun Chen, and Ningyu Zhang. Editing large language models: Problems, methods, and opportunities. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023b.
- Yunzhi Yao, Ningyu Zhang, Zekun Xi, Mengru Wang, Ziwen Xu, Shumin Deng, and Huajun Chen. Knowledge circuits in pretrained transformers. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 118571–118602. Curran Associates, Inc., 2024a. doi: 10.52202/079017-3765. URL https://proceedings.neurips.cc/paper_files/paper/2024/file/d6df31b1be98e04be48af8bedb95b499-Paper-Conference.pdf.
- Zijun Yao, Weijian Qi, Liangming Pan, Shulin Cao, Linmei Hu, Weichuan Liu, Lei Hou, and Juanzi Li. Seakr: Self-aware knowledge retrieval for adaptive retrieval augmented generation, 2024b. URL <https://arxiv.org/abs/2406.19215>.
- Evelyn Yee, Alice Li, Chenyu Tang, Yeon Ho Jung, Ramamohan Paturi, and Leon Bergen. Dissociation of faithful and unfaithful reasoning in llms. *arXiv preprint arXiv:2405.15092*, 2024.
- Howard Yen, Tianyu Gao, Minmin Hou, Ke Ding, Daniel Fleischer, Peter Izsak, Moshe Wasserblat, and Danqi Chen. Helmet: How to evaluate long-context language models effectively and thoroughly. *arXiv preprint arXiv:2410.02694*, 2024.
- Edward Yeo, Yuxuan Tong, Morry Niu, Graham Neubig, and Xiang Yue. Demystifying long chain-of-thought reasoning in llms, 2025. URL <https://arxiv.org/abs/2502.03373>.
- Gal Yona, Roei Aharoni, and Mor Geva. Narrowing the knowledge evaluation gap: Open-domain question answering with multi-granularity answers, 2024. URL <https://arxiv.org/abs/2401.04695>.
- Zheng-Xin Yong, M. Farid Adilazuarda, Jonibek Mansurov, Ruochen Zhang, Niklas Muenighoff, Carsten Eickhoff, Genta Indra Winata, Julia Kreutzer, Stephen H. Bach, and Alham Fikri Aji. Crosslingual reasoning through test-time scaling, 2025. URL <https://arxiv.org/abs/2505.05408>.
- Hao Yu, Aoran Gan, Kai Zhang, Shiwei Tong, Qi Liu, and Zhaofeng Liu. Evaluation of retrieval-augmented generation: A survey. In *CCF Conference on Big Data*, pages 102–120. Springer, 2024a.

- Jifan Yu, Xiaozhi Wang, Shangqing Tu, Shulin Cao, Daniel Zhang-Li, Xin Lv, Hao Peng, Zijun Yao, Xiaohan Zhang, Hanming Li, et al. Kola: Carefully benchmarking world knowledge of large language models. *arXiv preprint arXiv:2306.09296*, 2023.
- Le Yu, Bowen Yu, Haiyang Yu, Fei Huang, and Yongbin Li. Language models are super mario: Absorbing abilities from homologous models as a free lunch. In *Forty-first International Conference on Machine Learning*, 2024b.
- Yijiong Yu. Llms do not think step-by-step in implicit reasoning, 2024. URL <https://arxiv.org/abs/2411.15862>.
- Mohammad Zbeeb, Hasan Abed Al Kader Hammoud, and Bernard Ghanem. Reasoning vectors: Transferring chain-of-thought capabilities via task arithmetic, 2025. URL <https://arxiv.org/abs/2509.01363>.
- Ji Zeng, Dayuan Fu, Tiantian Mi, Yumin Zhuang, Yaxing Huang, Xuefeng Li, Lyumanshan Ye, Muhang Xie, Qishuo Hua, Zhen Huang, Mohan Jiang, Hanning Wang, Jifan Lin, Yang Xiao, Jie Sun, Yunze Wu, and Pengfei Liu. daVinci-Dev: Agent-native mid-training for software engineering, 2026. URL <https://arxiv.org/abs/2601.18418>.
- Chi Zhang, Mengqi Zhang, Xiaotian Ye, Runxi Cheng, Zisheng Zhou, Ying Zhou, Pengjie Ren, and Zhumin Chen. Spectral characterization and mitigation of sequential knowledge editing collapse, 2026a. URL <https://arxiv.org/abs/2601.11042>.
- Jiajie Zhang, Nianyi Lin, Lei Hou, Ling Feng, and Juanzi Li. Adaptthink: Reasoning models can learn when to think. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 3716–3730, 2025a.
- Jifan Zhang, Ziyue Luo, Jia Liu, Ness Shroff, and Robert Nowak. Gpt-4o as the gold standard: A scalable and general purpose approach to filter language model pretraining data, 2025b. URL <https://arxiv.org/abs/2410.02755>.
- Jinghan Zhang, Junteng Liu, Junxian He, et al. Composing parameter-efficient modules with arithmetic operation. *Advances in Neural Information Processing Systems*, 36: 12589–12610, 2023.
- Mengqi Zhang, Xiaotian Ye, Qiang Liu, Pengjie Ren, Shu Wu, and Zhumin Chen. Uncovering overfitting in large language model editing, 2025c. URL <https://arxiv.org/abs/2410.07819>.
- Ningyu Zhang, Yunzhi Yao, Bozhong Tian, Peng Wang, Shumin Deng, Mengru Wang, Zekun Xi, Shengyu Mao, Jintian Zhang, Yuansheng Ni, et al. A comprehensive study of knowledge editing for large language models. *arXiv preprint arXiv:2401.01286*, 2024a.
- Ruichen Zhang, Rana Muhammad Shahroz Khan, Zhen Tan, Dawei Li, Song Wang, and Tianlong Chen. The quest for efficient reasoning: A data-centric benchmark to cot distillation, 2026b. URL <https://arxiv.org/abs/2505.18759>.

- Ruiqi Zhang, Licong Lin, Yu Bai, and Song Mei. Negative preference optimization: From catastrophic collapse to effective unlearning. In *Conference on Language Modeling (COLM)*, 2024b.
- Zhaoyang Zhang, Shuli Jiang, Yantao Shen, Yuting Zhang, Dhananjay Ram, Shuo Yang, Zhuowen Tu, Wei Xia, and Stefano Soatto. Reinforcement-aware knowledge distillation for llm reasoning, 2026c. URL <https://arxiv.org/abs/2602.22495>.
- Zhexin Zhang, Junxiao Yang, Yida Lu, Pei Ke, Shiyao Cui, Chujie Zheng, Hongning Wang, and Minlie Huang. From theft to bomb-making: The ripple effect of unlearning in defending against jailbreak attacks, 2025d. URL <https://arxiv.org/abs/2407.02855>.
- Chengshuai Zhao, Zhen Tan, Pingchuan Ma, Dawei Li, Bohan Jiang, Yancheng Wang, Yingzhen Yang, and Huan Liu. Is chain-of-thought reasoning of llms a mirage? a data distribution lens, 2025a. URL <https://arxiv.org/abs/2508.01191>.
- James Xu Zhao, Bryan Hooi, and See-Kiong Ng. Test-time scaling in reasoning models is not effective for knowledge-intensive tasks yet, 2025b. URL <https://arxiv.org/abs/2509.06861>.
- Jun Zhao, Zhihao Zhang, Luhui Gao, Qi Zhang, Tao Gui, and Xuanjing Huang. Llama beyond english: An empirical study on language capability transfer, 2024a. URL <https://arxiv.org/abs/2401.01055>.
- Qinghua Zhao, Xueling Gong, Xinyu Chen, Zhongfeng Kang, and Xinlu Li. A layer-wise analysis of supervised fine-tuning, 2026a. URL <https://arxiv.org/abs/2604.11838>.
- Siyan Zhao, Zhihui Xie, Mengchen Liu, Jing Huang, Guan Pang, Feiyu Chen, and Aditya Grover. Self-distilled reasoner: On-policy self-distillation for large language models, 2026b. URL <https://arxiv.org/abs/2601.18734>.
- Tianzhe Zhao, Jiaoyan Chen, Shuxiu Zhang, Haiping Zhu, Qika Lin, and Jun Liu. Exploring knowledge conflicts for faithful llm reasoning: Benchmark and method. *arXiv preprint arXiv:2604.11209*, 2026c.
- Yu Zhao, Alessio Devoto, Giwon Hong, Xiaotang Du, Aryo Pradipta Gema, Hongru Wang, Kam-Fai Wong, and Pasquale Minervini. Steering knowledge selection behaviours in llms via sae-based representation engineering, 2024b. URL <https://arxiv.org/abs/2410.15999>.
- Yu Zhao, Xiaotang Du, Giwon Hong, Aryo Pradipta Gema, Alessio Devoto, Hongru Wang, Xuanli He, Kam-Fai Wong, and Pasquale Minervini. Analysing the residual stream of language models under knowledge conflicts, 2024c. URL <https://arxiv.org/abs/2410.16090>.
- Yufeng Zhao, Junnan Liu, Hongwei Liu, Dongsheng Zhu, Yuan Shen, Songyang Zhang, and Kai Chen. Dissecting tool-integrated reasoning: An empirical study and analysis, 2025c. URL <https://arxiv.org/abs/2508.15754>.

- Zhongyu Zhao, Menghang Dong, Rongyu Zhang, Wenzhao Zheng, Yunpeng Zhang, Huanrui Yang, Dalong Du, Kurt Keutzer, and Shanghang Zhang. Factorllm: Factorizing knowledge via mixture of experts for large language models, 2024d. URL <https://arxiv.org/abs/2408.11855>.
- Boyuan Zheng, Michael Y. Fatemi, Xiaolong Jin, Zora Zhiruo Wang, Apurva Gandhi, Yueqi Song, Yu Gu, Jayanth Srinivasa, Gaowen Liu, Graham Neubig, and Yu Su. Skillweaver: Web agents can self-improve by discovering and honing skills, 2025. URL <https://arxiv.org/abs/2504.07079>.
- Ce Zheng, Lei Li, Qingxiu Dong, Yuxuan Fan, Zhiyong Wu, Jingjing Xu, and Baobao Chang. Can we edit factual knowledge by in-context learning? In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023a.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in Neural Information Processing Systems*, 36: 46595–46623, 2023b.
- Shu Zhong, Mingyu Xu, Tenglong Ao, and Guang Shi. Understanding transformer from the perspective of associative memory, 2025. URL <https://arxiv.org/abs/2505.19488>.
- Zexuan Zhong, Zhengxuan Wu, Christopher D. Manning, Christopher Potts, and Danqi Chen. MQuAKE: Assessing knowledge editing in language models via multi-hop questions. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- Chenxi Zhou, Pengfei Cao, Jiang Li, Bohan Yu, Jinyu Ye, Jun Zhao, and Kang Liu. From signal degradation to computation collapse: Uncovering the two failure modes of llm quantization. *arXiv preprint arXiv:2604.19884*, 2026a.
- Hongyi Zhou, Kai Ye, Erhan Xu, Jin Zhu, Ying Yang, Shijin Gong, and Chengchun Shi. Demystifying group relative policy optimization: Its policy gradient is a u-statistic, 2026b. URL <https://arxiv.org/abs/2603.01162>.
- Yang Zhou, Pengfei Cao, Yubo Chen, Qingbin Liu, Dianbo Sui, Xi Chen, Kang Liu, and Jun Zhao. M2Edit: Locate and edit multi-granularity knowledge in multimodal large language model. In Christos Christodoulopoulos, Tanmoy Chakraborty, Carolyn Rose, and Violet Peng, editors, *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 29029–29042, Suzhou, China, November 2025a. Association for Computational Linguistics. ISBN 979-8-89176-332-6. doi: 10.18653/v1/2025.emnlp-main.1478. URL <https://aclanthology.org/2025.emnlp-main.1478/>.
- Zeqi Zhou, Fang Wu, Shayan Talaei, Haokai Zhao, Cheng Meixin, Tinson Xu, Amin Saberi, and Yejin Choi. When to trust context: Self-reflective debates for context reliability, 2025b. URL <https://arxiv.org/abs/2506.06020>.
- Chen Zhu, Ankit Singh Rawat, Manzil Zaheer, Srinadh Bhojanapalli, Daliang Li, Felix Yu, and Sanjiv Kumar. Modifying memories in transformer models. *arXiv preprint arXiv:2012.00363*, 2020.

- Kaijie Zhu, Jiaao Chen, Jindong Wang, Neil Zhenqiang Gong, Diyi Yang, and Xing Xie. Dyval: Graph-informed dynamic evaluation of large language models. *arXiv e-prints*, pages arXiv–2309, 2023.
- Siqi Zhu, Xuyan Ye, Hongyu Lu, Weiye Shi, and Ge Liu. The many faces of on-policy distillation: Pitfalls, mechanisms, and fixes, 2026. URL <https://arxiv.org/abs/2605.11182>.
- Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J. Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J. Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.